

Project Online Data Migration

Project & Resources V2

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Document Details

Version History

Version	Date	Author	Details
0.1	17 th March 2026	John Price	First draft of the procedural instructions.
1	7 th April 2026	John Price	Completion of Project Online migration steps.
2	9 th April 2026	John Price	Completion of SharePoint Online migration steps.
3	14 th April 2026	John Price	Updates based on Altus Partner feedback.
4	15 th April 2026	John Price	Updates based on Altus Partner feedback.
5	16 th April 2026	John Price	Updates based on Altus Partner feedback.
6	20 th April 2026	John Price	Updates based on Altus Partner feedback.
7	22 nd April 2026	John Price	Addition of the upcoming changes section.
8	24 th April 2026	John Price	Addition of the following changes: 1. Clarifications regarding the ClientID that needs to be set for these scripts (-Run-Migration.ps1, ExportAllProjects.ps1, and ExportLookupTableData.ps1) 2. Added in syntax for alternative ODATA filters that can be used to extract information from Project Online.
9	27 th April 2026	John Price	Addition of troubleshooting steps for renamed project that hasn't been published.
10	1 st May 2026	John Price	Updates based on Altus Partner feedback.
11	1 st May 2026	John Price	Prerequisites and troubleshooting moved to separate documents.
12	4 th May 2026	John Price	Updated task custom field migration process section.
13	29 th May 2026	John Price	Updated to include the task custom field process from Project Online to Altus (bypassing MS Project Desktop).
14	11 th June 2026	John Price	Correction to quick steps on Page 102 (Section 9.2) and addition of video links for 9.1 and 9.2.
15	12 th June 2026	John Price	Added a post-migration steps showing how to share the Altus app with custom security roles.
16	15 th June 2026	John Price	Two changes: 1. Pages 18 - 19 Clearer details on the two migration paths for task custom fields. 2. Pages 27 - 28 More detailed examples of how to use the ProjectOnlineOdataFilter.
17	8 th July 2026	John Price	Update external links.
18	30 th July 2026	John Price	Added section 10 covering how to migrate calendars and resource exceptions to Altus.

Document Purpose

The purpose of this document is to provide the reader with detailed information about how to migrate project schedules from Project Online to Altus.

Document Type

This is a procedural 'follow the bouncing-ball' type of document. It is highly detailed and is designed for the user to follow the steps to complete the end-to-end data migration. There are certain areas where scripts need to be modified. Examples of these are provided to assist the reader with the script modifications.

Frequently Asked Questions

We have created a separate document covering frequently asked questions. This can be accessed via this link and covers all migration process:

[Project Online Data Migration - Frequently Asked Questions](#)

Migration Overview

Introduction

On the 5th September 2025 Microsoft announced that it had brought forward the original retirement of Project Online from 2030 to 30th September 2026.

From this date, Project Online will be shut down and no longer accessible. All Project Web App (PWA) access, data, and integrations will stop working.

What is being retired?

Project Online will be fully decommissioned. This includes the SharePoint Online project subsites where users store list-based information such as Risks, Issues, Actions, and Project Documentation.

What will remain available?

The following applications will remain available to new and existing customers:

- Project Desktop (Stand-alone and M365 versions).
- [Project Server Subscription Edition](#).

Why is Microsoft retiring Project Online?

Microsoft has stated that the Project Online legacy architecture (built on older SharePoint Online services) limits the ability to deliver modern cloud, AI, and integration features. Innovation investment is now focused on newer platforms built on the Microsoft Power Platform, Dataverse, and AI-enabled capabilities (such as Planner Premium/Project Manager Agent).

The call to action

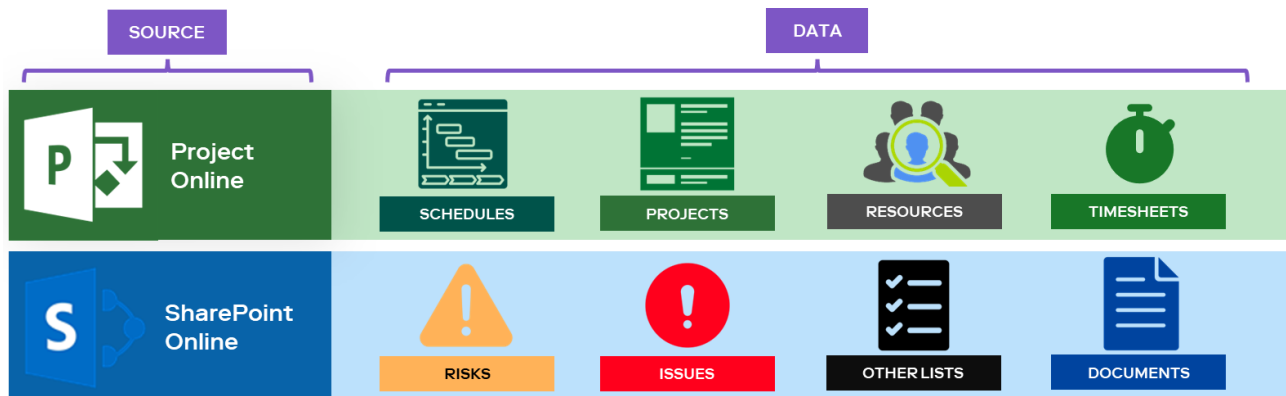
It is imperative that Partners and Customers work quickly and diligently to migrate their data out of Project Online as soon as possible to avoid loss of valuable corporate data. The process covered in this document will assist in that endeavour.

Altus Migration Solution

To assist new and existing Altus customers, we have created scripts and an approach to enable customers to retain their project online data when migrating to Altus PPM.

Data that will be migrated

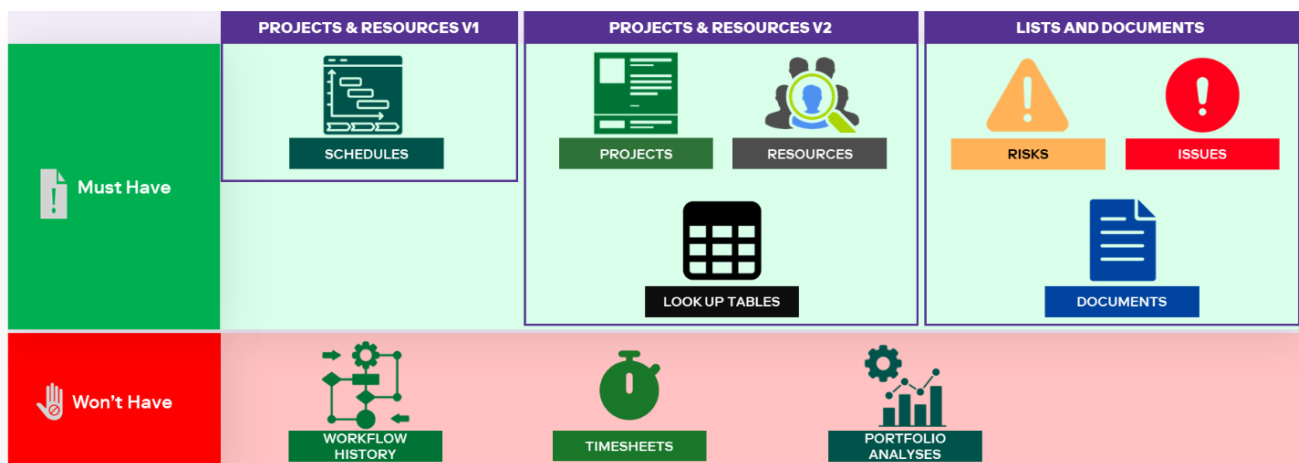
The following data has been identified for data migration as part of the retirement of Project Online:



This process enables the user to migrate the following data to Altus PPM:

Source	Data	Details
Project Online	Projects	Including custom field data and lookup tables.
	Resources	Including custom field data and lookup tables.
	Schedules	Including custom field data, resource rates, and baselines.
SharePoint Online	List Items	Including risks, issues, and custom lists.
	Documents	Export project document for upload to Teams or SharePoint.

The diagram below articulates what the scripts will and won't migrate:



Data that won't be migrated

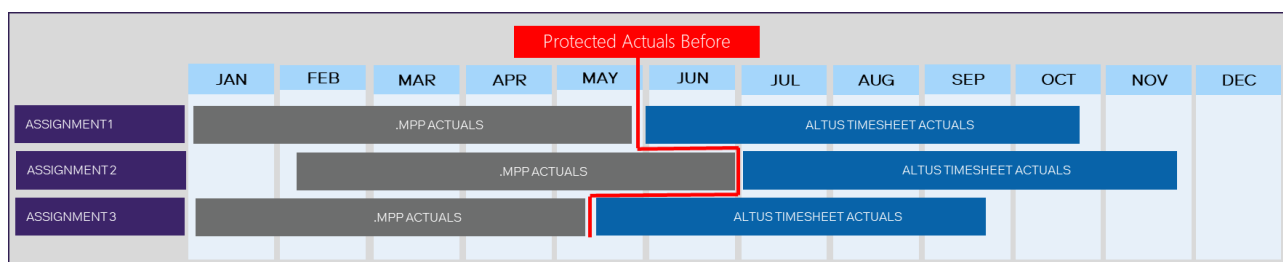
The following data won't be migrated:

- **Workflow History:** Data covering the progression of projects through a controlled lifecycle.
- **Timesheets:** Data captured in the Project Online timesheet and published to schedules.
- **Portfolio Analyses:** Data captured when portfolio modelling is created in Project Online.

Timesheets Data

The Altus approach is to ensure that Project Desktop actuals are not overwritten when approved Altus timesheets are imported. As such, we have devised an approach using a date to honour 'protected actuals' that will work by:

- Ensuring that actuals in the .MPP schedule file are not overwritten, we have implemented logic to prevent this from occurring.
- Using an assignment-level date field referred to as Protected Actuals Before.
- This is the date when approved Altus timesheets actuals are first imported to the .MPP file.
- This ensures that existing historical actuals are protected and not overwritten when approved timesheet actuals are imported to the .MPP file.
- Note, this date is set at the assignment-level rather than the task-level. This ensures that actuals are protected at the appropriate level and cater for timesheet actuals being processed by different resources at different dates.
- This date enables Partners and Customers to avoid migrating timesheets as this date will preserve historical actuals in the MPP.



Workflow History

This captures the history of project approvals as they progress through stage gates within the project lifecycle. Given the relatively low frequency of these transactions, the small data volumes involved, and variations across project lifecycles, it was determined that this would be best addressed as a post-migration activity. The effort required to align projects to the appropriate stage following migration is minimal.

Portfolio Analyses

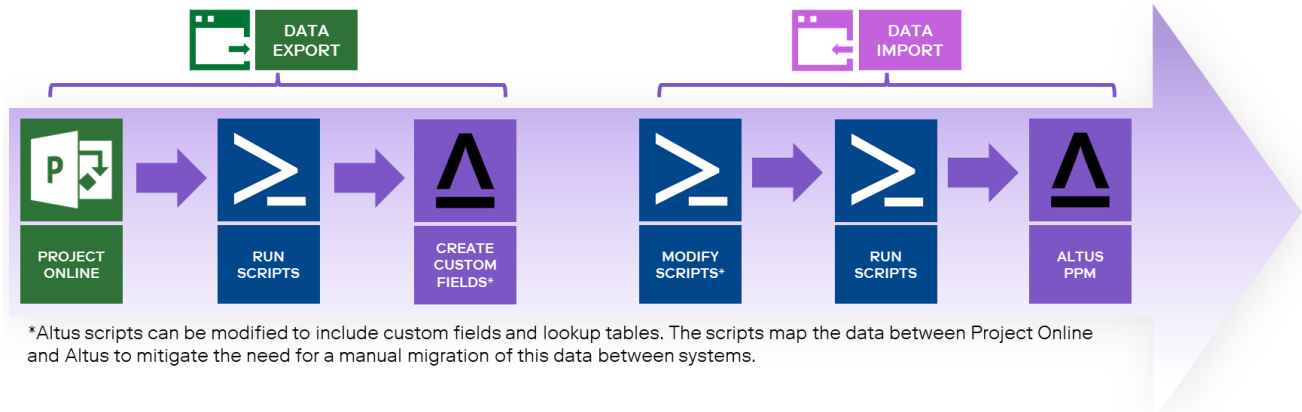
The Portfolio Analyser has been a core feature of Project Server since 2010 and is also available in Project Online. It enables organisations to assess which projects can be delivered within budget and resource constraints, as well as to model "what-if" scenarios.

Altus provides an equivalent capability through its Strategic Prioritisation module. While the migration of Portfolio Analyser data is not supported within the standard data export scripts provided by

Microsoft, the effort required to re-establish this information post-migration is minimal. As such, the overall impact is limited.

Migration Approach

The diagram below articulates the end-to-end process for exporting data from Project Online and publishing it to Altus.



Migration Outcomes

At the culmination of the migration, the following will be migrated to Altus:

- **Enterprise Resources** are created (Referred to as Bookable Resources).
- **Projects** (Created in the Initiatives section of Altus).
- **Schedules** (Migrated to the tasks section within an Altus Project).
- **Lists Data** is migrated to the relevant Altus register.
- **Documents** are exported.

Projects & Resources Data Migrated

The tables below articulate the out-of-the-box (OOTB) data that is migrated to Altus. These can be extended by modifying the Altus data migration scripts (examples are provided later in this document).

.MPP Table	Altus Table
Task Entry (<i>Task Level 0</i>)	sensei_project
Resource Entry	sensei_bookableresource
Task Entry	sensei_task

Below are the OOTB fields that are migrated from Project Online to Altus. This can be extended by updating the scripts to map custom fields.

Projects

.MPP Field	Altus Field	Details	Example
Task Name	sensei_name	Where the TaskID = 0	Adelaide Fitness Project
N/A	sensei_projecttype	Default value set	Major Project
N/A	sensei_externalprojectid	Set the GUID = Project Online GUID	ProjectDesktop_d14bae12-39b3-e911-b07b-00155d9c3f18

Resources

.MPP Field	Altus Field	Details	Example
Name	sensei_name	Firstname Surname	Lucius Fox
User Logon Account	sensei_user	Firstname Surname	Lucius Fox
Max. Units	sensei_targetutilization	Default value set	100
Type	sensei_resourcetype	Default value set	Named
Role	sensei_primaryrole	Default value set	Unclassified
Calendar GUID	Sensei_enterprisecalendar	Default value set	Default Calendar

Tasks

.MPP Field	Altus Field	Example Data
Task Name	Name	Identify Goals and Objectives
Resource Names	Assigned To	Project Manager
Start	Start	2-12-19
Finish	Finish	6-12-19
% Complete	Percent Complete	25%
Work	Effort (Hours)	40
Remaining Work	Effort Remaining (Hours)	30
Actual Work	Effort Completed (Hours)	10
Duration	Duration (Days)	5 days
Type	Task Type	Fixed Units
Task Mode	Manually Schedule	Yes
Notes	Notes	Goals were discussed
Deadline	Due Date	6-12-19
Summary	Summary	No
Milestone	Milestone	No
Locked	Locked	No

SharePoint List Data Migrated

The tables below articulate the OOTB data that is migrated to Altus. These can be extended by modifying the Altus data migration scripts (examples are provided later in this document).

SharePoint List	Altus Table
Risks	sensei_risk
Issues	sensei_issue

Risks

SharePoint Field	Altus Field	Example Data
Title	sensei_name	Requirements Not Clearly Defined
Description	sensei_description	Project requirements are incomplete or ambiguous, leading to rework and delays.
DueDate	sensei_duedate	30/04/2026 0:00
Status	statuscode	(1) Active
Category	sensei_category	(2) Category2
Probability	sensei_likelihood	50%
Impact	sensei_consequence	5
Exposure	sensei_riskscore	2.5
Cost	sensei_cost	\$10,000.00
Mitigation_x0020_plan	sensei_mitigationplan	Conduct workshops with stakeholders, document requirements, and obtain formal sign-off.
Contingency_x0020_plan	sensei_contingencyplan	Rebaseline scope and timeline if major requirement changes emerge.
Trigger_x0020_Description	sensei_triggerdescription	When the due date is lapsed, a decision will be require.
Trigger	sensei_trigger	Date
Owner	sensei_riskowner	John Price
AssignedTo	sensei_assignedto	Marc Soester

Issues

SharePoint Field	Altus Field	Example Data
Title	sensei_name	Resource Overallocation
Discussion	sensei_description	Key developer assigned to multiple projects causing delays.
DueDate	sensei_duedate	24/04/2026 00:00
Status	statuscode	(1) Active
Category	sensei_category	(3) Category3
Priority	sensei_priority	(2) Medium
AssignedTo	sensei_assignedto	John Price
Cost_x0020_Impact	sensei_costimpact	\$10,000.00
Resolution	sensei_resolution	Rebalance workload and secure additional support.

Custom Lists

The scripts can be extended to enable custom SharePoint list data to be migrated to Altus. This works by reading the lists from the associated SharePoint sub-sites that are attached to Project Online projects. All column types within these custom lists can be migrated.

Constraints

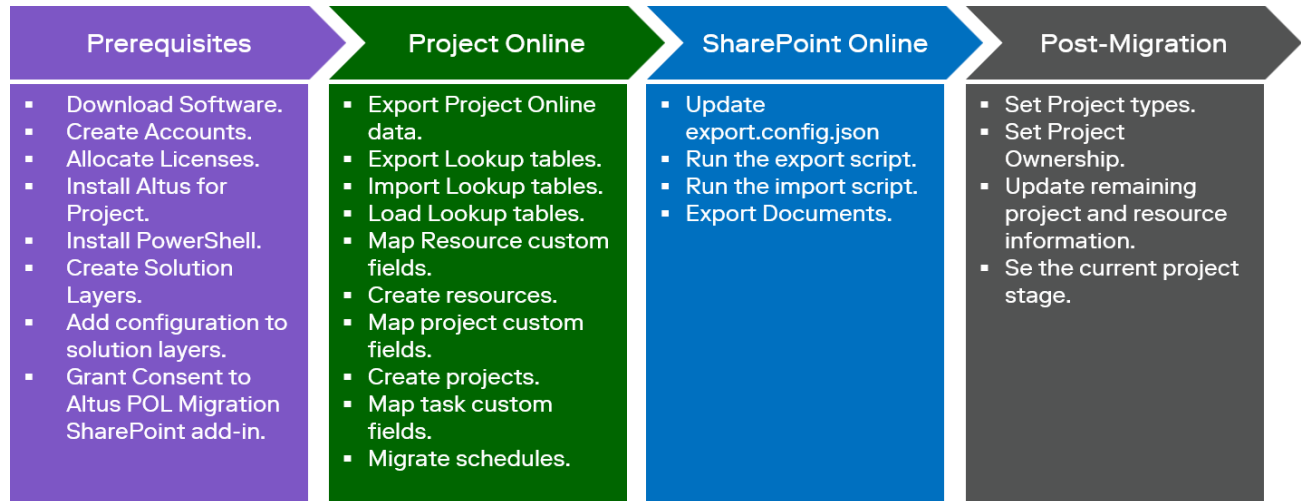
The Altus migration solution for Projects, Resources and Schedules has the following constraints that the user should be aware of prior to running the process.

Constraint	Details
Bookable Resources	The resource creation process will only create new Bookable Resources in Altus where there is a match to a system user in Dataverse. <u>Impact:</u> ▪ If there is no corresponding system user, a bookable resource won't be created in Altus. ▪ If the bookable resource isn't created, the assignments won't be created in Altus, which then results in tasks being unassigned in the Altus project. <u>Call to action:</u> ▪ You will need to ensure that at a minimum system user records exist in the Power Platform environment where Altus has been deployed. ▪ To do this, please follow these steps documented by Microsoft: Create users - Power Platform Microsoft Learn
Non-Text Lookup tables	The Microsoft export scripts that are used to export data from Project Online don't output any data for non-text lookup tables. T <u>Impact:</u> ▪ You won't be able to use the Altus scripts to import this data into Altus as the source files are blank. <u>Call to action:</u> ▪ As such, you will have to manually migrate these.

Migration Process

Overview

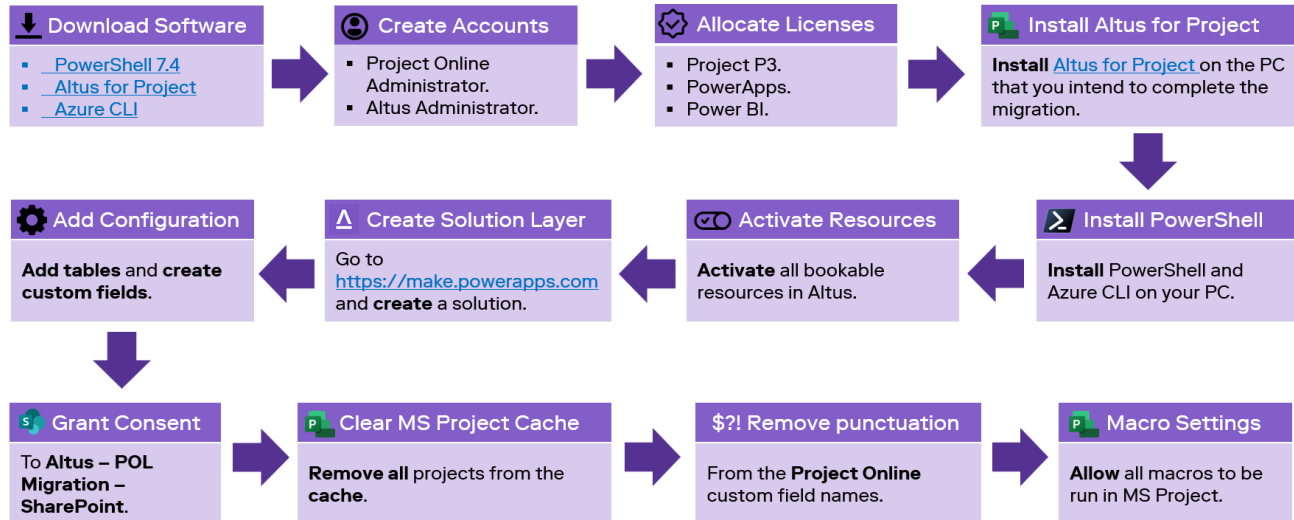
The migration process has three key steps that need to be completed:



1. **Prerequisites** | These must be completed before the migration can be started.
2. **Project Online data migration** | Export Project Online data and import to Altus.
3. **SharePoint Online data migration** | Export SharePoint Online data and import to Altus.
4. **Post-Migration** | These are steps to complete once the data migration scripts have been run.

Prerequisites

To run the schedule migration process, you will need to ensure the following is in place prior to running the data migration scripts:



Prerequisites	Details
1. Download Software	Download PowerShell and Altus for Project.
2. Create Accounts	Create Project Online and Altus account for data export and import.
3. Allocate Licenses	Allocate the required licenses to the accounts.
4. Install Altus for Project	Install Altus for Project so that schedules can be connected and published.
5. Install PowerShell and Azure CLI	Install PowerShell so that the migration scripts can be run.
6. Activate Resources	Activate all bookable resources in Altus.
7. Create Solution Layers	Create Solution layers for lookup tables and custom fields.
8. Add configuration to a Solution Layer	Add new custom fields to existing tables. Create new lookup tables and add these to the solution layer.
9. Grant consent to “Altus – POL Migration – SharePoint”	The Altus data migration process requires permissions to the SharePoint Online site collection to get the required data from both Project Online and SharePoint Online.
10. Clear cache in MS Project	You will need to ensure you have cleaned out your cache in MS Project to avoid possible issues with MS Project.
11. Remove punctuation from Project Online custom fields	There is a known issue with the Microsoft export scripts (which Altus have modified to export data from Project Online) which prevents the process from exporting data where custom fields have punctuation in the title. These fields will need to be updated.
12. Change Macros settings in MS Project	To prevent the scripts from getting stuck during the migration process, it is recommended that you change the macro settings in MS Project.

All details about the prerequisites can be access in this document: [\[POL\] Prerequisites](#)

Project Online Data Migration

Summary

Once all the prerequisites have been met, the data from Project Online can be exported.

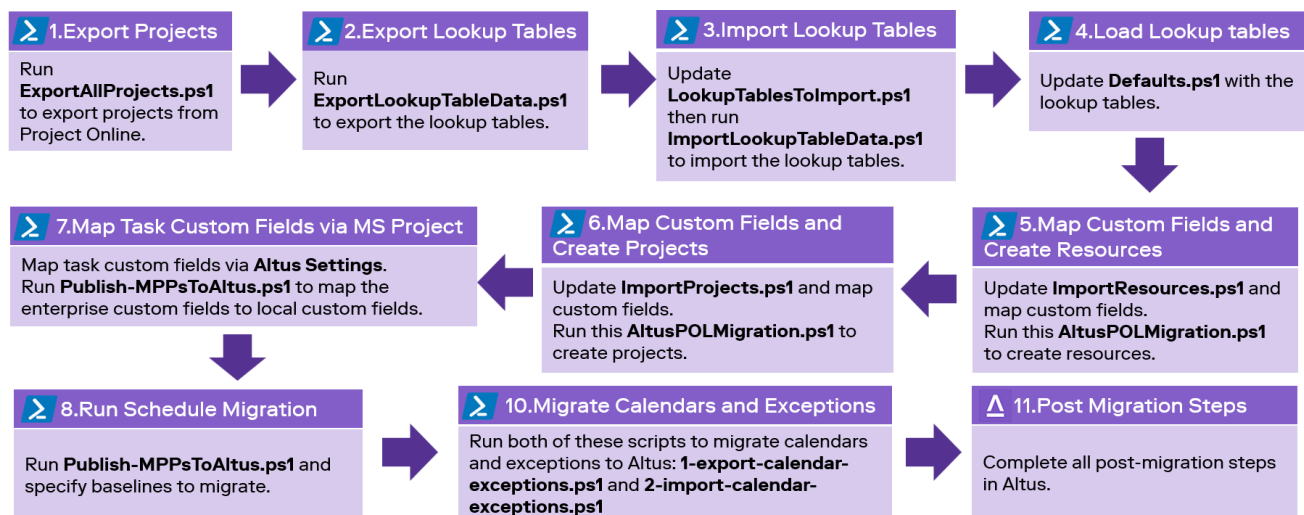
There are two pathways for the migration of data from Project Online to Altus:

1. Migrate task custom fields to Microsoft Project.
2. Migrate task custom fields to Altus.

The tables below articulate these pathways:

Migrate Task Custom Fields to Microsoft Project

Follow this process if you wish to retain the task custom fields in Altus, noting the limitations covered on [this page](#) of this document.

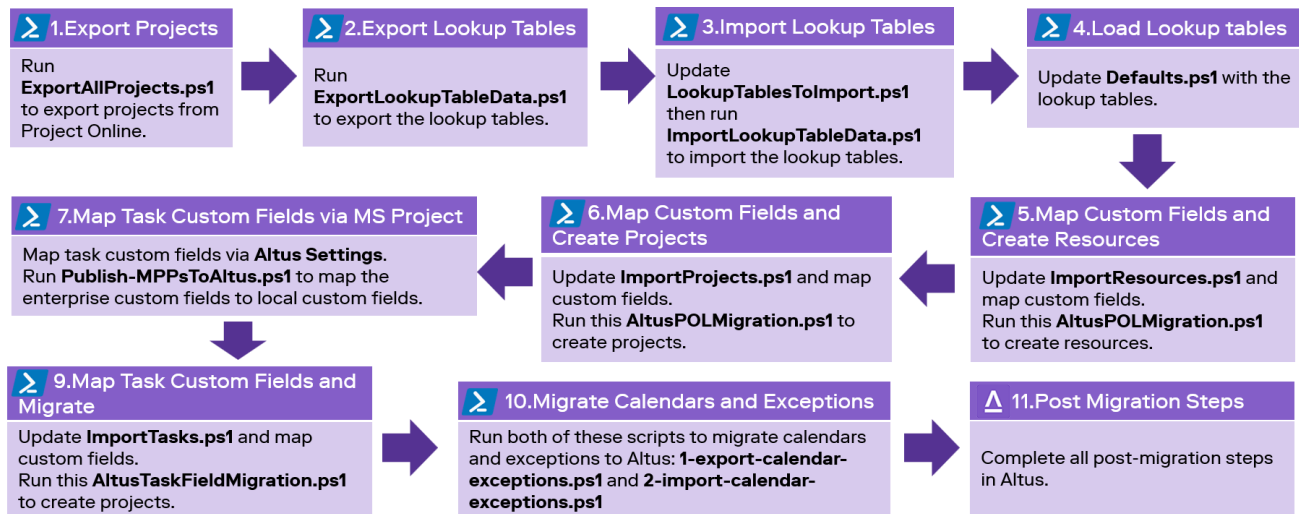


Step	Details
1. Export Project Online Data	Run this script to export data: ExportAllProjects.ps1
2. Export Lookup Tables	Update and run this script to map Lookup tables for migration: LookupTablesToImport.ps1
3. Import Lookup tables	Update and run this script to migrate the lookup tables: .\ImportLookupTableData.ps1
4. Load Lookup tables	Update Defaults.ps1 to load lookup tables for projects.
5. Map Resources Custom Fields and Create Resources	Update and run this script to map project custom fields for migration: ImportResources.ps1
6. Map Project Custom Fields and Create Projects	Update and run this script to map project custom fields for migration: ImportProjects.ps1
7. Map Task Custom Fields	Open Altus Settings and map the Project Schedule Task fields to Altus task custom fields. Note You only need to run this part of the process if you are mapping to local MS Project desktop custom fields.
8. Run Schedule Migration	Run this script: Publish-MPPsToAltus.ps1

Step	Details
10. Migrate Calendars and Exceptions	The final script to run is to migrate the enterprise calendars and exceptions. This will migrate calendar and resource exceptions to Altus.
11. Post Migration Steps	Complete the post migration steps documented.

Migrate Task Custom Fields to Altus

If you wish to migrate task custom fields directly to Altus and bypass Microsoft Project, please follow this process detailed below:



Step	Details
1. Export Project Online Data	Run this script to export data: ExportAllProjects.ps1
2. Export Lookup Tables	Update and run this script to map Lookup tables for migration: LookupTablesToImport.ps1
3. Import Lookup tables	Update and run this script to migrate the lookup tables: .\ImportLookupTableData.ps1
4. Load Lookup tables	Update Defaults.ps1 to load lookup tables for projects.
5. Map Resources Custom Fields and Create Resources	Update and run this script to map project custom fields for migration: ImportResources.ps1
6. Map Project Custom Fields and Create Projects	Update and run this script to map project custom fields for migration: ImportProjects.ps1
8. Run Schedule Migration	Run this script: Publish-MPPsToAltus.ps1
9. Map Tasks Custom Fields and Migrate Data	Migrate task level custom fields directly to Altus (including lookup tables). Note You only need to run this part of the process if you are migrating custom fields directly to Altus.
10. Migrate Calendars and Exceptions	The final script to run is to migrate the enterprise calendars and exceptions. This will migrate calendar and resource exceptions to Altus.
11. Post Migration Steps	Complete the post migration steps documented.

Before you start this process, you will need to have:

- Complete the [\[POL\] Prerequisites](#)
- Create the relevant tables and custom fields.
- Add the fields to the relevant forms.

1. Export Project Online Data

This process will download all projects and schedules from the source Project Online environment:

1. Download the scripts.
2. Run the export scripts.

To complete the schedule export process, you will need to follow these steps:

1. Download the scripts

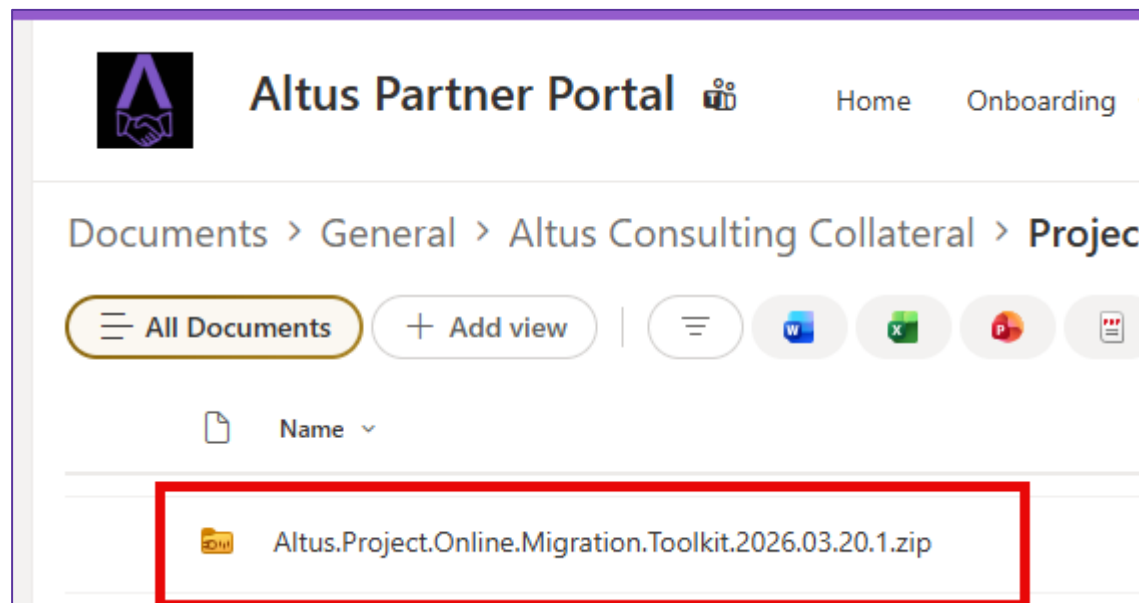
Quick Summary:

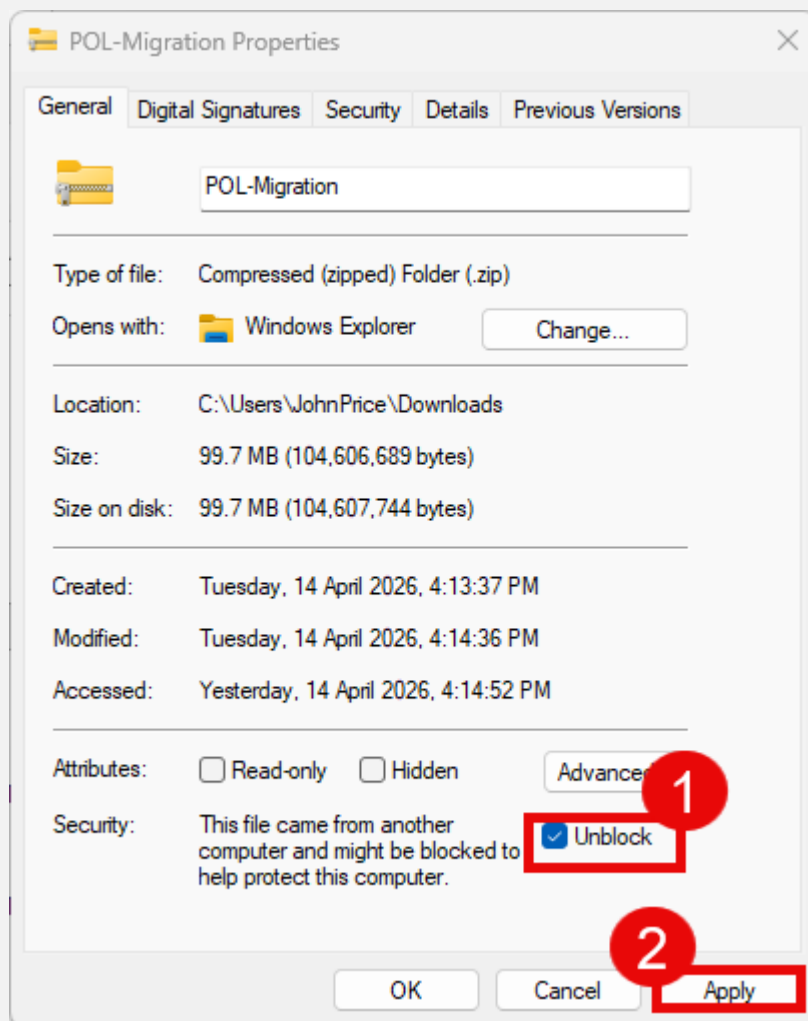
1. **Open** this link: [Project Online Migration](#).
2. **Download** this .ZIP folder: [Altus.Project.Online.Migration.Toolkit.2026.05.26.1.zip](#).
3. **Unblock** the zip file.
4. **Extract** the scripts to your **desktop**.
5. Move files to the folder where you intend to run them from.

1. Download the scripts

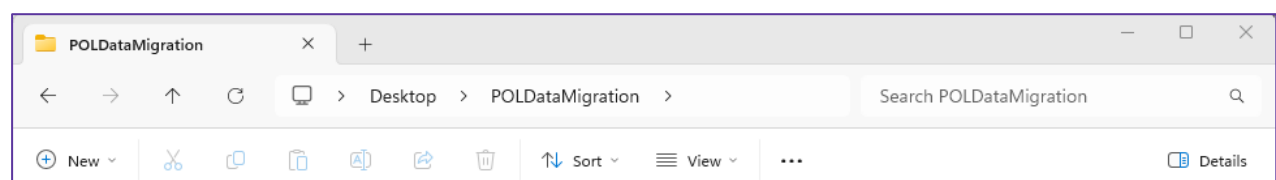
1. **Open** this link: [Project Online Migration](#).

2. **Download** this .ZIP folder: [Altus.Project.Online.Migration.Toolkit.2026.05.26.1.zip](#).



1. Download the scripts**3. Unblock** the zip file.

Note: If you get errors running the script after this step, you might need to unblock individual script files.

4. Extract the scripts to your **desktop**:Example

C:\Users\JohnPrice\Desktop\POLDataMigration

5. Move files to the folder where you intend to run them from.

Once you have extracted the scripts to your desktop, you can then move them to a location that you wish to use to run them from.

2. Run the Export Scripts

Quick Summary:

Before running the export scripts, you will need to get the following:

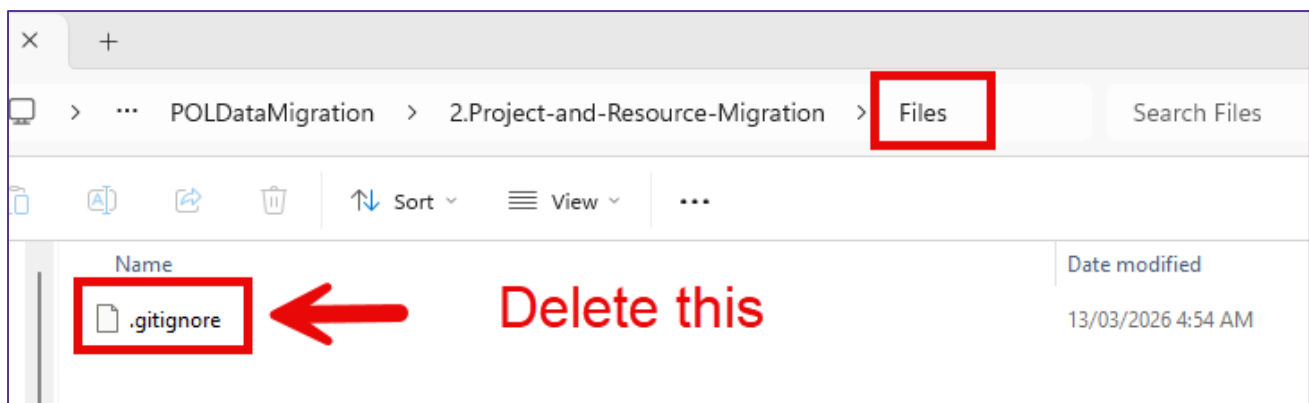
1. Gather the export scripts.
2. Open **SharePoint Online Management Shell**.
3. **Navigate** to the location when you extracted the scripts to.
4. Run the following command. You can use the Project Filter to specify specific projects to export. You can also use an asterisk "*" to find projects with a key word.
5. **Sign in** with the project administrator account to Project Online.

Video:

See link below to a video which covers the above steps with an example migration:

 [Project Online Data Migration - 1 Export Projects from Project Online.mp4](#)

Note: Before running the script, you might need to delete the contents from the file folder. Some users have reported seeing this file. If this appears, please remove it from the file folder otherwise the export script will error.



ClientId

To run the export script, we utilise this ClientID to extract the data from Project Online:

d3590ed6-52b3-4102-aeff-aad2292ab01c

Yes, d3590ed6-52b3-4102-aeff-aad2292ab01c is the well-known Azure AD application ID for Microsoft Office (the Office desktop client). It's a first-party Microsoft application registered in Azure AD, commonly used in scripts that need to authenticate to SharePoint/Office 365 resources using interactive login via MSAL.

This doesn't need to be updated in the scripts and is here as a point of reference.

This ClientID cannot be used in the following scripts (as they will require the ClientID setup in [this step](#)):

- ExportLookupTableData.ps1

- -Run-Migration.ps1

2. Run the Export Scripts

1. Before running the export scripts, you will need to get the following:

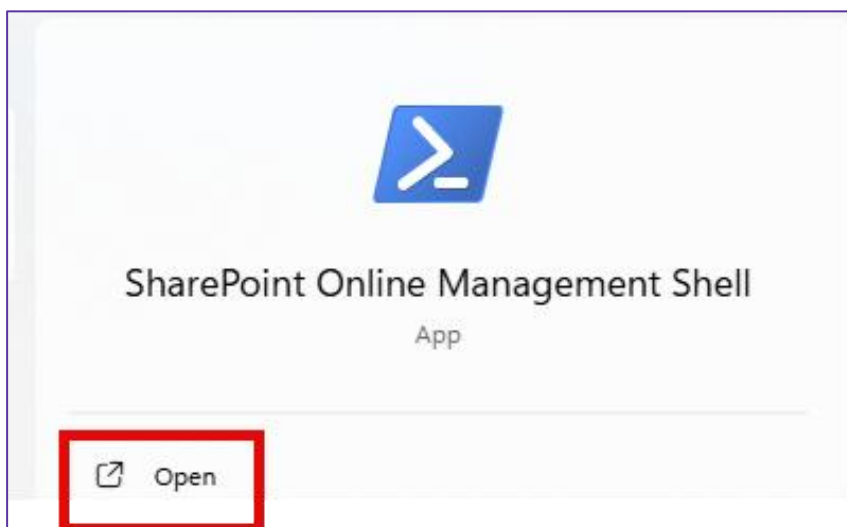
The output directory is the Files folder that exists in the downloaded scripts folder structure:

Property	Example
Project Online URL	https://senseicloud.sharepoint.com/sites/vNext
Script Location	cd C:\Users\JohnPrice\Desktop\POLDataMigration\0.Project-Online-Extraction
Files Location	cd C:\Users\JohnPrice\Desktop\POLDataMigration\2.Project-and-Resource-Migration\Files

Important

You will also need to ensure that you have opened **Project Desktop** on your computer as this is required to save the .MPP schedule files. Please don't connect to the Project Online environment, MS Project only needs to be open to save the .MPP files.

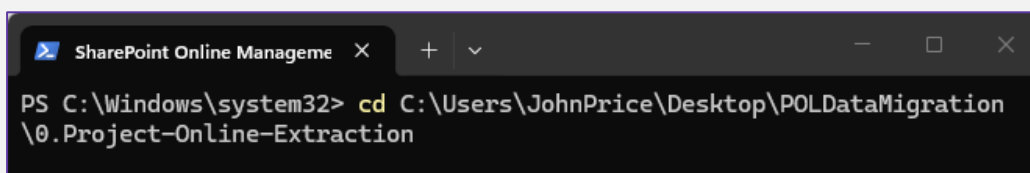
2. Open **SharePoint Online Management Shell**.



3. **Navigate** to the location when you extracted the scripts to.

Example:

cd C:\Users\JohnPrice\Desktop\POLDataMigration\0.Project-Online-Extraction

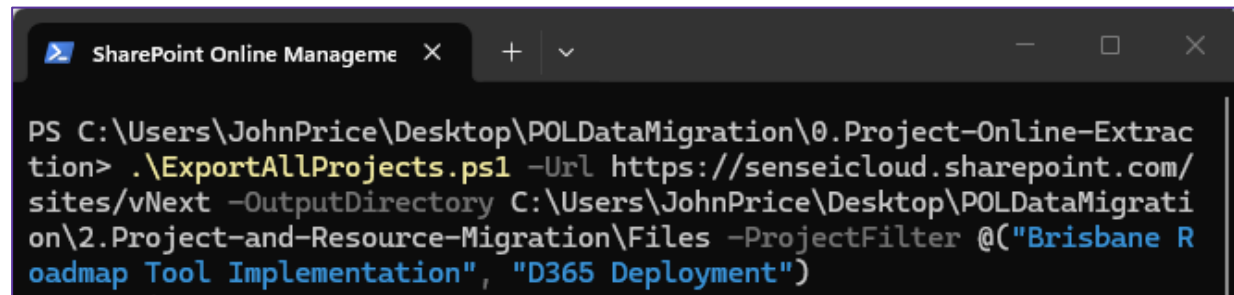


2. Run the Export Scripts

4. Run the following command. You can use the Project Filter to specify specific projects to export. You can also use an asterisk "*" to find projects with a key word:

Example:

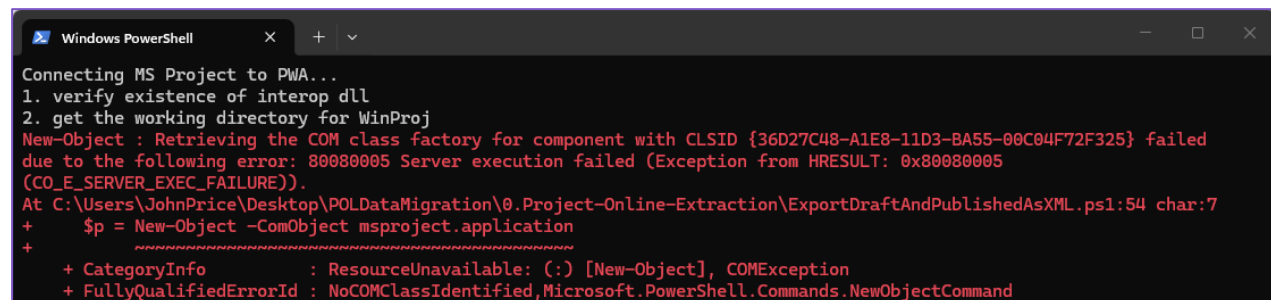
```
.\ExportAllProjects.ps1 -Url https://senseicloud.sharepoint.com/sites/vNext -OutputDirectory C:\Users\JohnPrice\Desktop\POLDataMigration\2.Project-and-Resource-Migration\Files -ProjectFilter @("Brisbane Roadmap Tool Implementation", "D365 Deployment")
```



```
PS C:\Users\JohnPrice\Desktop\POLDataMigration\0.Project-Online-Extraction> .\ExportAllProjects.ps1 -Url https://senseicloud.sharepoint.com/sites/vNext -OutputDirectory C:\Users\JohnPrice\Desktop\POLDataMigration\2.Project-and-Resource-Migration\Files -ProjectFilter @("Brisbane Roadmap Tool Implementation", "D365 Deployment")
```

Note: If the Output Directory name has any spaces, you will need to escape these in double quotes.

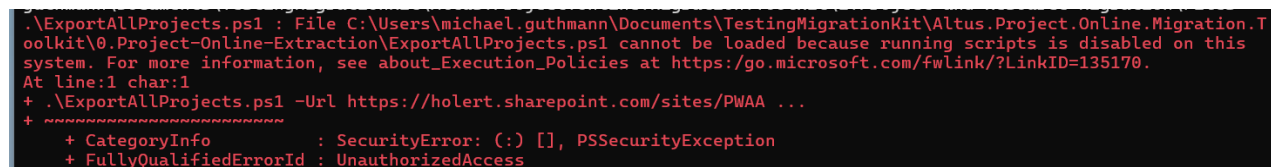
If you receive this message when running the export script you may need to close and reopen Project Desktop. If this doesn't resolve the issue, you will need to reboot your PC:



```
Connecting MS Project to PWA...
1. verify existence of interop dll
2. get the working directory for WinProj
New-Object : Retrieving the COM class factory for component with CLSID {36D27C48-A1E8-11D3-BA55-00C04F72F325} failed
due to the following error: 80080005 Server execution failed (Exception from HRESULT: 0x80080005
(CO_E_SERVER_EXEC_FAILURE)).
At C:\Users\JohnPrice\Desktop\POLDataMigration\0.Project-Online-Extraction\ExportDraftAndPublishedAsXML.ps1:54 char:7
+ $p = New-Object -ComObject msproject.application
+ ~~~~~
+ CategoryInfo          : ResourceUnavailable: (:) [New-Object], COMException
+ FullyQualifiedErrorId : NoCOMClassIdentified,Microsoft.PowerShell.Commands.NewObjectCommand
```

Note: If you get this error message, you will need to run this command first:

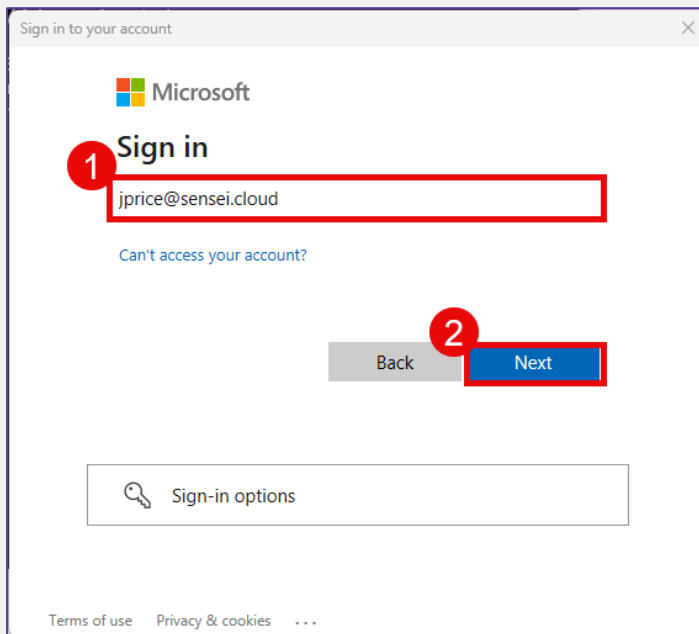
Set-ExecutionPolicy -ExecutionPolicy Bypass -Scope Process -Force



```
.\ExportAllProjects.ps1 : File C:\Users\michael.guthmann\Documents\TestingMigrationKit\Altus.Project.Online.Migration.Toolkit\0.Project-Online-Extraction\ExportAllProjects.ps1 cannot be loaded because running scripts is disabled on this
system. For more information, see about_Execution_Policies at https://go.microsoft.com/fwlink/?LinkID=135170.
At line:1 char:1
+ .\ExportAllProjects.ps1 -Url https://holert.sharepoint.com/sites/PWAA ...
+ ~~~~~
+ CategoryInfo          : SecurityError: (:) [], PSSecurityException
+ FullyQualifiedErrorId : UnauthorizedAccess
```

2. Run the Export Scripts

5. Sign in with the project administrator account to Project Online:



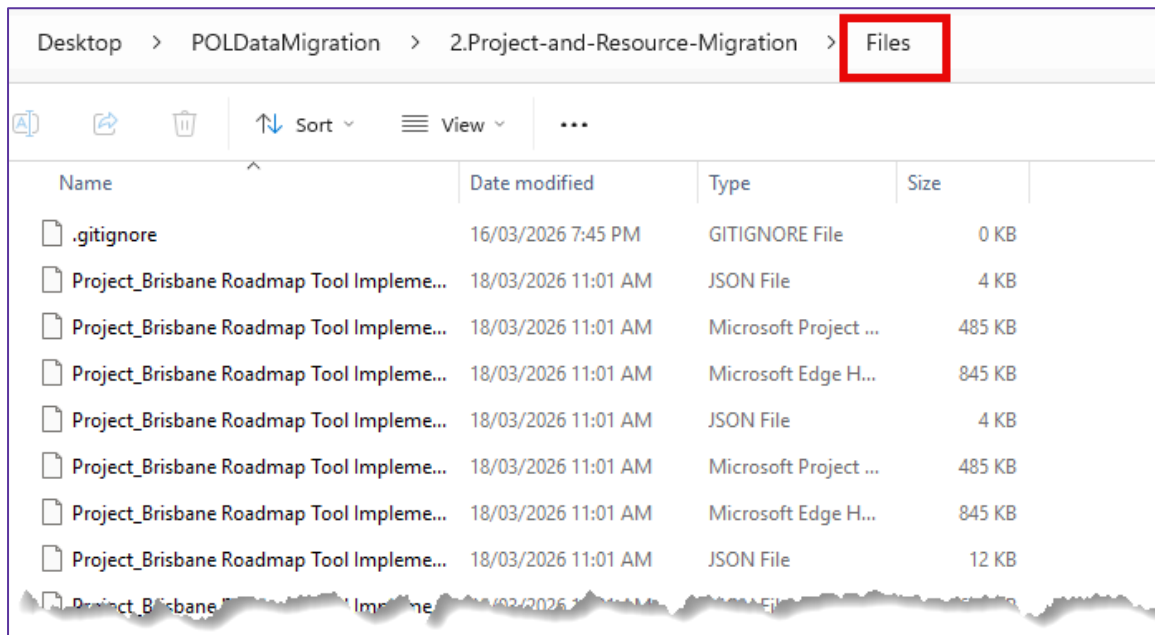
Script will run and export each project and it's corresponding .MPP in sequence:

```

SharePoint Online Manageme  X  +  v  -  □  X
2. open file from draft store - Brisbane Roadmap Tool Implementation
3. save file as xml - Brisbane Roadmap Tool Implementation
4. save file as mpp - Brisbane Roadmap Tool Implementation
5. close file - Brisbane Roadmap Tool Implementation
Exporting published schedule in MPP/XML formats...
1. make sure initialize the interop dll - Brisbane Roadmap Tool Implem
entation
2. open file from published store - Brisbane Roadmap Tool Implementati
on
3. save file as xml - Brisbane Roadmap Tool Implementation
4. save file as mpp - Brisbane Roadmap Tool Implementation
5. close file - Brisbane Roadmap Tool Implementation
Exporting draft data for project: Brisbane Roadmap Tool Implementation
...
Exporting published data for project: Brisbane Roadmap Tool Implementa
tion...
Exporting reporting data for project: Brisbane Roadmap Tool Implementa
tion...
...Exporting tasks...
...Exporting assignments...
...Exporting resources...
...Exporting project baselines...
...Exporting task timephased data...
...Exporting assignment timephased data...
...Exporting task baseline timephased data...
...Exporting assignment baseline timephased data...
Export complete!
PS C:\Users\JohnPrice\Desktop\POLDataMigration\0.Project-Online-Extrac
tion>

```

Once complete, you can access the files from this location:



Filtering projects to be exported

The export script can be updated to utilise different filters. Some examples of these have been provided below for reference:

Operators

These can be used to determine how fields are filtered:

Operator	Syntax
Equals	eq
Not Equals	ne
Greater Than	gt
Greater Than or Equal To	ge
Less Than	lt
Less Than or Equal To	le

Example filters

Filter Value	Syntax
Text	EnterpriseProjectTypeName eq 'Major Project'
Numbers	ProjectFixedCost gt 0
Guids	EnterpriseProjectTypeId eq guid'7473ef8c-9f25-e911-afb0-00155da06b17'
Booleans	ProjectEnterpriseFeatures eq true
Date	ProjectStartDate gt datetime'2024-09-01T00:00:00Z'

Example in PowerShell

PowerShell | Start Date filter

This only exports project with a start date greater than 1st April 2026.

```
.\ExportAllProjects.ps1 -Url https://<clientname>.sharepoint.com/sites/pwa -OutputDirectory <folder name>\2.Project-and-Resource-Migration\Files -ProjectOnlineOdataFilter "ProjectStartDate gt datetime'2026-04-01T00:00:00Z'"
```

PowerShell | Project Status filter

This only exports projects with a Project Status = Active

```
.\ExportAllProjects.ps1 -Url https://<clientname>.sharepoint.com/sites/pwa -OutputDirectory <folder name>\2.Project-and-Resource-Migration\Files -ProjectOnlineOdataFilter "ProjectStatus eq 'Active'"
```

Example in ODATA feed

You can also filter this in ODATA to check the results before exporting this data:

```
https://<clientname>.sharepoint.com/sites/pwa/_api/ProjectData/Projects?$filter=ProjectStartDate%20gt%20datetime%272025-01-01T00:00:00%27
```

```
https://<clientname>.sharepoint.com/sites/pwa/_api/ProjectData/Projects?$filter=ProjectStatus%20eq%27Active%27
```

2. Export Lookup tables

Once the data has been exported from Project Online and the custom fields created, the lookup tables can be mapped.

Before running this script, you will need to decide which ClientID you are going to use. This is detailed in [this document](#).

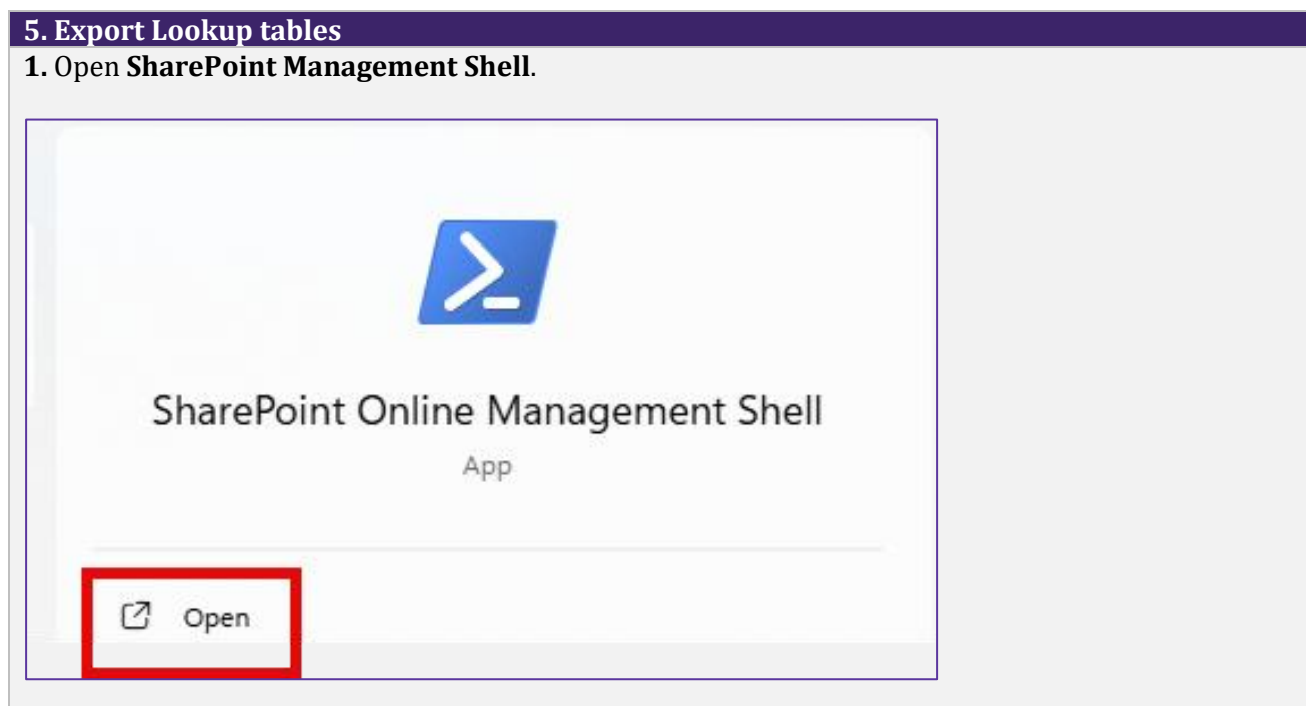
Quick Summary:

1. Open **SharePoint Management Shell**.
2. Navigate to the location when you extracted the scripts to.
3. Run the following command: `.\ExportLookupTableData.ps1 -PwaUrl https://tenant.sharepoint.com/sites/pwa`

Video:

See link below to a video which covers the above steps with an example migration:

 [Project Online Data Migration - 2 Export Lookup tables.mp4](#)

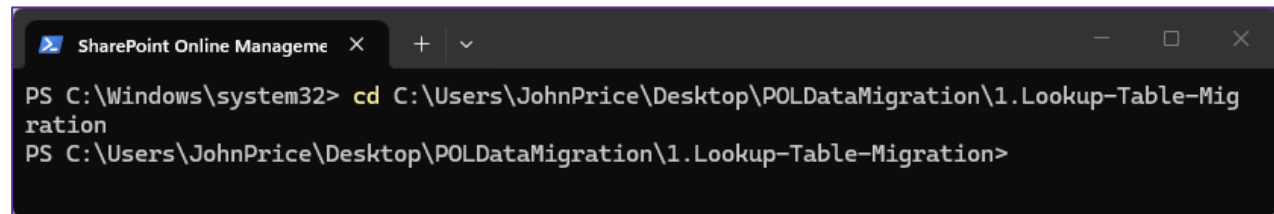


5. Export Lookup tables

2. Navigate to the **location** when you extracted the scripts to.

Example:

```
cd C:\Users\JohnPrice\Desktop\POLDataMigration\1.Lookup-Table-Migration
```



```
PS C:\Windows\system32> cd C:\Users\JohnPrice\Desktop\POLDataMigration\1.Lookup-Table-Migration
PS C:\Users\JohnPrice\Desktop\POLDataMigration\1.Lookup-Table-Migration>
```

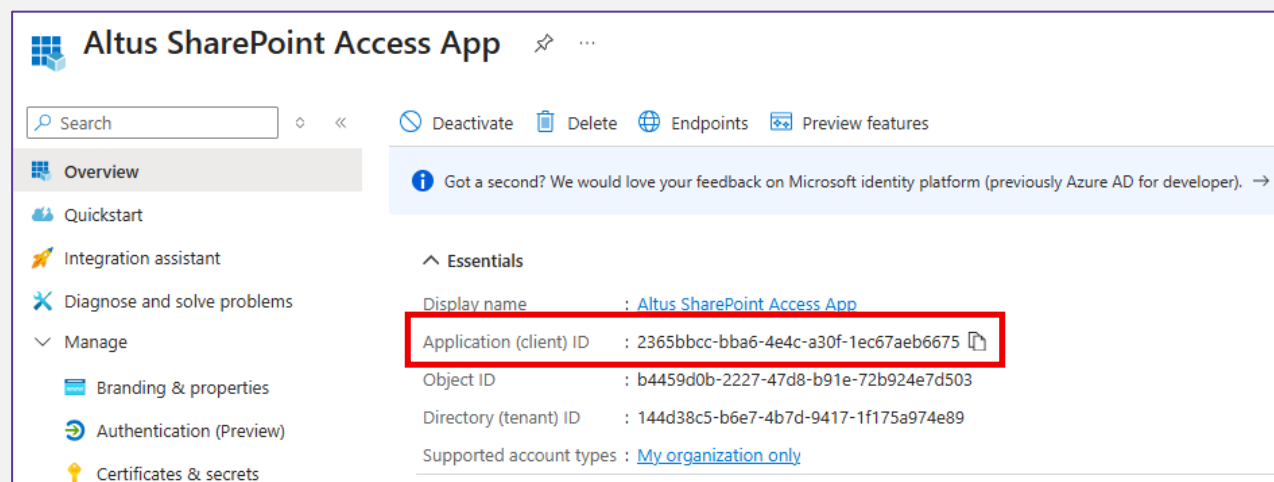
3. Run the following command and pass the following parameter:

Example:

```
.\ExportLookupTableData.ps1 -PwaUrl "https://senseicloud.sharepoint.com/sites/vNext" -ClientId "30b4ad0b-d939-4cd4-bb6d-fa2d39fb4694"
```

Note: If you aren't using the Altus Sharepoint app ClientID, you will need to grab this from the application you have configured in [this document](#).

This is where you source the client ID from:



Altus SharePoint Access App

Search

Deactivate Delete Endpoints Preview features

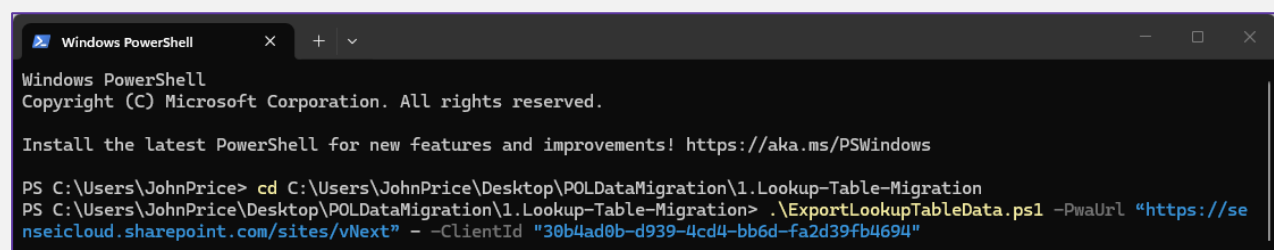
Got a second? We would love your feedback on Microsoft identity platform (previously Azure AD for developer). →

Overview Quickstart Integration assistant Diagnose and solve problems Manage

Branding & properties Authentication (Preview) Certificates & secrets

Essentials

Display name	: Altus SharePoint Access App
Application (client) ID	: 2365bbcc-bba6-4e4c-a30f-1ec67aeb6675
Object ID	: b4459d0b-2227-47d8-b91e-72b924e7d503
Directory (tenant) ID	: 144d38c5-b6e7-4b7d-9417-1f175a974e89
Supported account types	: My organization only



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

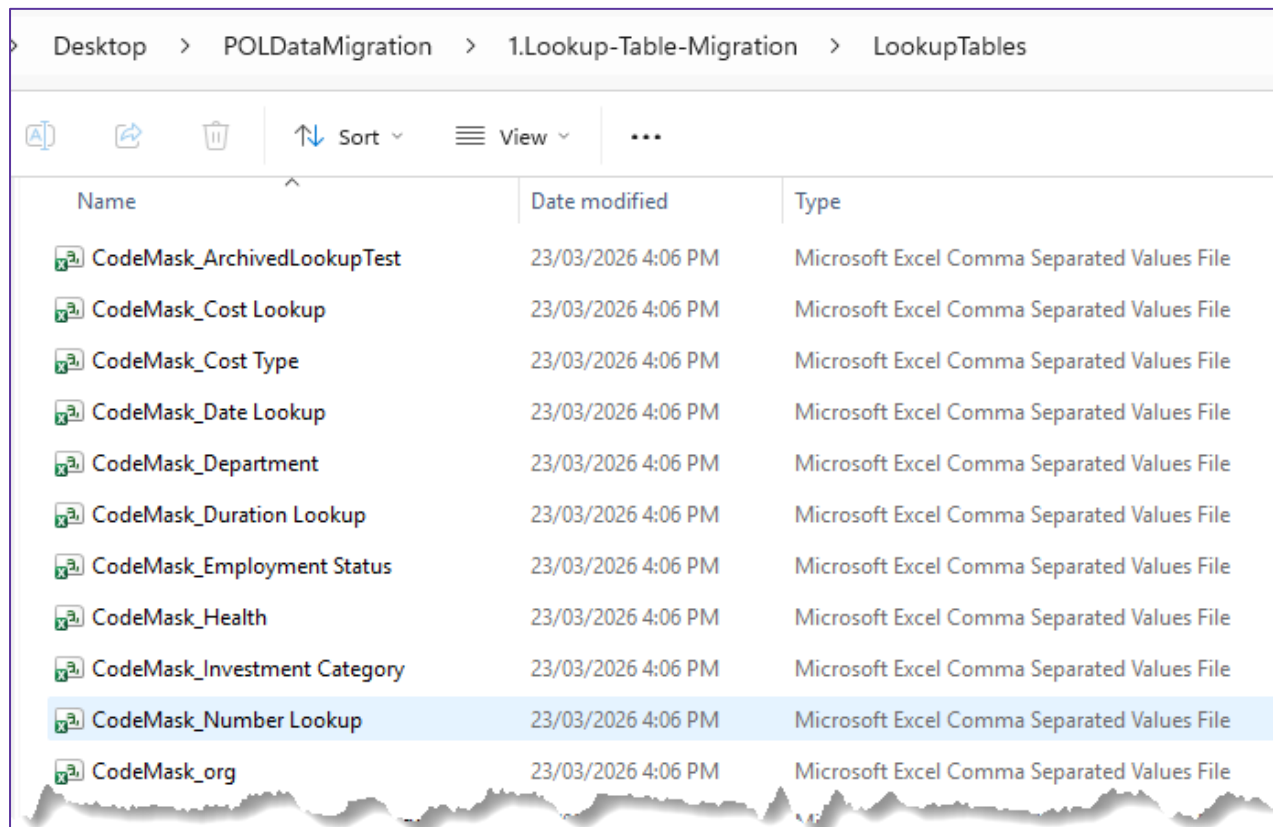
PS C:\Users\JohnPrice> cd C:\Users\JohnPrice\Desktop\POLDataMigration\1.Lookup-Table-Migration
PS C:\Users\JohnPrice\Desktop\POLDataMigration\1.Lookup-Table-Migration> .\ExportLookupTableData.ps1 -PwaUrl "https://senseicloud.sharepoint.com/sites/vNext" -ClientId "30b4ad0b-d939-4cd4-bb6d-fa2d39fb4694"
```

Lookup tables will be exported to this location:

Example:

C:\Users\JohnPrice\Desktop\POLDataMigration\1.Lookup-Table-Migration\LookupTables

In this format: Comma Separated Values (.CSV)



Desktop > POLDataMigration > 1.Lookup-Table-Migration > LookupTables			
Sort View ...			
Name	Date modified	Type	
CodeMask_ArchivedLookupTest	23/03/2026 4:06 PM	Microsoft Excel Comma Separated Values File	
CodeMask_Cost Lookup	23/03/2026 4:06 PM	Microsoft Excel Comma Separated Values File	
CodeMask_Cost Type	23/03/2026 4:06 PM	Microsoft Excel Comma Separated Values File	
CodeMask_Date Lookup	23/03/2026 4:06 PM	Microsoft Excel Comma Separated Values File	
CodeMask_Department	23/03/2026 4:06 PM	Microsoft Excel Comma Separated Values File	
CodeMask_Duration Lookup	23/03/2026 4:06 PM	Microsoft Excel Comma Separated Values File	
CodeMask_Employment Status	23/03/2026 4:06 PM	Microsoft Excel Comma Separated Values File	
CodeMask_Health	23/03/2026 4:06 PM	Microsoft Excel Comma Separated Values File	
CodeMask_Investment Category	23/03/2026 4:06 PM	Microsoft Excel Comma Separated Values File	
CodeMask_Number Lookup	23/03/2026 4:06 PM	Microsoft Excel Comma Separated Values File	
CodeMask_org	23/03/2026 4:06 PM	Microsoft Excel Comma Separated Values File	

3. Import Lookup tables

Before running this part of the process, you need to have:

1. Created the lookup tables in Altus as per the requirements specified in the [03 - PO to Altus Migration - Scope and Field Mapper] document.
2. Exported the lookup tables.

The import process has two sub-steps:

1. Configure the script to map the lookup tables.
2. Run the script to import lookup tables.

1. Configure the script to map the lookup tables

Quick Summary:

1. **Open** the **solution layer**.
2. Open the **tables** section.
3. Select the first lookup table > Properties.
4. Get the table and field details.
5. Repeat steps 3 and 4 for all lookup tables.
6. Open the solution layer and get the information about the lookup tables.
7. Open this script: **ImportLookupTableData.ps1** in edit mode.
8. Update the relevant parts of the script.

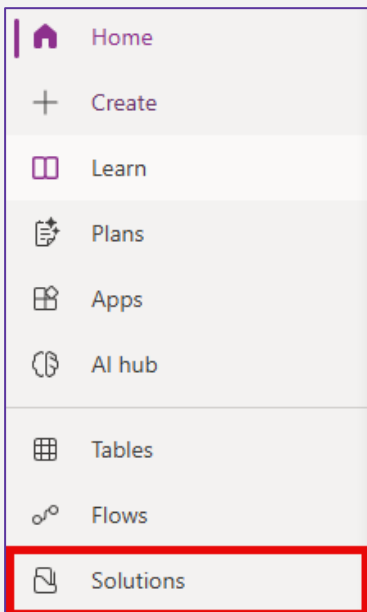
Video:

See link below to a video which covers the above steps with an example migration:

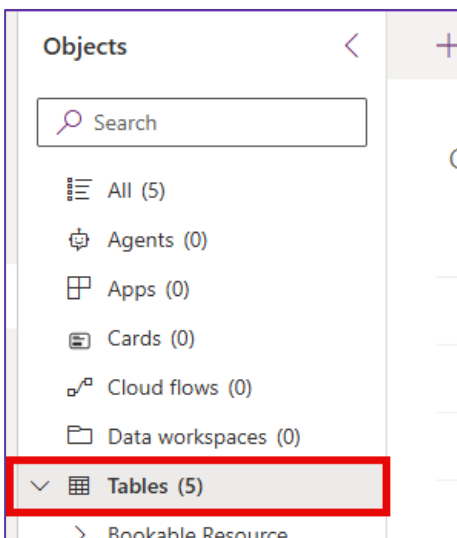
 [Project Online Data Migration - 3 Import Lookup tables.mp4](#)

1. Configure the script to map the lookup tables.

1. Open the **solution layer**.



2. Open the **tables** section.



1. Configure the script to map the lookup tables.**3. Select the first lookup table > Properties**

The screenshot displays the Microsoft Dynamics 365 interface. On the left, a navigation pane lists various entities: Agents (0), Apps (0), Cards (0), Cloud flows (0), Data workspaces (0), and Tables (5). Under 'Tables (5)', 'Bookable Resource' and 'EmploymentStatus' are listed. 'EmploymentStatus' is highlighted with a red box and a red circle containing the number 1. On the right, the 'Table properties' section is visible. It contains a table with the following data:

Name	Primary column	Description
EmploymentStatus	Name	Used to capture the employment status for all na bookable resources.
Type	Last modified	
Standard	6 days ago	

Below the table properties, there is a section titled 'EmploymentStatus columns and data'. A red box and a red circle containing the number 2 highlight the 'Properties' tab in the top right corner of the table properties section.

1. Configure the script to map the lookup tables.

4. Get the table and field details.

- Table:
 - Logical Name
- Fields:
 - Name
 - Description
 - Parent

Edit table

Use tables to hold and organize your data. Previously called entities

[Learn more](#)

Properties

Primary column

Display name *

EmploymentStatus

Plural name *

EmploymentStatuses

Description

Used to capture the employment status for all named bookable resources.

☐ Enable attachments (including notes and files) ¹

Advanced options ^

Schema name *

ce_EmploymentStatus

Logical name

ce_employmentstatus

You will need to expand the **Advanced options** section so that you can get the **Logical name**.

5. Repeat steps 3 and 4 for all lookup tables.

1. Configure the script to map the lookup tables.

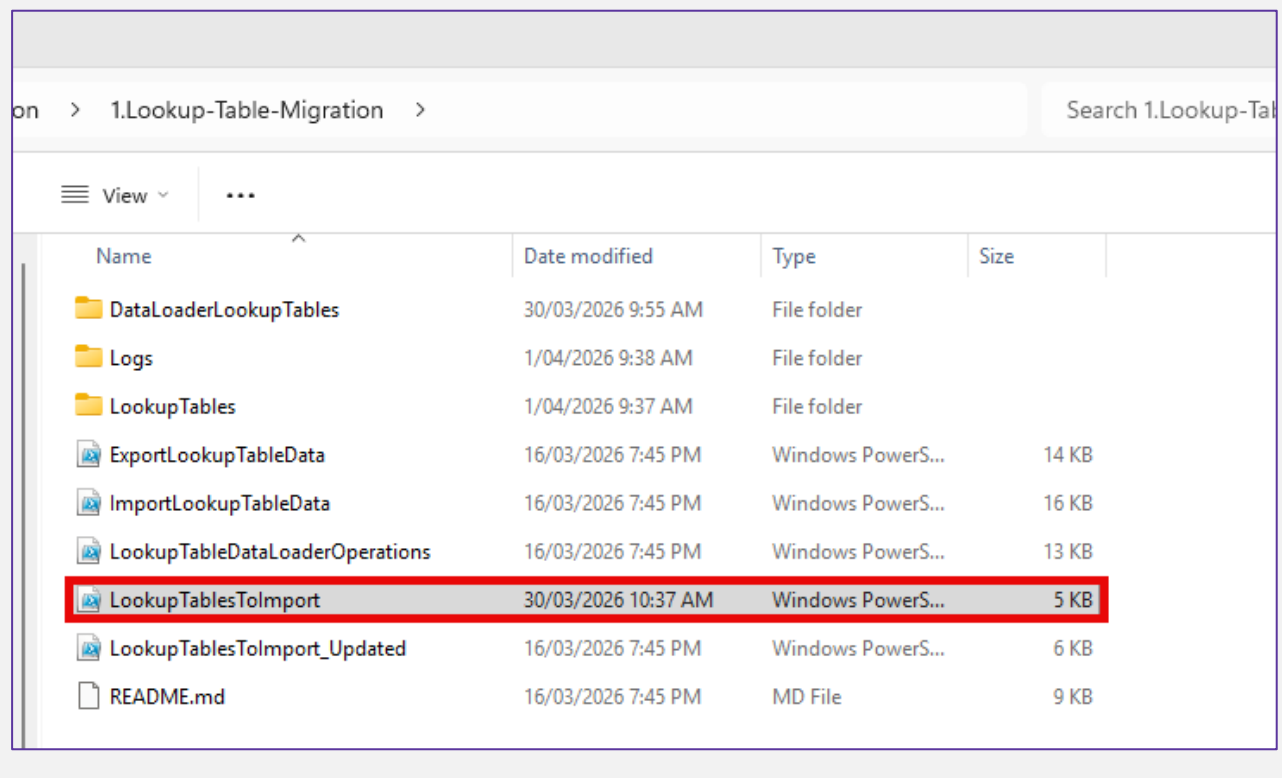
6. Open the solution layer and get the information about the lookup tables. Below is an example of both flat and hierarchical (self-referential) lookup tables:

Note: You will need to add the solution layer prefix (in the example below this is **ce**):

Also, you will need to remove all capitalisation of letters, otherwise the script will error.

Table Logical Name (Display Name)	Table Collection Name (Plural Name)	Name Field (Not display name)	Parent Field Name
ce_employmentstatus	ce_employmentstatuses	ce_name	N/A
ce_rbs	ce_rbses	ce_name	ce_parent
ce_examplemultiselectlookup	ce_examplemultiselectlookup	ce_name	ce_parent
ce_skills	ce_skillses	ce_name	ce_parent
ce_teamname	ce_teamnames	ce_name	N/A

7. Open this script: **LookupTablesToImport.ps1** in edit mode.



1. Configure the script to map the lookup tables.

8. Update these parts of the script:

1. **Name of** .CSV file for the lookup table
2. **Dataverse** table details (from the step 6).

```

$LookupTableFileLocation = (Join-Path (Split-Path -Parent $MyInvocation.M
$LookupTableMappings = @(
#   EXAMPLE FLAT MAPPING
#   @{
#       SourceFile           = "LookupTable Department.csv"
#       MapAsHierarchy       = $false
#       DataverseTable       = @{ TableLogicalName   = "cr62c_department"
#                               TableCollectionName = "cr62c_departments"
#                               NameField           = "cr62c_name"
#       }
#   }
#   ...
#   EXAMPLE SELF REFERENTIAL HIERARCHY MAPPING
#   @{
#       SourceFile           = "LookupTable Skills.csv"
#       MapAsHierarchy       = $true
#       DataverseTable       = @{ TableLogicalName   = "cr62c_skills"
#                               TableCollectionName = "cr62c_skillses"
#                               NameField           = "cr62c_name"
#                               PrevLevelLookupField = "cr62c_parentskill"
#       }
#   }
# }
)

```

See attached file for example of updated script:

Note: You will need to uncomment the script before running it. To do this, remove all hashtags/pound symbols (#) from the script.

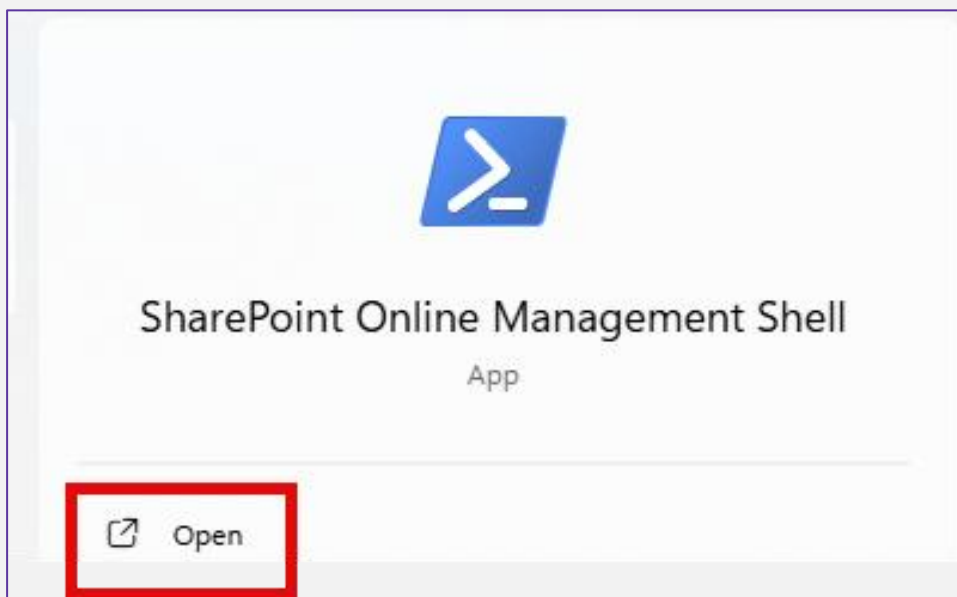
2. Run the script to Import lookup tables

Quick Summary:

1. Open SharePoint Management Shell.
2. Navigate to the location when you extracted the scripts.
3. Run the following command: ImportLookupTableData.ps1.
4. Type in the Altus environment URL.
5. Login to Altus with your credentials.
6. Once done, the script will run and import the lookup tables.

2. Run the script to Import lookup tables

1. Open SharePoint Management Shell

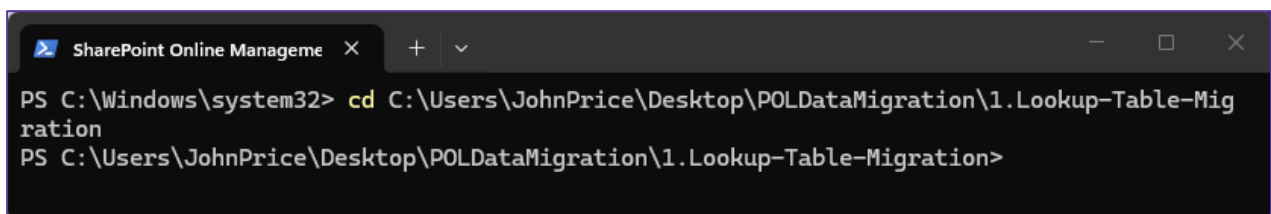


2. Navigate to the **location** when you extracted the scripts:

cd + script location

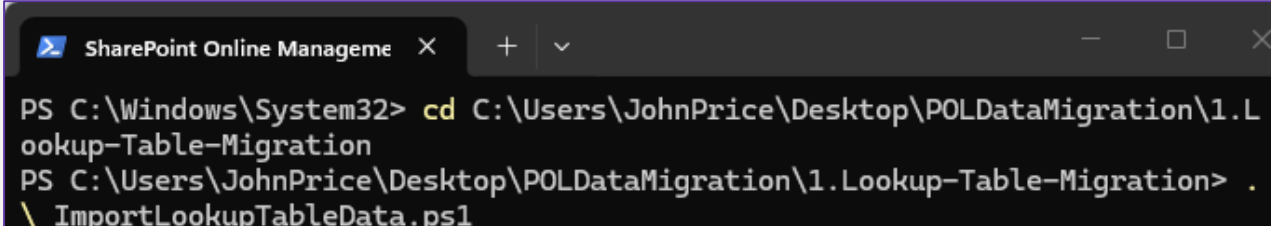
Example:

cd C:\Users\JohnPrice\Desktop\POLDataMigration\1.Lookup-Table-Migration



2. Run the script to Import lookup tables

3. Run the following command: **.\ImportLookupTableData.ps1**



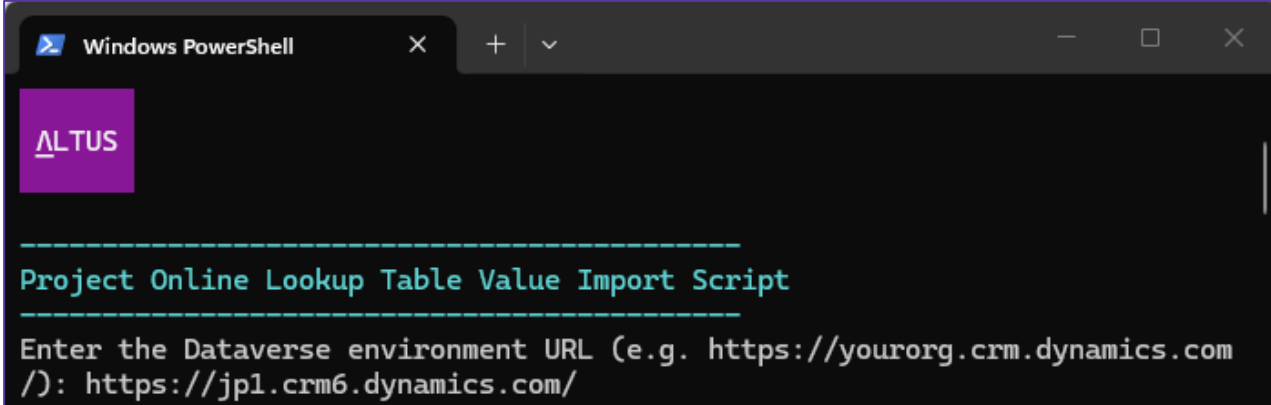
```
PS C:\Windows\System32> cd C:\Users\JohnPrice\Desktop\POLDataMigration\1.L  
ookup-Table-Migration  
PS C:\Users\JohnPrice\Desktop\POLDataMigration\1.Lookup-Table-Migration> .  
\ ImportLookupTableData.ps1
```

4. Type in the Altus environment URL:

<https://Altus.crm6.dynamics.com/>

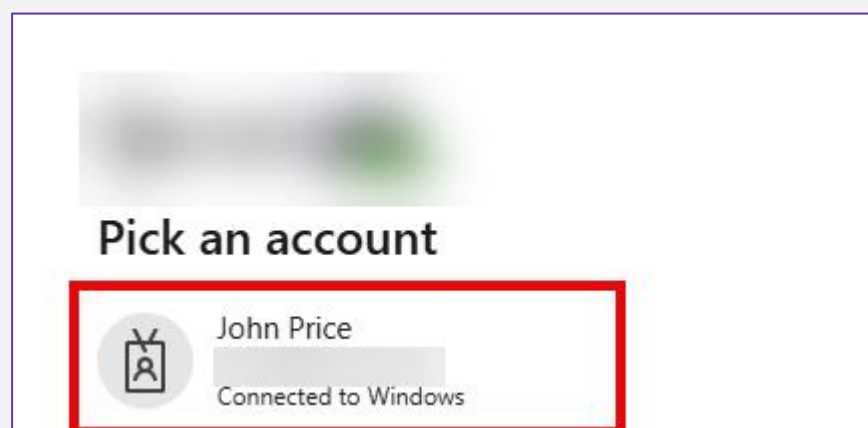
Example:

<https://jp1.crm6.dynamics.com/>



```
ALTUS  
  
-----  
Project Online Lookup Table Value Import Script  
-----  
Enter the Dataverse environment URL (e.g. https://yourorg.crm.dynamics.com  
/): https://jp1.crm6.dynamics.com/
```

5. Login to Altus with your credentials:



2. Run the script to Import lookup tables

6. Once done, the script will run and import the lookup tables.

Any errors will be noted in the associated log file (where applicable).

```

Windows PowerShell
OK: SUCCESS - C:\Users\JohnPrice\Desktop\POLDataMigration\1.Lookup-Table-Migration\DataLoaderLookupTables\Data.xml imported successfully
Duration: 00:00:10

Import Summary:
-----
Total Lookup Table Files Processed: 4
Successful Imports: 1
Successful Records: 38
Failed Imports: 0
Total Errors: 0

Lookup table data import completed.
PS C:\Users\JohnPrice\Desktop\POLDataMigration\1.Lookup-Table-Migration>
  
```

Example of an import lookup table

Below is a screenshot of a lookup table that has been imported into Altus:

CustomerEnhancements > Tables > RBS

Table properties		
Name	Primary column	Description
RBS	Name	Hierarchical Resource Breakdown Structure (RBS) from Project Online for the associated resource.
Type	Last modified	
Standard	6 days ago	

RBS columns and data

Name * ↑ ↓	Parent	Description
Corporate		
Corporate.Construction - Manage...	Corporate	
Corporate.Construction - Manage...	Corporate.Construction - Manage...	
Corporate.Finance - Management	Corporate	

4. Load Lookup tables

Before mapping the project and resource custom fields, you will need to map the lookup tables by updating the **Default.ps1** script.

Quick Summary:

1. Open 2.Project-and-Resource-Migration then the script: **Default.ps1**
2. Update the lookup tables section in the script.
3. Update the choice fields section in the script.

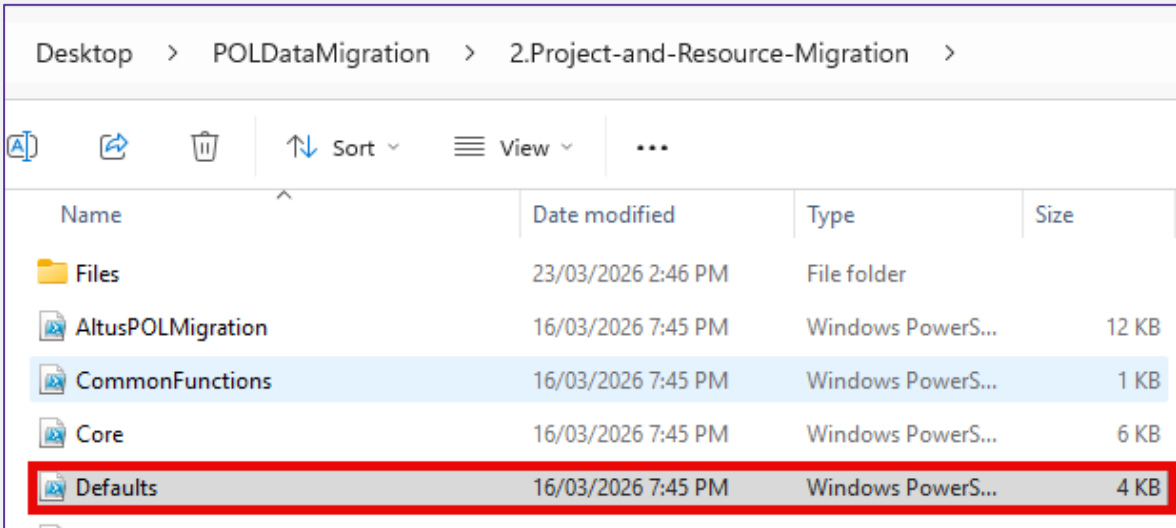
Video:

See link below to a video which covers the above steps with an example migration:

 [Project Online Data Migration - 4 Load Lookup tables.mp4](#)

4. Load lookup tables

1. Open 2.Project-and-Resource-Migration then the script: **Default.ps1**



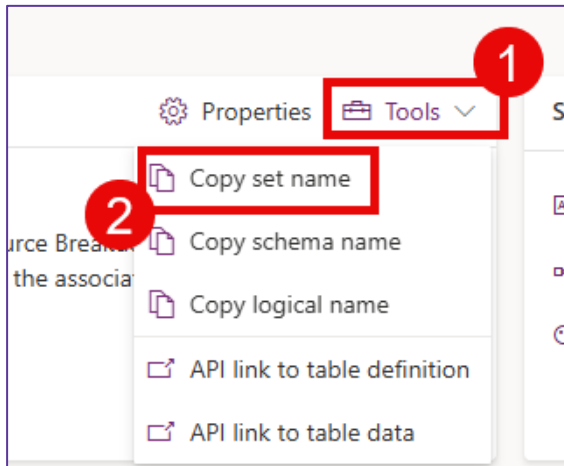
Name	Date modified	Type	Size
Files	23/03/2026 2:46 PM	File folder	
AltusPOLMigration	16/03/2026 7:45 PM	Windows PowerS...	12 KB
CommonFunctions	16/03/2026 7:45 PM	Windows PowerS...	1 KB
Core	16/03/2026 7:45 PM	Windows PowerS...	6 KB
Defaults	16/03/2026 7:45 PM	Windows PowerS...	4 KB

2. Update the **lookup tables** section in the script.

The parts highlighted in Yellow have been updated. Note, you will need to uncomment the script also.

To get the **Table collection name** you will need to:

1. Select **Tools**
2. **Copy Set Name**



LOOKUP TABLES TO LOAD FOR REFERENCE

Note: These lookup tables must be pre-loaded before running the Project and Resource migration to ensure proper mapping of lookup field values.

```
$LookupTablesToLoad = @({
```

```
  #(Examples below)
```

```
  @{
```

```
    Lookup = "lookup_RBS" # Name of the lookup table that you will use internally to this script to identify the dataset
```

```
    DataverseTable = @{
```

```
      TableLogicalName = "ce_rbs"
```

```
      TableCollectionName = "ce_rbses"
```

```
      NameField = "ce_name"
```

```
    }
```

```
  },
```

```
  @{
```

```
    Lookup = "lookup_EmploymentStatus" # Name of the lookup table that you will use internally to this script to identify the dataset
```

```
    DataverseTable = @{
```

```
      TableLogicalName = "ce_employmentstatus"
```

```
      TableCollectionName = "ce_employmentstatuses"
```

```
      NameField = "ce_name"
```

```
    }
```

```
  },
```

```
  @{
```

```
    Lookup = "lookup_TeamName" # Name of the lookup table that you will use internally to this script to identify the dataset
```

```
    DataverseTable = @{
```

```
      TableLogicalName = "ce_teamname"
```

```
      TableCollectionName = "ce_teamnames"
```

```
      NameField = "ce_name"
```

4. Load lookup tables

```

    }
  },
  @{
    Lookup = "lookup_Skills" # Name of the lookup table that you will use internally to this script to
    identify the dataset
    DataverseTable = @{
      TableLogicalName = "ce_skills"
      TableCollectionName = "ce_skillses"
      NameField = "ce_name"
    }
  }
}
)

```

3. Update the **choice fields** section in the script.

The parts highlighted in Yellow have been updated. Note, you will need to uncomment the script also.

```

# CHOICE FIELDS TO LOAD FOR REFERENCE
# Note: These choice fields must be pre-loaded before running the Project and Resource migration to
ensure correct mapping of choice field values.
$ChoiceFieldsToLoad = @(
  @{
    Name = "choice_YesNo" # Name of the choice field that you will use internally to this script to
    identify the dataset
    IsGlobalChoice = $true # Set to $true if this is a global choice field, otherwise $false for local
    choice fields
    DataverseTable = @{
      TableLogicalName = "sensei_project" # Logical/internal name of the Dataverse table
      ChoiceField = "ce_examplechoicefield" # Logical/internal name of the choice field in the
      Dataverse table
    }
  }
)
)

```

Below is an example of lookup table that has been loaded from Project Online to Altus:

Project Online

Edit Lookup table: Employment Status ⓘ

Name

Type a unique name for the lookup table.

* Name:

Lookup Table

Edit the lookup table

Level	Value	Description
1	New	
1	Existing	
1	Not Set	

Display order for lookup table:

☒ By row number

☐ Sort ascending

☐ Sort descending

Altus

CustomerEnhancements > Tables > EmploymentStatus

Table properties

Properties Tools ▾

Name	Primary column	Description
EmploymentStatus	Name	Used to capture the employment status for all named bookable resources.
Type	Last modified	
Standard	3 weeks ago	

EmploymentStatus columns and data

EmploymentStatus* ▾	Name* ↑ ▾	Description ▾
e8bc618d-22b6-ec11-b15d-0015...	Existing	
e7bc618d-22b6-ec11-b15d-0015...	New	
07db7046-25b6-ec11-a528-0015...	Not Set	

5. Map Resource Custom Fields and Create Resources

Before running this process, you need to have created all the required custom fields in the target Altus environment. The scripts migrate data only (no configuration).

Field Type Mapping

The script will map fields as follows:

Project Online Field Type	Dataverse Field Type	Dataverse Format
Cost	Currency	N/A
Date (Date only)	Date and time	Date only
Date (Date and Time)	Date and time	Date and time
Duration	Whole Number	Duration
Flag	Choice	Yes/no
Number (Whole)	Whole Number	None
Number (Decimal)	Decimal	None
Text (Single line of Text)	Single line of text	Text

Example Mapping

The example below will map the following fields between Project Online and Altus. The letters before the underscore (“_”) denotes the solution layer name.

Project Online Field Name	Altus Internal Name	Dataverse Format
Resource Example Cost Field	ce_resourceexamplecostfield	Currency
Resource Example Date Field	ce_resourceexampledatefield	Date and time > Date only
Resource Example Duration Field	ce_resourceexampledurationfield	Whole Number > Duration
Resource Example Flag Field	ce_resourceexamplechoicefield	Choice > YesNoChoice
Resource Example Number Field	ce_resourceexamplenumbersfield	Number > Decimal
Resource Example text field	ce_resourceexampletextfield	Text > Multiple lines of text
Resource Example Lookup Table Field	ce_resourceexamplelookuptablefield	Text > Lookup

1. Map Resource Custom Fields

The ImportResources.ps1 script needs to be updated to map custom fields between systems.

The parts of the script that are highlighted below are as follows. Only these parts of the script block need to be updated.

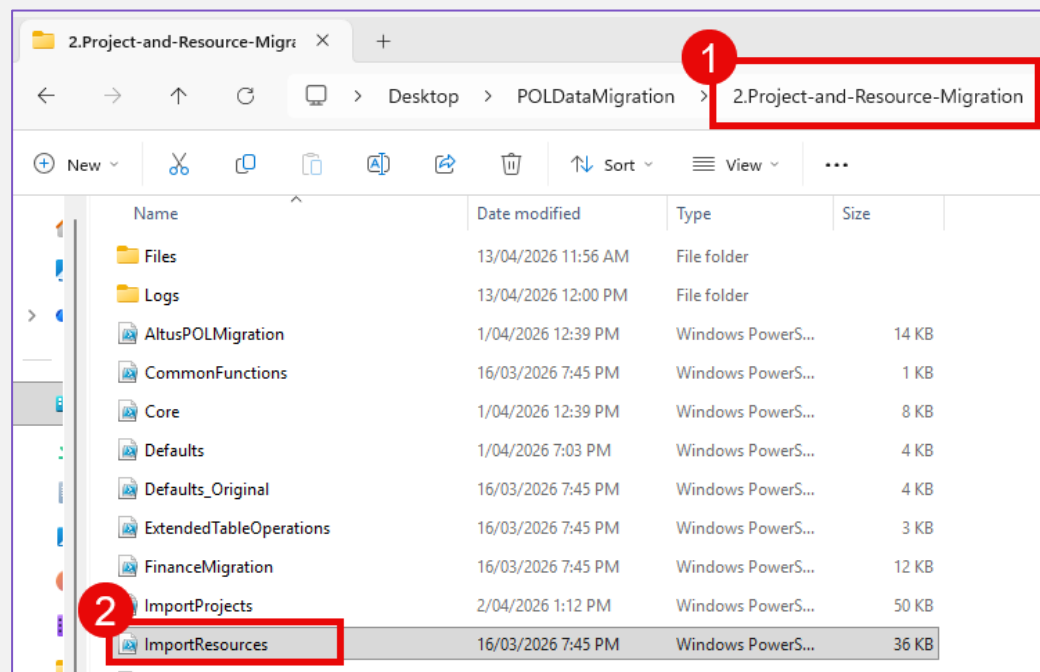
Attribute	Details
Project Online Field	Display name for the field as it appears in Project Online.
Altus Field	This is the Logical Name for the Altus field.
Default.ps1 script reference	Reference to another Altus script, specifically the alias.
Additional Altus variables	Altus references to related tables.

Quick Summary:

1. Open this folder, then edit this script: **ImportResources.ps1**.
2. Map all **Text** fields.
3. Map all **Date** fields.
4. Map all **Number** fields.
5. Map all **Duration** fields.
6. Map all **Cost** fields.
7. Map all **Flag** fields (Flag to Flag).
8. Map all **Flag** fields (Choice to Flag).
9. Map all **Lookup** fields (Lookup to Choice).
10. Map all **Lookup** fields (Single-select).
11. Map all **Lookup** fields (Multi-select).

1. Map Resource Custom Fields

1. Open this folder, then edit this script: **ImportResources.ps1**



1. Map Resource Custom Fields

2. Map all **Text** fields.

```
# --- Enterprise Custom Fields (from $ProjectResource.CustomFields[]) ---
$textValue = Get-ResourceCustomFieldTextValue -ProjectResource $ProjectResource -
CustomFieldName 'Resource Example text field'
if ($textValue) { $resourceBody['ce_resourceexampletextfield'] = [string]$textValue }
```

3. Map all **Date** fields.

```
$dateValue = Get-ReportingCustomFieldTextValue -ProjectResource $ProjectResource -
CustomFieldName 'Resource Example Date Field'
if ($dateValue) { $resourceBody['ce_resourceexampledatefield'] = ($dateValue -as
[datetime]).ToString("yyyy-MM-dd") }
```

4. Map all **Number** fields.

```
$numberValue = Get-ResourceCustomFieldTextValue -ProjectResource $ProjectResource -
CustomFieldName 'Resource Example Number Field'
if ($numberValue) { $resourceBody['ce_resourceexamplenumberfield'] =
[int][double]$numberValue }
```

5. Map all **Duration** fields.

```
$durationValue = Get-ResourceCustomFieldTextValue -ProjectResource $ProjectResource -
CustomFieldName 'Resource Example Duration Field'
if ($durationValue) { $resourceBody['ce_resourceexampledurationfield'] =
[int][double]$durationValue }
```

6. Map all **Cost** fields.

```
$costValue = Get-ResourceCustomFieldTextValue -ProjectResource $ProjectResource -
CustomFieldName 'Resource Example Cost Field'
if ($costValue) { $resourceBody['ce_resourceexamplecostfield'] = ($costValue -as [decimal]) }
```

7. Map all **Flag** fields (Flag to Flag).

```
$cfValue = Get-ResourceCustomFieldTextValue -ProjectResource $ProjectResource -
CustomFieldName 'Resource Example Flag Field'
if ($cfValue) {
    if ($cfValue -eq 'true') { $ynValue = $true } else { $ynValue = $false } # must be set as True and
False not 1 and 0.
    $resourceBody['ce_resourceexampleflagfield'] = $ynValue
}
```

1. Map Resource Custom Fields

8. Map all **Flag** fields (Lookup to Flag).

```
$cfValue = Get-ResourceCustomFieldTextValue -ProjectResource $ProjectResource -
CustomFieldName 'Resource Example Choice Field'
if ($cfValue) {
    if ($cfValue -eq 'Yes') { $ynValue = $true } else { $ynValue = $false }
    $resourceBody['ce_resourceexampleflagfield2'] = $ynValue
}
```

9. Map all **Lookup** fields (Lookup to Choice).

```
Set-ResourceChoiceField -ResourceBody $resourceBody -ProjectResource $ProjectResource -
POLCustomFieldName 'Resource Example Choice Field' -ChoiceName 'choice_YesNo' -
DataverseFieldName 'ce_resourceexamplechoicefield'
```

10. Map all **Lookup** fields (Single-select).

```
Set-ResourceLookupField -ResourceBody $resourceBody -ProjectResource $ProjectResource -
POLCustomFieldName 'Resource Example Lookup Field' -LookupName 'lookup_EmploymentStatus' -
DataverseFieldName 'ce_ResourceExampleLookupField'
```

Note: The Dataverse field name in this example is case sensitive.

11. Map all **Lookup** fields (Multi-select). Additional variables are highlighted in **Pink**.

```
Set-ResourceManyToManyRelationship -ResourceBody $resourceBody -ProjectResource
$ProjectResource -POLCustomFieldName 'Resource Example Multiselect Lookup Field' -LookupName
'lookup_MultiValueTable' -RelationshipName
'ce_sensei_bookableresource_ce_ExampleMultiSelectLookup_ce_ExampleMultiSelectLookup' -
RelatedTableCollectionName 'ce_examplemultiselectlookups' -RelatedTablePrimaryKey
'ce_examplemultiselectlookupid'
```

The following additional pieces of information need to be set for multi-select lookup fields:

Variable	Value
- RelationshipName	ce_sensei_bookableresource_ce_ExampleMultiSelectLookup_ce_ExampleMultiSelectLookup
- RelatedTableCollectionName	ce_examplemultiselectlookups
- RelatedTablePrimaryKey	ce_examplemultiselectlookupid

Attached below is an example script containing the code blocks above.



ImportResources.zip

Variables for Multi-Select lookup fields

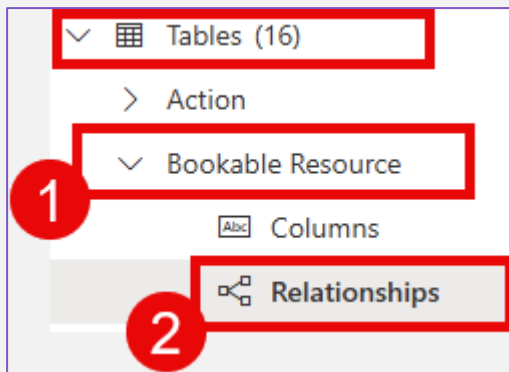
When migrating multi-select lookup fields, you'll need to get some additional information from Dataverse regarding the relationship. The steps below are a continuation of the examples above.

Quick Summary:

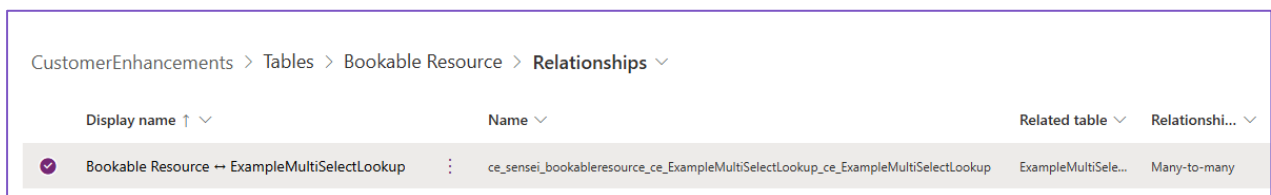
1. Open the table where the multiselect lookup field is > Relationships.
2. Open the many-to-many relationship for the lookup table:
3. Copy the Relationship name including the solution layer prefix.
4. Open the multi-select lookup table > columns.
5. Copy the name for the unique identifier field.
6. Open the table > properties.
7. Copy the plural name.

Variables for Multi-Select lookup fields

1. Open the **table** where the multiselect lookup field is > **Relationships**.



2. Open the **many-to-many** relationship for the lookup table:



Variables for Multi-Select lookup fields

3. Copy the **Relationship name** including the solution layer prefix.

Many-to-many

Choose the **Related table** to create your relationship. [Learn more](#)

Current (Many)

Table *

Bookable Resource

* — *

Related (Many)

Table *

ExampleMultiSelectLookup

☒ Searchable

Relationship name *

ce_ sensei_bookableresource_ce_ExampleMultiSelectLookup_ce_ExampleMultiSelectLookup

This is required for the script.

4. Open the multi-select lookup table > columns.

Tables (16)

- > Action
- > Bookable Resource
- > DecisionCategory
- > EmploymentStatus
- > ExampleMultiSelectLoo...

Columns

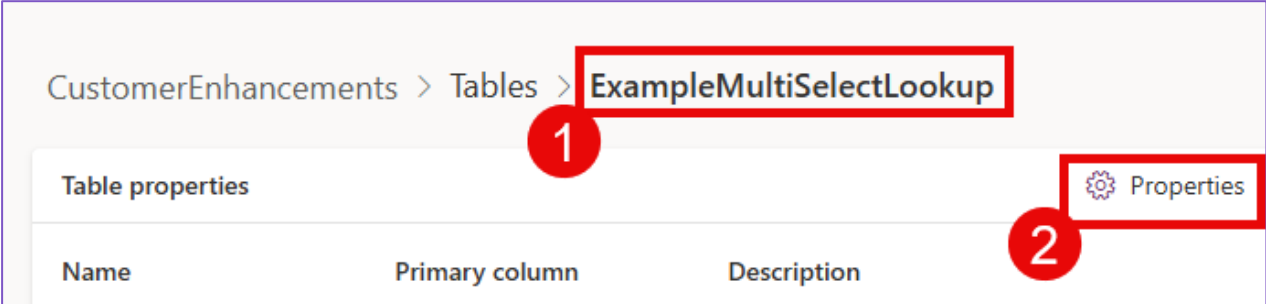
5. Copy the **name** for the **unique identifier** field.

CustomerEnhancements > Tables > ExampleMultiSelectLookup > Columns

Display name ↑	Name	Data type
☒ ExampleMultiSelectLookup	ce_ExampleMultiSelectLookupId	Unique identifier

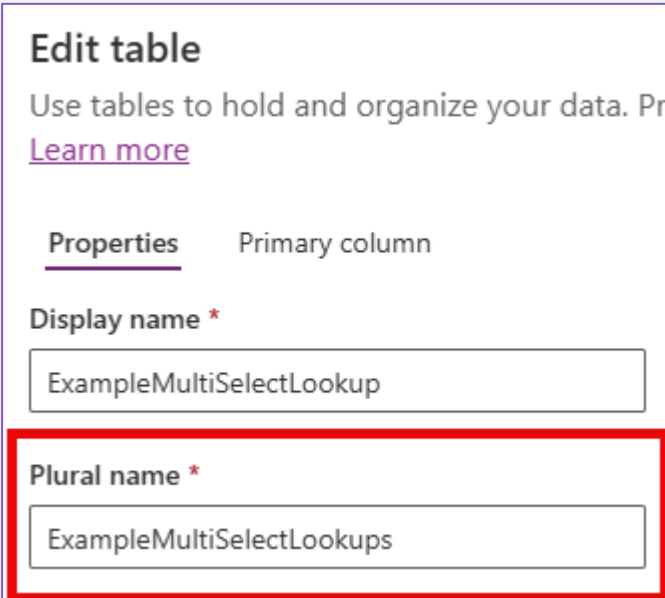
Variables for Multi-Select lookup fields

6. Open the table > properties.



The screenshot shows a breadcrumb navigation path: "CustomerEnhancements > Tables > ExampleMultiSelectLookup". A red box and the number "1" highlight the "ExampleMultiSelectLookup" link. Below the breadcrumb is a "Table properties" section. On the right side of this section, there is a "Properties" link with a gear icon. A red box and the number "2" highlight this "Properties" link. Below the "Table properties" section is a table with three columns: "Name", "Primary column", and "Description".

7. Copy the plural name.



The screenshot shows the "Edit table" form. It has a title "Edit table" and a subtitle "Use tables to hold and organize your data. [Pr](#)". Below the subtitle is a link "[Learn more](#)". There are two tabs: "Properties" (which is selected and underlined) and "Primary column". Under the "Properties" tab, there are two fields. The first field is labeled "Display name *" and contains the text "ExampleMultiSelectLookup". The second field is labeled "Plural name *" and contains the text "ExampleMultiSelectLookups". A red box highlights the "Plural name" field and its label.

2. Create Resources

Quick Summary:

1. Open SharePoint Online Management Shell.
2. Navigate to the location of this script.
3. Run the following command: `.\AltusPOLMigration.ps1`
4. Enter the Target Altus environment URL.
5. Select the Microsoft account to login with.
6. Select mode of operation.
7. Select **2: Import Resources (Named and Generic)**.

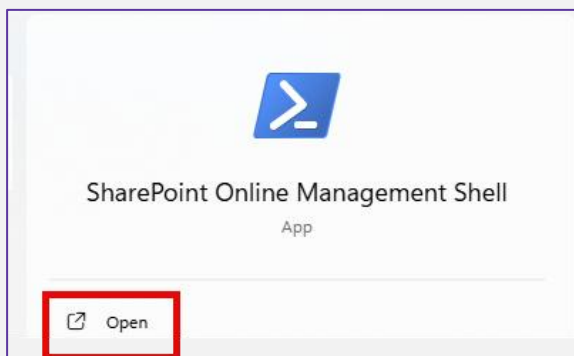
Video:

See link below to a video which covers the above steps with an example migration:

 [Project Online Data Migration - 5.2 Create Resources.mp4](#)

2. Create Resources

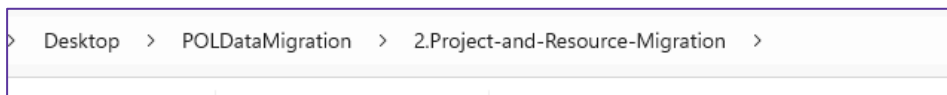
1. Open SharePoint Online Management Shell.



2. Navigate to the location of this script: `.\AltusPOLMigration.ps1`

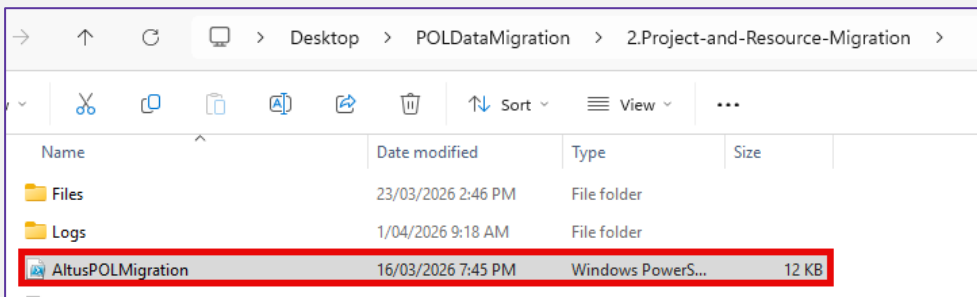
Example:

`cd c:\Users\JohnPrice\Desktop\POLDataMigration\2.Project-and-Resource-Migration`



2. Create Resources

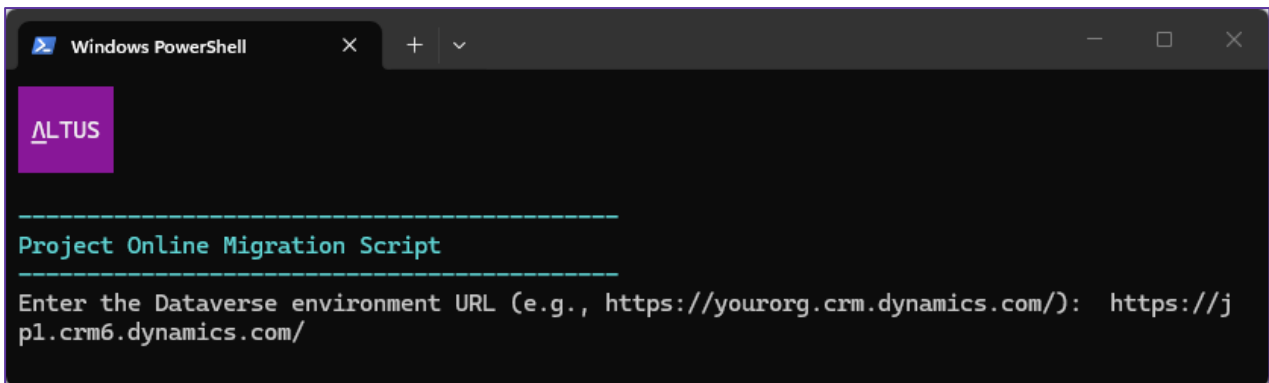
3. Run the following command: **.\AltusPOLMigration.ps1**



4. Enter the Target Altus environment URL.

Example:

<https://jp1.crm6.dynamics.com/>

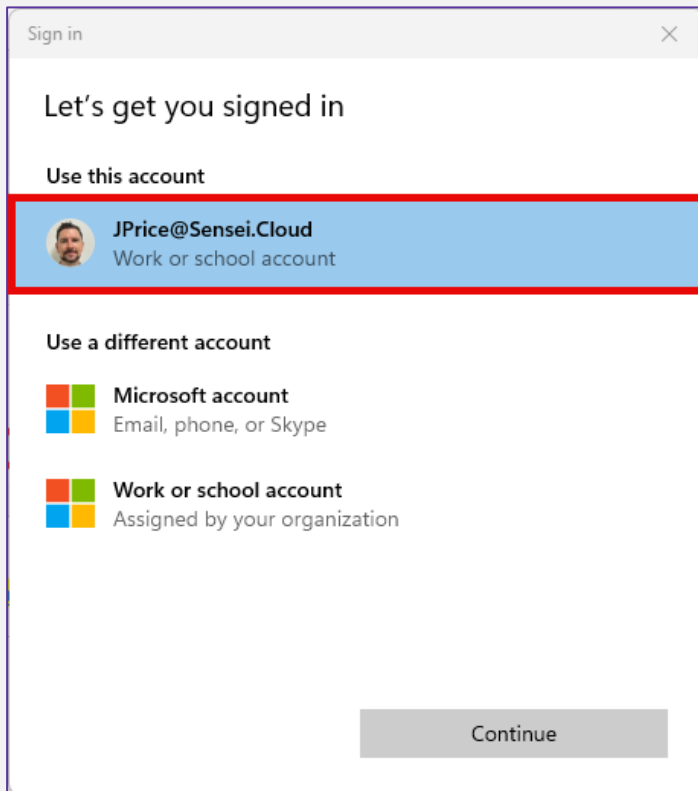


2. Create Resources

5. Select the Microsoft account to login with.

Example:

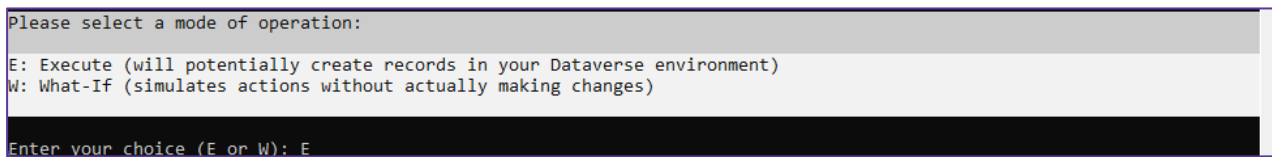
jprice@sensei.cloud



6. Select mode of operation:

E: Execute = Commit records to Dataverse

W: What-If = Simulate actions with creating records in Dataverse



7. Select 2: Import Resources (Named and Generic).

OOTB Data Mapping

The table below shows how the fields from the .MPP file are mapped to Altus.

.MPP Table	Altus Table
Resource Entry	sensei_bookableresource

The following OOTB fields are mapped as follows:

.MPP Field	Altus Field	Details	Example
Name	sensei_name	Firstname Surname	Lucius Fox
User Logon Account	sensei_user	Firstname Surname	Lucius Fox
Max. Units	sensei_targetutilization	Default value set	100
Type	sensei_resourcetype	Default value set	Named
Role	sensei_primaryrole	Default value set	Unclassified
Calendar GUID	Sensei_enterprisecalendar	Default value set	Default Calendar

Examples of Imported Resources

Below are examples of imported resources:

Named Resources

For a named resource, the following Altus OOTB fields will be set:

Resource Field	Value Set
Resource Type	Named
User	Matching System Username
Name	Resource Name from the .MPP file Resource Sheet
Primary Role	Unclassified
Enterprise Calendar	Default Calendar
Target Utilisation	100

Lucius Fox - Saved
 Bookable Resource

General Calendar Exceptions Rates Allocation Related ▾

Resource Type * Named ▾

User * Lucius Fox (Available) ×

Name * Lucius Fox

Department --- ▾

Primary Role * Unclassified ×

Enterprise Calendar * Default Calendar ×

Target Utilisation (%) 100

Form assist

Generic Resources

For a generic resource, the following Altus fields will be set:

Resource Field	Value Set
Resource Type	Generic
Name	Resource Name from the .MPP file Resource Sheet

Change Manager - Saved
Bookable Resource

General | Calendar Exceptions | Rates | Related ▾

Resource Type * Generic ▾

Name * Change Manager

Department --- ▾

Custom Fields

If you have mapped any custom fields they will appear for the selected resource. See example below for comparison between Project Online and Altus:

Project Online:

PMO **Project Centre** Project Details

Edit Resource: Chee Ho Tai

Type

Type: 
☐ Budget
☐ Generic

User Authentication

Enter the user account that the resource will need to logon.

* User logon account:

Resource Custom Fields

- 1 Resource Example Choice Field:
 
- 2 Resource Example Cost Field:
- 3 Resource Example Date Field:
 
- 4 Resource Example Duration Field:
- 5 Resource Example Flag Field:
 
- 6 Resource Example Lookup Field:
 
- 7 Resource Example Multiselect Lookup Field:
 
- 8 Resource Example Number Field:
- 9 Resource Example text field:

Altus:

Chee Ho Tai - Saved

Bookable Resource

General

Calendar Exceptions

Rates

Allocation

Related

Resource Type

*

Named

▼

User

*

 Chee Ho Tai (Busy) ×

🔍

Name

*

Chee Ho Tai

Department

▼

Primary Role

*

 Unclassified ×

🔍

Custom Fields

1

Resource Example Choice Field

Yes

▼

2

Resource Example Cost Field

\$123.00

3

Resource Example Date Field

13/04/2026

⋮

4

Resource Example Duration Field

1.33 hours

▼

9

Resource Example Text Field

Text field text

8

Resource Example Number Field

111

6

Resource Example Lookup Field

 New ×

🔍

5

Resource Example Flag Field

No

▼

Custom Fields

7

Multiselect Lookup

⋮

☐

Name ↑ ▼

Created On ▼

☐

Low Organisational Impact

10/04/2026 ...

☐

No Organisational Impact

10/04/2026 ...

Rows: 2

6. Map Project Custom Fields and Create Projects

Before running this process, you need to have created all the required custom fields in the target Altus environment. The scripts migrate data only (no configuration).

Field Type Mapping

The script will map fields as follows:

Project Online Field Type	Dataverse Field Type	Dataverse Format
Cost	Currency	N/A
Date (Date only)	Date and time	Date only
Date (Date and Time)	Date and time	Date and time
Duration	Whole Number	Duration
Flag	Choice	Yes/no
Number (Whole)	Whole Number	None
Number (Decimal)	Decimal	None
Text (Single line of Text)	Single line of text	Text
Text (Multiple lines of Text)	Multiple lines of text	Text

Example Mapping

The example below will map the following fields between Project Online and Altus. The letters before the underscore ("_") denotes the solution layer name.

Project Online Field Name	Altus Internal Name	Dataverse Format
Example Cost Field	ce_examplecostfield	Currency
Example Date Field	ce_exampledatefield	Date and time > Date only
Example Duration Field	ce_exampledurationfield	Whole Number > Duration
Example Flag Field	ce_examplechoicefield	Choice > YesNoChoice
Example Number Field	ce_examplenumberfield	Number > Decimal
Example Multi line text field	ce_examplemultilinetextfield	Text > Multiple lines of text
Example Lookup Table Field	ce_examplelookuptablefield	Text > Lookup

1. Map Project Custom Fields

The ImportProjects.ps1 script needs to be updated to map custom fields between systems. I

The parts of the script that are highlighted below are as follows. Only these parts of the script block need to be updated.

Attribute	Details
Project Online Field	Display name for the field as it appears in Project Online.
Altus Field	This is the Logical Name for the Altus field.
Default.ps1 script reference	Reference to another Altus script, specifically the alias.
Additional Altus variables	Altus references to related tables.

Quick Summary:

1. Open this folder, then edit this script: **ImportProjects.ps1**.
2. Map all **Text** fields.
3. Map all **Date** fields.
4. Map all **Number** fields.
5. Map all **Duration** fields.
6. Map all **Cost** fields.
7. Map all **Flag** fields (Flag to Flag).
8. Map all **Flag** fields (Choice to Flag).
9. Map all **Lookup** fields (Lookup to Choice).
10. Map all **Lookup** fields (Single-select).
11. Map all **Lookup** fields (Multi-select).

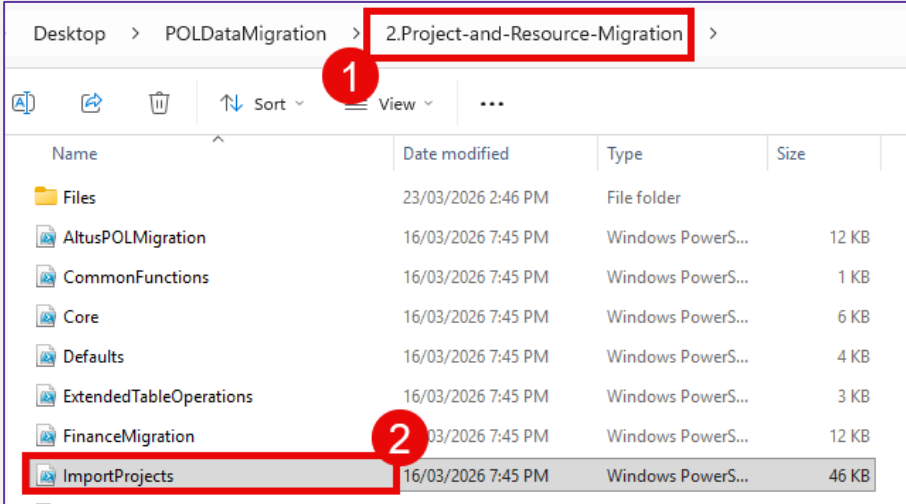
Video:

See link below to a video which covers the above steps with an example migration:

 [Project Online Data Migration - 6.1 Map Custom Fields.mp4](#)

1. Map Project Custom Fields

1. Open this folder, then edit this script: **ImportProjects.ps1**



1. Map Project Custom Fields

2. Map all **Text** fields.

```
# --- Enterprise Custom Fields (from $ReportingProject.CustomFields[]) ---
$textValue = Get-ReportingCustomFieldTextValue -ReportingProject $ReportingProject -
CustomFieldName 'Example Multi line text field'
if ($textValue) { $Project['ce_examplemultinetextfield'] = [string]$textValue }
```

3. Map all **Date** fields.

```
$dateValue = Get-ReportingCustomFieldTextValue -ReportingProject $ReportingProject -
CustomFieldName 'Example Date Field'
if ($dateValue) { $Project['ce_exampledatefield'] = ($dateValue -as [datetime]).ToString("yyyy-
MM-dd") }
```

4. Map all **Number** fields.

```
$numberValue = Get-ReportingCustomFieldTextValue -ReportingProject $ReportingProject -
CustomFieldName 'Example Number Field'
if ($numberValue) { $Project['ce_examplenumberfield'] = [int][double]$numberValue }
```

5. Map all **Duration** fields.

```
$durationValue = Get-ReportingCustomFieldTextValue -ReportingProject $ReportingProject -
CustomFieldName 'Example Duration Field'
if ($durationValue) { $Project['ce_exampledurationfield'] = [int][double]$durationValue }
```

6. Map all **Cost** fields.

```
$costValue = Get-ReportingCustomFieldTextValue -ReportingProject $ReportingProject -
CustomFieldName 'Example Cost Field'
if ($costValue) { $Project['ce_examplecostfield'] = ($costValue -as [decimal]) }
```

7. Map all **Flag** fields (Flag to Flag).

```
$cfValue = Get-ReportingCustomFieldTextValue -ReportingProject $ReportingProject -
CustomFieldName 'Example Flag Field'
if ($cfValue) {
    if ($cfValue -eq 'true') { $ynValue = $true } else { $ynValue = $false } # must be set as True and
False not 1 and 0.
    $Project['ce_exampleflagfield'] = $ynValue
}
```

8. Map all **Flag** fields (Lookup to Flag).

```
$cfValue = Get-ReportingCustomFieldTextValue -ReportingProject $ReportingProject -
CustomFieldName 'Example Choice Field'
if ($cfValue) {
    if ($cfValue -eq 'Yes') { $ynValue = $true } else { $ynValue = $true }
    $Project['ce_exampleflagfield2'] = $ynValue
}
```

1. Map Project Custom Fields

9. Map all **Lookup** fields (Lookup to Choice).

Set-ProjectChoiceField -ProjectBody \$Project -ReportingProject \$ReportingProject -
POLCustomFieldName 'Example Choice Field' -ChoiceName 'choice_YesNo' -
DataverseFieldName 'ce_examplechoicefield'

10. Map all **Lookup** fields (Single-select).

Set-ProjectLookupField -ProjectBody \$Project -ReportingProject \$ReportingProject -
POLCustomFieldName 'Example Lookup Field' -LookupName 'lookup_TeamName' -
DataverseFieldName 'ce_ExampleLookupField'

Note: The Dataverse field name in this example is case sensitive.

11. Map all **Lookup** fields (Multi-select). Additional variables are highlighted in **Pink**.

Set-ProjectManyToManyRelationship -ProjectBody \$Project -ReportingProject \$ReportingProject -
POLCustomFieldName 'Example Multiselect Lookup Field' -LookupName 'lookup_MultiValueTable' -
RelationshipName 'ce_ExampleMultiSelectLookup_sensei_project_sensei_project' -
RelatedTableCollectionName 'ce_examplemultiselectlookups' -RelatedTablePrimaryKey
'ce_examplemultiselectlookupid'

The following additional pieces of information need to be set for multi-select lookup fields:

Variable	Value
-RelationshipName	ce_sensei_bookableresource_ce_ExampleMultiSelectLookup_ce_ExampleMultiSelectLookup
- RelatedTableCollectionName	ce_examplemultiselectlookups
- RelatedTablePrimaryKey	ce_examplemultiselectlookupid

Attached below is an example of the above script:



ImportProjects.zip

Variables for Multi-Select lookup fields

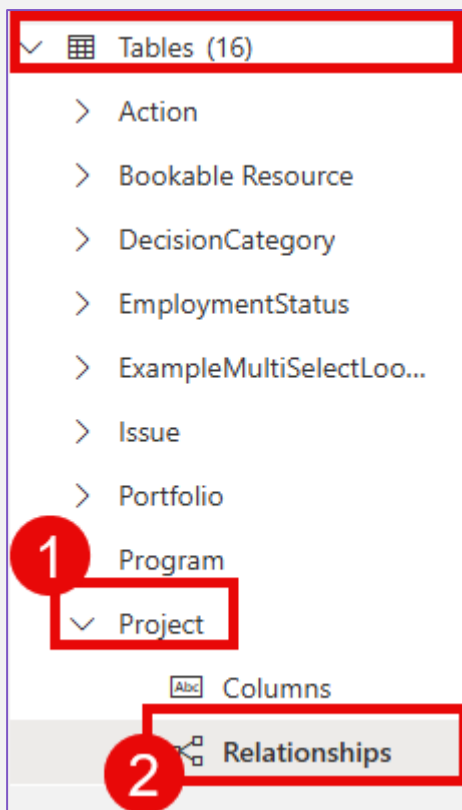
When migrating multi-select lookup fields, you'll need to get some additional information from Dataverse regarding the relationship. The steps below are a continuation of the examples above.

Quick Summary:

8. Open the table where the multiselect lookup field is > Relationships.
9. Open the many-to-many relationship for the lookup table:
10. Copy the Relationship name including the solution layer prefix.
11. Open the multi-select lookup table > columns.
12. Copy the name for the unique identifier field.
13. Open the table > properties.
14. Copy the plural name.

Variables for Multi-Select lookup fields

1. Open the **table** where the multiselect lookup field is > **Relationships**.



2. Open the **many-to-many** relationship for the lookup table:

CustomerEnhancements > Tables > Project > Relationships ▾			
Display name ↑ ▾	Name ▾	Related table ▾	Relationship type ▾
✓ ExampleMultiSelectLookup ↔ ...	ce_ExampleMultiSelectLookup_sensei_project_sensei_p...	Project	Many-to-many

Variables for Multi-Select lookup fields

3. Copy the **Relationship name** including the solution layer prefix.

Many-to-many

Choose the **Related table** to create your relationship. [Learn more](#)

Current (Many)

Table *

Project

* — *

Related (Many)

Table *

ExampleMultiSelectLookup

☒ Searchable

Relationship name *

ce_ ExampleMultiSelectLookup_sensei_project_sensei_project

Relationship table name *

This is required for the script.

4. Open the multi-select lookup table > columns.

Tables (16)

- > Action
- > Bookable Resource
- > DecisionCategory
- > EmploymentStatus
- > ExampleMultiSelectLoo...

Columns

5. Copy the **name** for the **unique identifier** field.

CustomerEnhancements > Tables > ExampleMultiSelectLookup > Columns

Display name ↑	Name	Data type
☒ ExampleMultiSelectLookup	ce_ExampleMultiSelectLookupId	Unique identifier

Variables for Multi-Select lookup fields

6. Open the table > properties.

The screenshot shows a breadcrumb trail: "CustomerEnhancements > Tables > ExampleMultiSelectLookup". A red box labeled "1" highlights the "ExampleMultiSelectLookup" link. Below the breadcrumb is a "Table properties" section. A red box labeled "2" highlights the "Properties" link, which is preceded by a gear icon.

7. Copy the plural name.

The screenshot shows the "Edit table" form. It has a title "Edit table" and a subtitle "Use tables to hold and organize your data. Pr". Below the subtitle is a link "Learn more". There are two tabs: "Properties" (selected) and "Primary column". Under the "Properties" tab, there are two fields: "Display name *" with the value "ExampleMultiSelectLookup" and "Plural name *" with the value "ExampleMultiSelectLookups". A red box highlights the "Plural name" field and its value.

2. Create Projects

Quick Summary:

8. Open SharePoint Online Management Shell.
9. Navigate to the location of this script.
10. Run the following command: `.\AltusPOLMigration.ps1`
11. Enter the Target Altus environment URL.
12. Select the Microsoft account to login with.
13. Select mode of operation.
14. Select **3: Import Projects**.

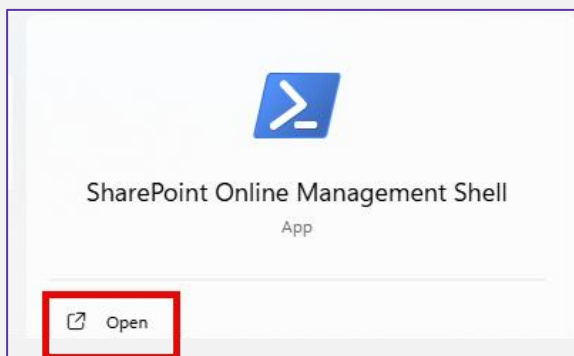
Video:

See link below to a video which covers the above steps with an example migration:

 [Project Online Data Migration - 6.2 Create Projects.mp4](#)

2. Create Projects

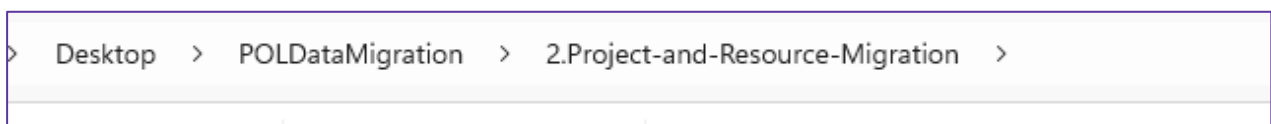
1. Open SharePoint Online Management Shell.



2. Navigate to the location of this script: `.\AltusPOLMigration.ps1`

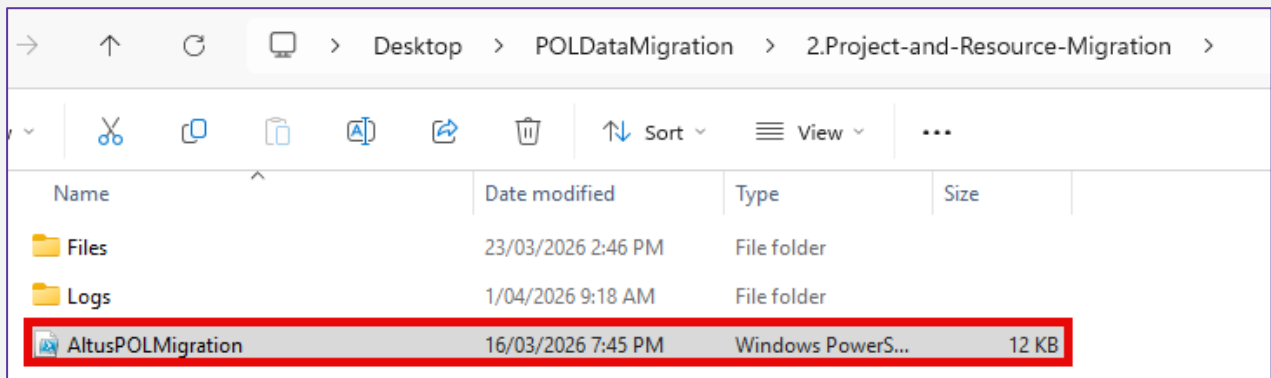
Example:

`cd c:\Users\JohnPrice\Desktop\POLDataMigration\2.Project-and-Resource-Migration`



2. Create Projects

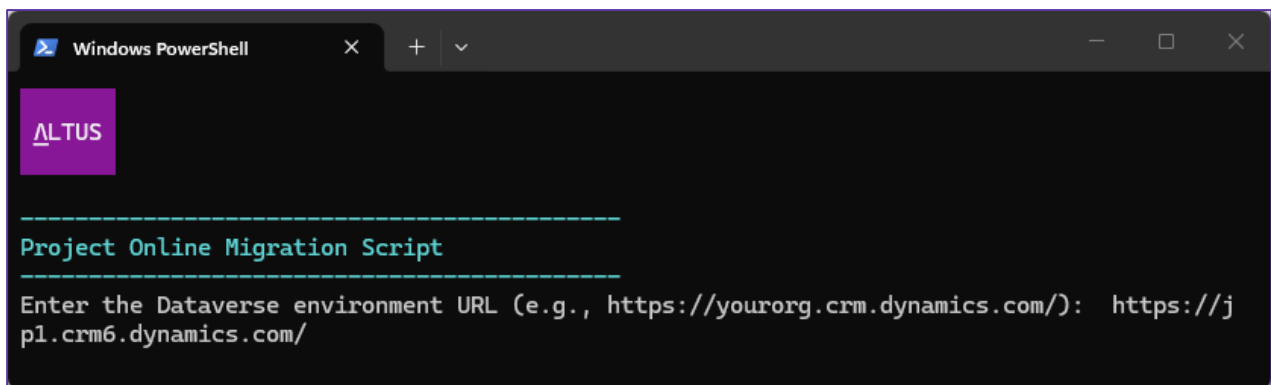
3. Run the following command: **.\AltusPOLMigration.ps1**



4. Enter the Target Altus environment URL.

Example:

<https://jp1.crm6.dynamics.com/>

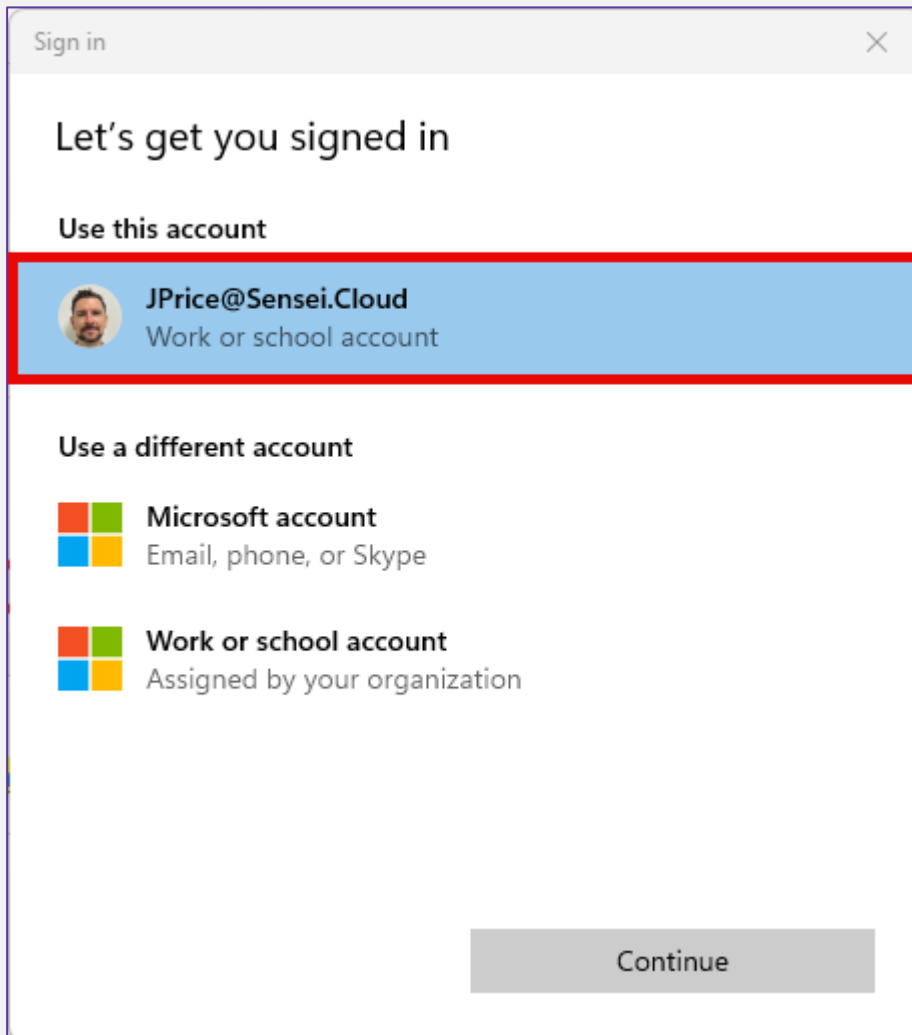


2. Create Projects

5. Select the Microsoft account to login with.

Example:

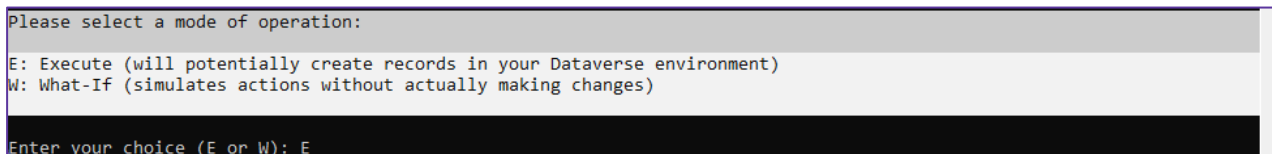
jprice@sensei.cloud



6. Select mode of operation:

E: Execute = Commit records to Dataverse

W: What-If = Simulate actions with creating records in Dataverse

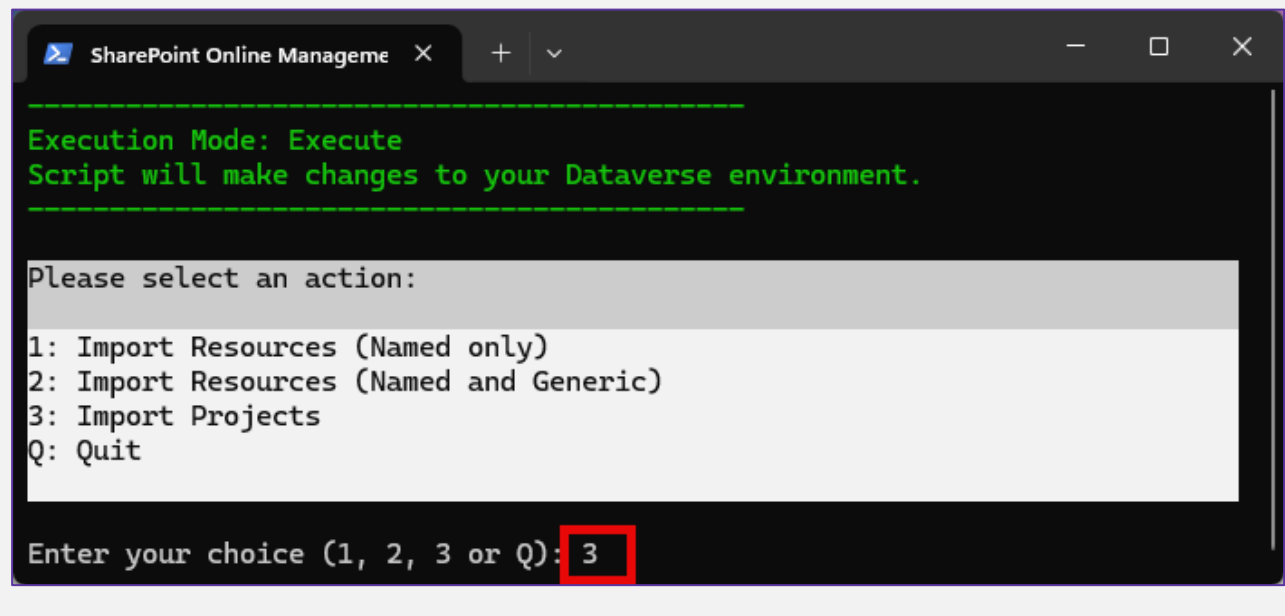


2. Create Projects

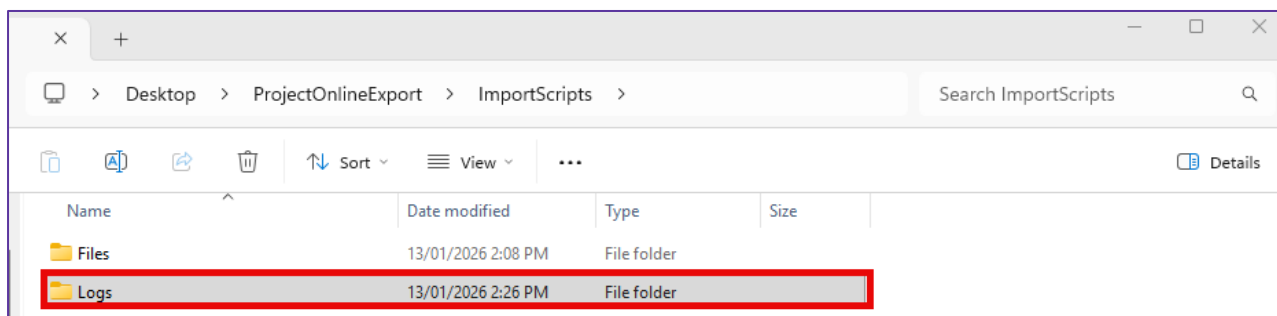
7. Select 3: Import Projects.

Note: You can only run one operation at a time. To Import Named Resources & Projects will require the script to be run twice.

When running the Projects Import, Microsoft Project Desktop will open for each project. Let this process run in the background, if you close Microsoft Project it will cause the script to fail.



Once complete, a log file will be generated in this folder:



OOTB Data Mapping

The table below shows how the OOTB fields from the .MPP file are mapped to Altus.

.MPP Table	Altus Table
Task Entry	sensei_project

The following OOTB fields are mapped as follows:

.MPP Field	Altus Field	Details	Example
Task Name	sensei_name	Where the TaskID = 0	Adelaide Fitness Project
N/A	sensei_projecttype	Default value set	Major Project
N/A	sensei_externalprojectid	Set the GUID = Project Online GUID	ProjectDesktop_d14bae12-39b3-e911-b07b-00155d9c3f18

Example of Imported Project

Below is an example of an imported Project.

Projects

In the projects view, the newly created project will appear as follows:

The icon denotes that it has a connected Microsoft Project Schedule.

In the project record, it will appear as follows:

The screenshot shows the 'Adelaide Office Refresh' project record in Altus. The project is marked as a 'Major Project' and is currently in the 'Planning' phase (less than 1 minute). The record is divided into two main sections: 'Summary' and 'Project Settings'. In the 'Summary' section, the 'Name' field is highlighted with a red box and contains the text 'Adelaide Office Refresh'. In the 'Project Settings' section, the 'Project Integration' field is highlighted with a red box and shows a green icon indicating a connected Microsoft Project Schedule. The 'Project Group' is set to 'No Group'.

Custom Fields

If you have mapped any custom fields they will appear for the selected project. See example below for comparison between Project Online and Altus:

Project Online:



Brisbane Roadmap Tool Implementation

Project Dashboard
Project Registers (Major)
Schedule

PMO
Project Centre
Project Details

Project Registers (Major)

Info Status: Checked-in Last Modified: 14/04/2026 9:51

Project Attributes

1	Example Choice Field demo for Project online data migration. POL lookup to dataverse choice.	No
2	Example Cost Field Demo field for the Project Online data migration.	\$1,234.56
3	Example Date Field Demo duration field for the project online data migration.	30/03/2026 8:00 AM
4	Example Duration Field Demo field for the project online data migration.	101d
5	Example Flag Field Demo field for the project online data migration.	Yes
6	Example Lookup Field Demo for the project online data migration.	Executive Team
7	Example Multi line text field Demo field for the project online data migration.	The project commenced in early Q2 with a focus on improving cross-functional collaboration and streamlining reporting processes across the organisation. Initial workshops identified several inefficiencies in how information was captured, shared, and actioned, particularly between delivery teams and senior stakeholders. As a result, a series of process improvements were proposed, including the introduction of standardised templates, clearer governance checkpoints, and enhanced visibility of key metrics. Throughout the implementation phase, feedback loops were established to ensure that proposed changes were both practical and aligned with user needs. While there was some initial resistance to adopting new ways of working, targeted training sessions and ongoing support helped to build confidence and drive engagement across teams. Early indicators suggest a measurable improvement in reporting accuracy and a reduction in duplicated effort. Looking ahead, the focus will shift towards embedding these changes into business-as-usual operations and identifying further opportunities for optimisation. Continuous improvement will remain a key priority, supported by regular reviews and stakeholder feedback to ensure sustained value delivery.
8	Example Multiselect Lookup Field Demo for the Project Online data migration.	No Organisational Impact, Low Organisational Impact
9	Example Number Field Demo field for the project online data migration.	5,678

Altus:

Brisbane Roadmap Tool Implementation - Saved

Project · Information ▾

Major Project
Active for 1 hour

<

Planning

Details Evaluation Key Dates Tasks Financials Information Tracking

Project Online Custom Fields

5 Example Flag Field2 Yes ▾

1 ExampleChoiceField No ▾

2 ExampleCostField \$1,234.56

4 ExampleDurationField 13.47 hours ▾

3 ExampleDateField 30/03/2026 📅

6 ExampleLookupField 🗑️ Executive Team × 🔍

9 ExampleNumberField 5,678.00

8 ExampleMultilineTextField The project commenced in early Q2 with a focus on improving cross-functional collaboration and streamlining

7 Example Multiselect lookup

Example Multiselect Lookup ⋮

☐ Name ↑ ▾

Created On ▾

☐ Low Organisational Impact

10/04/2026 2:05 ...

☐ No Organisational Impact

10/04/2026 2:05 ...

Rows: 2

7. Map Task Custom Fields

As part of the data migration to Altus, you have two options related to the migration of task level custom fields:

1. Mapping them to local custom fields in Microsoft Project Desktop.
2. Mapping them directly to Altus task custom fields.

If you wish to continue to manage custom fields in MS Project, continue in this section, noting that there are some limitations to be aware of.

If you wish to manage these in Altus going forward, please go to [this section](#) of the document.

Mapping to local MS Project custom fields

As part of the migration to Altus, all Project Online Enterprise Custom fields must be converted to Local Custom fields in MS Project Desktop.

Whilst Project Online provides customers with the ability to create an unlimited number of task-level custom fields, MS Project Desktop is limited to the following:

Note: If you don't have any task custom fields to map, you can skip this part of the process.

MS Project hard limits for Task Custom Fields

MS Project Desktop Local Custom Field Type	Number Available to Use
Date	9 (Date1 – Date9)
Finish	10 (Finish1 - Finish10)
Flag	20 (Flag1 - Flag20)
Number	20 (Number1 - Number20)
Start	10 (Start1 - Start10)
Text	26 (Text1 – Text26)
Outline Code	10 (Outline Code1 - Outline Code10)

The following local fields are reserved for use by Altus for Project including the details of why they have been reserved:

Task Local Field	Reserved For
Text 29	Task GUID
Text 30	Task Hash
Date 10	Protected Actuals Before date

Resource Local Field	Reserved For
Text 27	Resource Role
Text 28	Resource Type
Text 29	Resource GUID
Text 30	Resource Hash

The end-to-end process to migrate the task-level custom fields is as follows:

1. Create the task custom fields in Altus.

2. Set the mapping in Altus.
3. Connect to Altus via Microsoft Project.
4. Parameter and Run this script: **Publish-MPPsToAltus.ps1**

Note: The Microsoft script doesn't output any values for non-text-based lookup tables. As such, the process detailed below won't work for these types of lookup tables.

In this example, the following fields will be mapped from Project Online. Note, none of these fields will be included in the Altus baselines.

⚠ Task lookup fields will be broken by the migration process. The local custom fields will retain the data post-migration but these will be single line of text fields.

Project Online Enterprise Task Field	Project Online Local Task Field	Project Online Type
Task Example Date Field	Date1	Date
Task Example Flag Field	Flag1	Flag
Task Example Lookup Field	Text3	Text
Task Example Number Field	Number1	Number
Task Example Text Field	Text4	Text

Project Online Task Field	Altus Field Logical Name	Altus Field Type
Task Example Date Field	ce_taskexampledatefield	Date only
Task Example Flag Field	ce_taskexampleflagfield	Yes/no
Task Example Lookup Field	ce_taskexamplelookuptotextfield	Single line of text
Task Example Number Field	ce_taskexamplenumberfield	Whole number
Task Example Text Field	ce_taskexampletextfield	Single line of text

Note: Custom cost and duration fields are currently unsupported for mapping.

Task lookup fields

Note, the current process will convert Project Online task lookup fields to text fields (with no lookups) in Project Desktop.

As such, when users set data in these fields, this will be free text rather than lookup table values.

Investigation

We are currently investigating a two-way sync of task lookup fields between Altus and Project Desktop (similar to how Project Online used to do this).

This would provide a similar user experience to what Project Online users were used to. We will provide further updates on this when the investigation is complete.

Alternative method

If you need to maintain lookup table fields within the schedule, you can create task custom fields in Altus (without a corresponding field in Project Desktop).

This will enable you to control the values users can set for their schedules.

The user will then update this information in the Altus schedule directly.

1. Create the task custom field in Altus

Follow this process to create the task custom fields in Altus: [Create Altus Custom Fields](#).

2. Generate the .csv mapping file

Quick Summary:

1. Open **MS Project Desktop**.
2. Open **SharePoint Online Management Shell**.
3. Connect to the **3.Schedule-Migration** folder.
4. Run this script: **New-FieldMappingFile.ps1**
5. When you run the script, you will be prompted to map decimal fields accordingly:

Video:

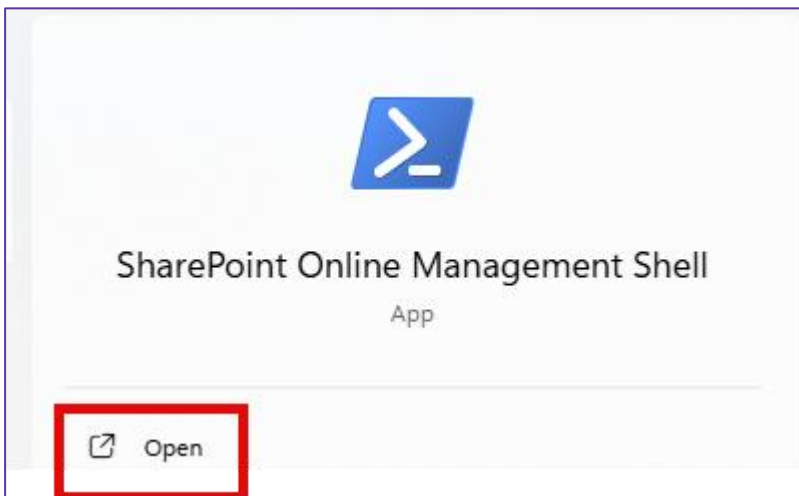
See link below to a video which covers the above steps with an example migration:

 [Project Online Data Migration - 7.2 Generate CSV mapping file](#)

2. Generate the .csv mapping file

1. Open **MS Project Desktop**.

2. Open **SharePoint Online Management Shell**.



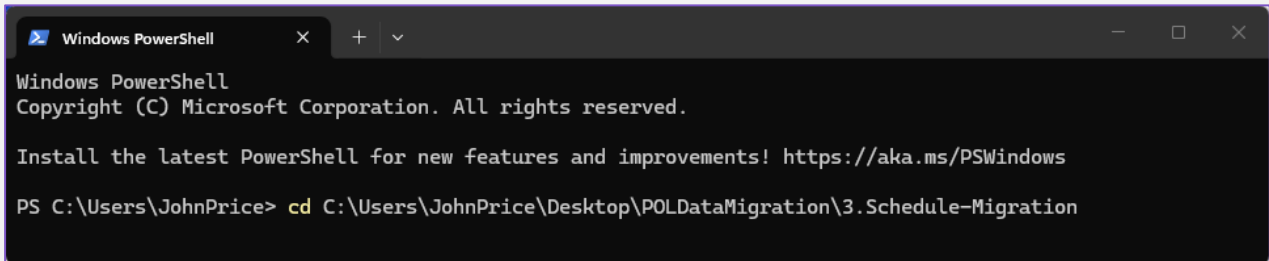
Note: Do not run as an Administrator as this will cause the script to error.

2. Generate the .csv mapping file

3. Connect to the **3.Schedule-Migration** folder.

Example:

```
cd C:\Users\JohnPrice\Desktop\POLDataMigration\3.Schedule-Migration
```



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\JohnPrice> cd C:\Users\JohnPrice\Desktop\POLDataMigration\3.Schedule-Migration
```

4. Run this script: **New-FieldMappingFile.ps1**

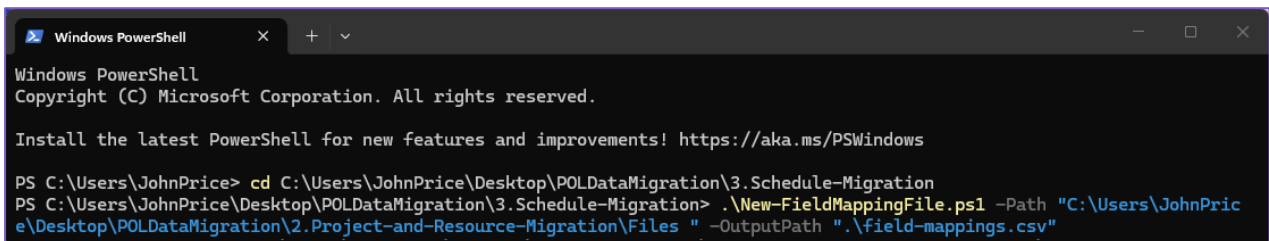
You will need to add the following to the script:

Parameter	Details
-Path	Location where the .MPP and .JSON files were exported to.
-OutputPath	Where you want the field-mappings.csv to be exported to.

By default, the field-mappings.csv will be exported to the same folder as the script is located.

Example

```
.\New-FieldMappingFile.ps1 -Path "C:\Users\JohnPrice\Desktop\POLDataMigration\2.Project-and-Resource-Migration\Files " -OutputPath ".\field-mappings.csv"
```



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

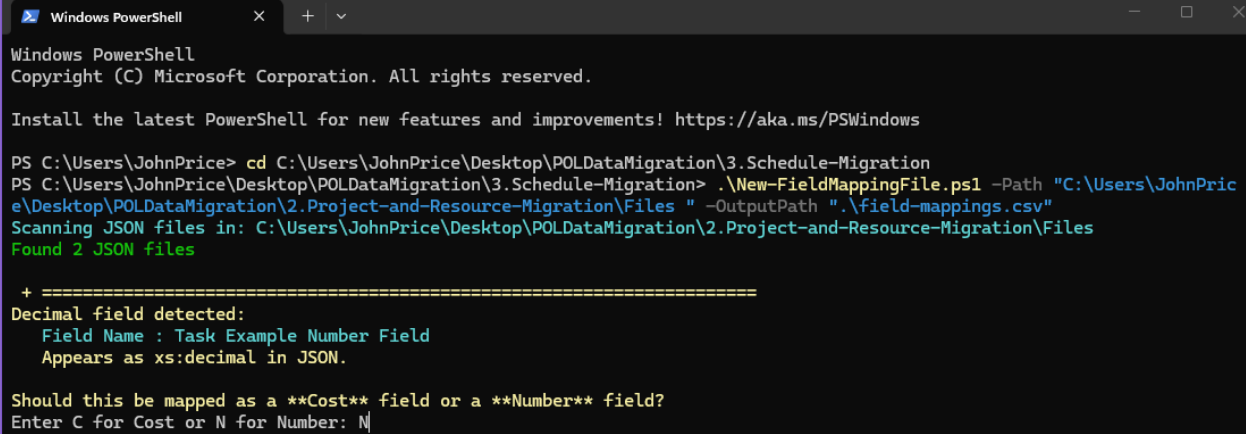
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\JohnPrice> cd C:\Users\JohnPrice\Desktop\POLDataMigration\3.Schedule-Migration
PS C:\Users\JohnPrice\Desktop\POLDataMigration\3.Schedule-Migration> .\New-FieldMappingFile.ps1 -Path "C:\Users\JohnPrice\Desktop\POLDataMigration\2.Project-and-Resource-Migration\Files " -OutputPath ".\field-mappings.csv"
```

2. Generate the .csv mapping file

5. When you run the script, you will be prompted to map decimal fields accordingly:

In this example, the number field will be mapped to a number field in MS Project.



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

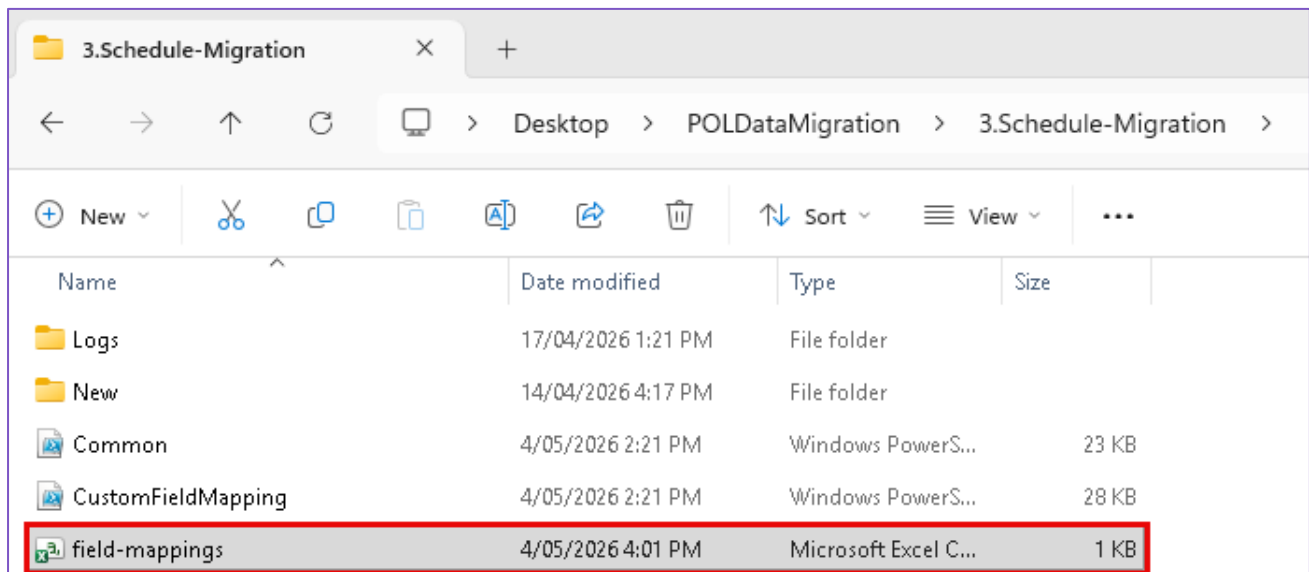
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\JohnPrice> cd C:\Users\JohnPrice\Desktop\POLDataMigration\3.Schedule-Migration
PS C:\Users\JohnPrice\Desktop\POLDataMigration\3.Schedule-Migration> .\New-FieldMappingFile.ps1 -Path "C:\Users\JohnPrice\Desktop\POLDataMigration\2.Project-and-Resource-Migration\Files" -OutputPath ".\field-mappings.csv"
Scanning JSON files in: C:\Users\JohnPrice\Desktop\POLDataMigration\2.Project-and-Resource-Migration\Files
Found 2 JSON files

+ =====
Decimal field detected:
  Field Name : Task Example Number Field
  Appears as xs:decimal in JSON.

Should this be mapped as a **Cost** field or a **Number** field?
Enter C for Cost or N for Number: N
```

Once this has been completed, the field-mapping.csv spreadsheet will be generated:



When you open this file, the following information will be displayed:

	A	B	C
1	CustomFieldGuid	CustomFieldName	LocalFieldTarget
2	0000e8d9-65f1-4769-9bd2-819d38036fcc	Health	Text1
3	0af68558-142e-f111-ab42-00155d582514	Task Example Text Field	Text2
4	2a259267-b1f2-e811-afa9-00155da0850d	Reportable	Text3
5	7d7b10f4-8c0b-f011-9293-00155d08d60e	Test Text Field	Text4
6	8e113873-b3f2-e811-afab-00155da4880c	Task Cost KPI	Text5
7	8ebf0790-182e-f111-ab37-00155d584f07	Task Example Lookup Field	Text6
8	907efb7a-b2f2-e811-afac-00155da4690d	Task Schedule KPI	Text7
9	a0a946de-b2f2-e811-afac-00155d9cad0c	Task Work KPI	Text8
10	bd52b19c-22b6-ec11-b15d-00155da0242a	Employment Status	Text9
11	ca95a3f0-6e30-f011-8187-00155d081102	task RAG	Number1
12	cbc9511a-182e-f111-ab40-00155d595912	Task Example Number Field	Number2
13	c1cd7930-142e-f111-ab42-00155d582514	Task Example Date Field	Date1
14	34f1390f-172e-f111-ab42-00155d58452b	Task Example Flag Field	Flag1

3. Set the field mapping in Altus

Quick Summary:

6. Open **Altus**.
7. Open **Settings**.
8. Select **Altus for Project Configuration**.
9. Select **New Field Mapping**.
10. Create the mapping as per the field mapping.csv
11. Repeat Steps 4 and 5 for all Task Custom fields you want to map.

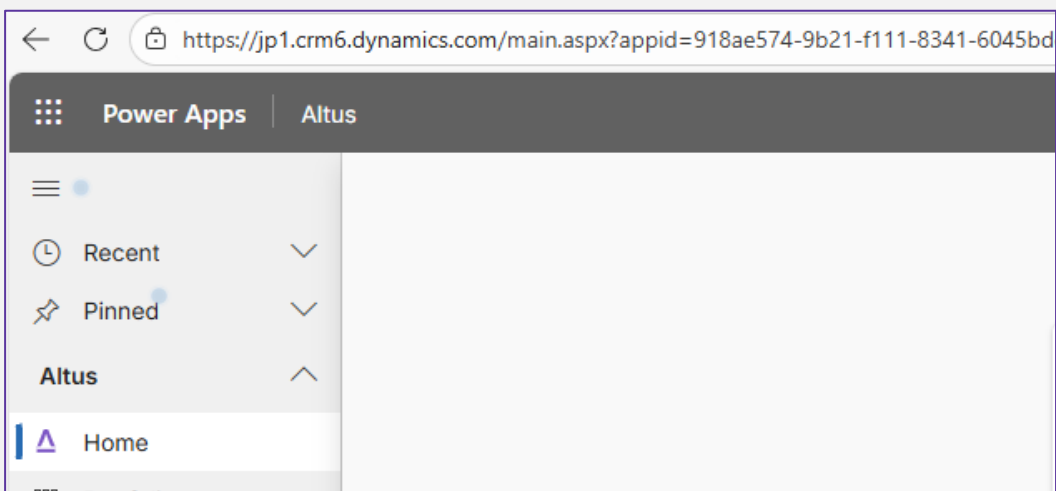
Video:

See link below to a video which covers the above steps with an example migration:

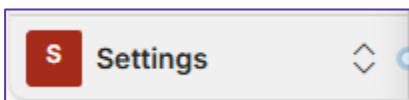
 [Project Online Data Migration - 7.3 Map Task Custom Fields in Altus.mp4](#)

2. Set the field mapping in Altus

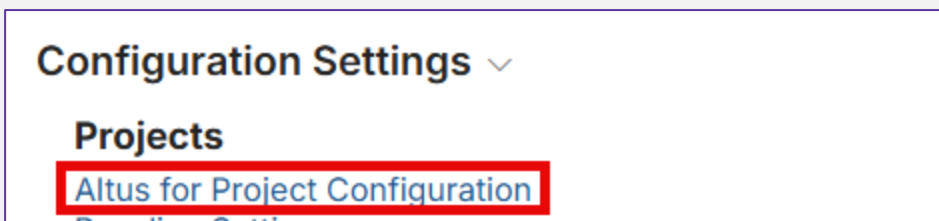
1. Open Altus.



2. Open Settings.



3. Select Altus for Project Configuration.



2. Set the field mapping in Altus

4. Select New Field Mapping.

projectDesktopConfig - Unsaved
Altus Config Settings

General Properties Related ▾

Deadline	   
Resource Name	   
Resource Units	   
Notes	   

Custom Field Mapping
This custom field mapping allows administrators the ability to map known custom fields to Altus fields, please note that standard mapping does take precedence.

Microsoft Project Field

Actions

+ New Field Mapping

5. Create the mapping as per the field mapping.csv

1. Specify the Microsoft Project Field.
2. Select the Altus Field.
3. Ok.

Field Mapping ✕

Field Mapping between Microsoft Project and Altus

Microsoft Project Field 1

Text2 sensei_task

The Field on Microsoft Project Table

Altus Field 2

TaskExampleTextField ×

The Field on Altus

OK 3 **Cancel**

6. Repeat Steps 4 and 5 for all Task Custom fields you want to map.

4. Run the script to map the enterprise task custom fields to local MS Project desktop custom fields

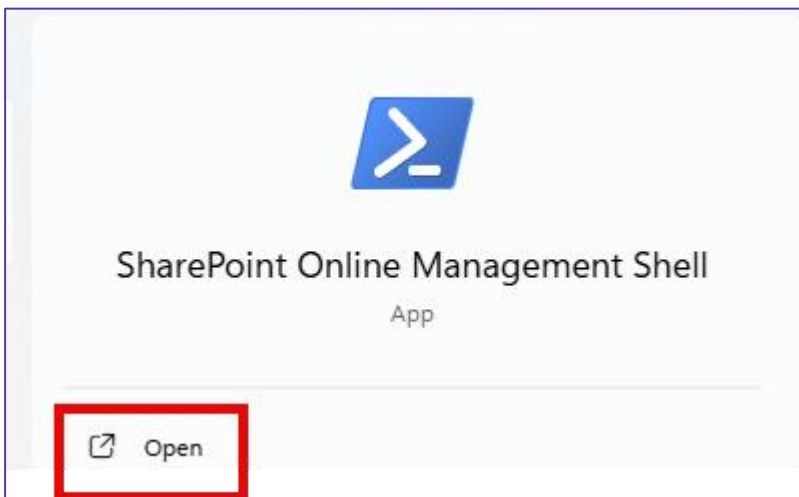
Quick Summary:

1. Open **MS Project Desktop**.
2. Open **SharePoint Online Management Shell**.
3. Connect to the **3.Schedule-Migration** folder.
4. Run this script: **Set-CustomFieldMapping.ps1**

4. Run the script to map the enterprise task custom fields to local MS Project desktop custom fields

1. Open **MS Project Desktop**.

2. Open **SharePoint Online Management Shell**.

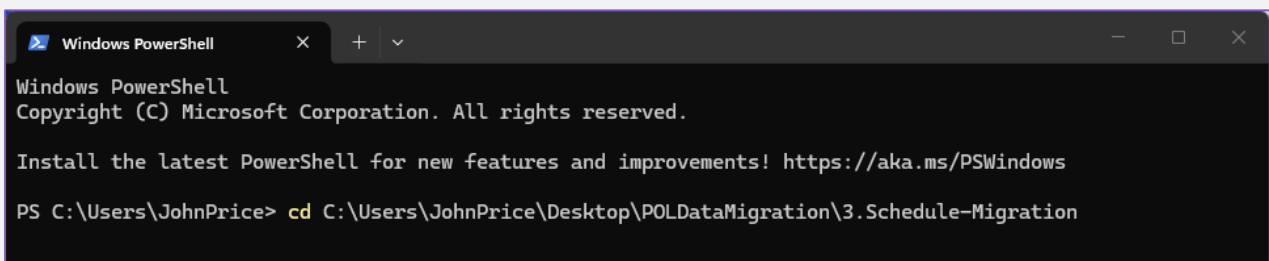


Note: Do not run as an Administrator as this will cause the script to error.

3. Connect to the **3.Schedule-Migration** folder.

Example:

`cd C:\Users\JohnPrice\Desktop\POLDataMigration\3.Schedule-Migration`



4. Run the script to map the enterprise task custom fields to local MS Project desktop custom fields

4. Run this script: **Set-CustomFieldMapping.ps1**

You will need to add the following to the script:

Parameter	Details
-Path	Location where the .MPP and .JSON files were exported to.
-MappingFile	Location where the 'field-mappings.csv' was saved to.

Example

```
.\Set-CustomFieldMapping.ps1 -MppFolder "C:\Users\JohnPrice\Desktop\POLDataMigration\2.Project-and-Resource-Migration\Files" -MappingFile "C:\Users\JohnPrice\Desktop\POLDataMigration\3.Schedule-Migration\field-mappings.csv"
```

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\JohnPrice> cd C:\Users\JohnPrice\Desktop\POLDataMigration\3.Schedule-Migration
PS C:\Users\JohnPrice\Desktop\POLDataMigration\3.Schedule-Migration> .\Set-CustomFieldMapping.ps1 -MppFolder "C:\Users\JohnPrice\Desktop\POLDataMigration\2.Project-and-Resource-Migration\Files" -MappingFile "C:\Users\JohnPrice\Desktop\POLDataMigration\3.Schedule-Migration\field-mappings.csv"
```

Once the process has been completed, the enterprise custom fields will be mapped to the local fields specified in the .csv file.

See below for an example of this process.

Task Example Date Field	Date1	Task Example Flag Field	Flag1	Task Example Number Field	Number2	Task Example Text Field	Text2	Task Example Lookup Field	Text6
NA	NA	No	No	0	0				
Thu 2/04/26	Thu 2/04/26	Yes	Yes	123	123	Identify Goals and Objectives	Identify Goals and Objectives		Yes
Fri 3/04/26	Fri 3/04/26	No	No	456	456	Develop Strategies and Plans	Develop Strategies and Plans		No
Sat 4/04/26	Sat 4/04/26	Yes	Yes	789	789	Research Previous Experience	Research Previous Experience		Yes
Thu 2/04/26	Thu 2/04/26	Yes	Yes	123	123	Identify Goals and Objectives	Identify Goals and Objectives		Yes
Fri 3/04/26	Fri 3/04/26	No	No	456	456	Develop Strategies and Plans	Develop Strategies and Plans		No
Sat 4/04/26	Sat 4/04/26	Yes	Yes	789	789	Research Previous Experience	Research Previous Experience		Yes
NA	NA	No	No	0	0				
NA	NA	No	No	0	0				
NA	NA	No	No	0	0				
Thu 2/04/26	Thu 2/04/26	Yes	Yes	123	123	Identify Goals and Objectives	Identify Goals and Objectives		Yes
Fri 3/04/26	Fri 3/04/26	No	No	456	456	Develop Strategies and Plans	Develop Strategies and Plans		No
Sat 4/04/26	Sat 4/04/26	Yes	Yes	789	789	Research Previous Experience	Research Previous Experience		Yes
Thu 2/04/26	Thu 2/04/26	Yes	Yes	123	123	Identify Goals and Objectives	Identify Goals and Objectives		Yes
Fri 3/04/26	Fri 3/04/26	No	No	456	456	Develop Strategies and Plans	Develop Strategies and Plans		No

8. Run Schedule Migration

There are two parts to the schedule migration process:

1. Connect to Altus via Microsoft Project.
2. Run the Schedule Publish Script.

Resource Rates

This script also migrates Resource Rates to Altus bookable resources.

Note: There is no action here for the user running the scripts; the resource rates will be migrated for matching resources.

Data Mapping

The table below shows how the OOTB fields from the .MPP file are mapped to Altus.

.MPP Table	Altus Table
Task Entry	sensei_task

The fields are mapped as follows:

.MPP Field	Altus Field	Example Data
Task Name	Name	Identify Goals and Objectives
Resource Names	Assigned To	Project Manager
Start	Start	2-12-19
Finish	Finish	6-12-19
% Complete	Percent Complete	25%
Work	Effort (Hours)	40
Remaining Work	Effort Remaining (Hours)	30
Actual Work	Effort Completed (Hours)	10
Duration	Duration (Days)	5 days
Type	Task Type	Fixed Units
Task Mode	Manually Schedule	Yes
Notes	Notes	Goals were discussed
Deadline	Due Date	6-12-19
Summary	Summary	No
Milestone	Milestone	No
Locked	Locked	No

1. Connect to Altus via Microsoft Project

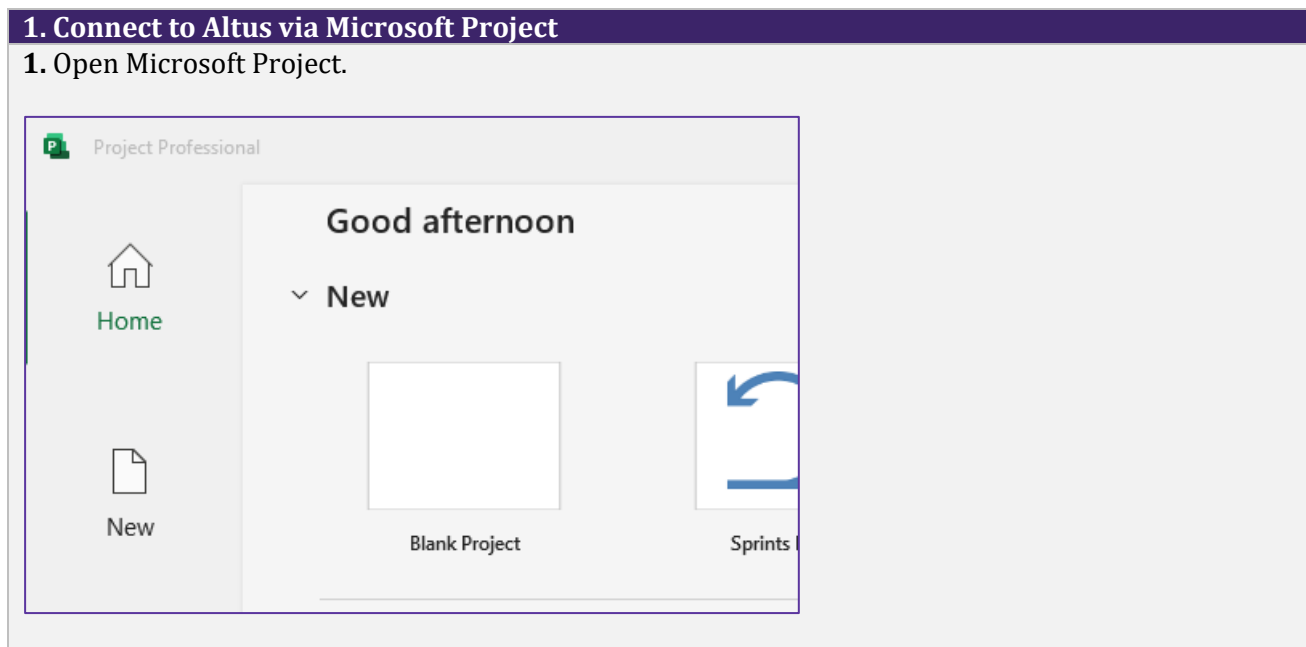
Quick Summary:

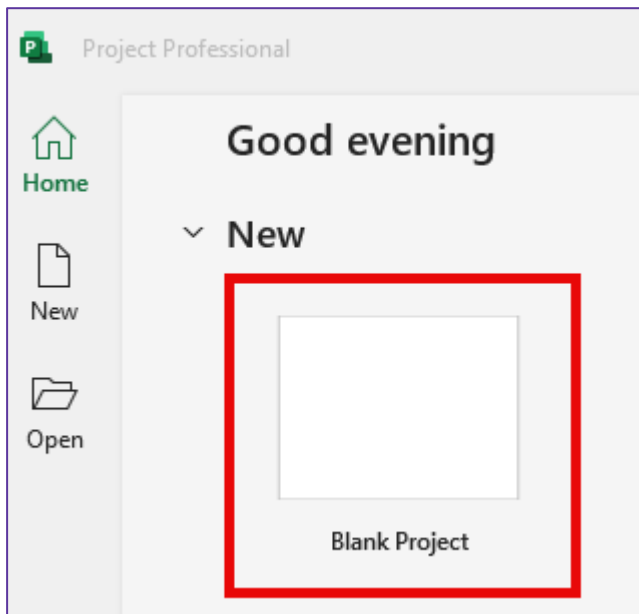
1. Open Microsoft Project.
2. Select Blank Project.
3. Save the Project.
4. Select the Altus Tab > Connect to Altus.
5. Sign in using your Microsoft Account.
6. Select the **Altus environment to connect to** > Next.
7. Once the list of projects loads you can click on Cancel.

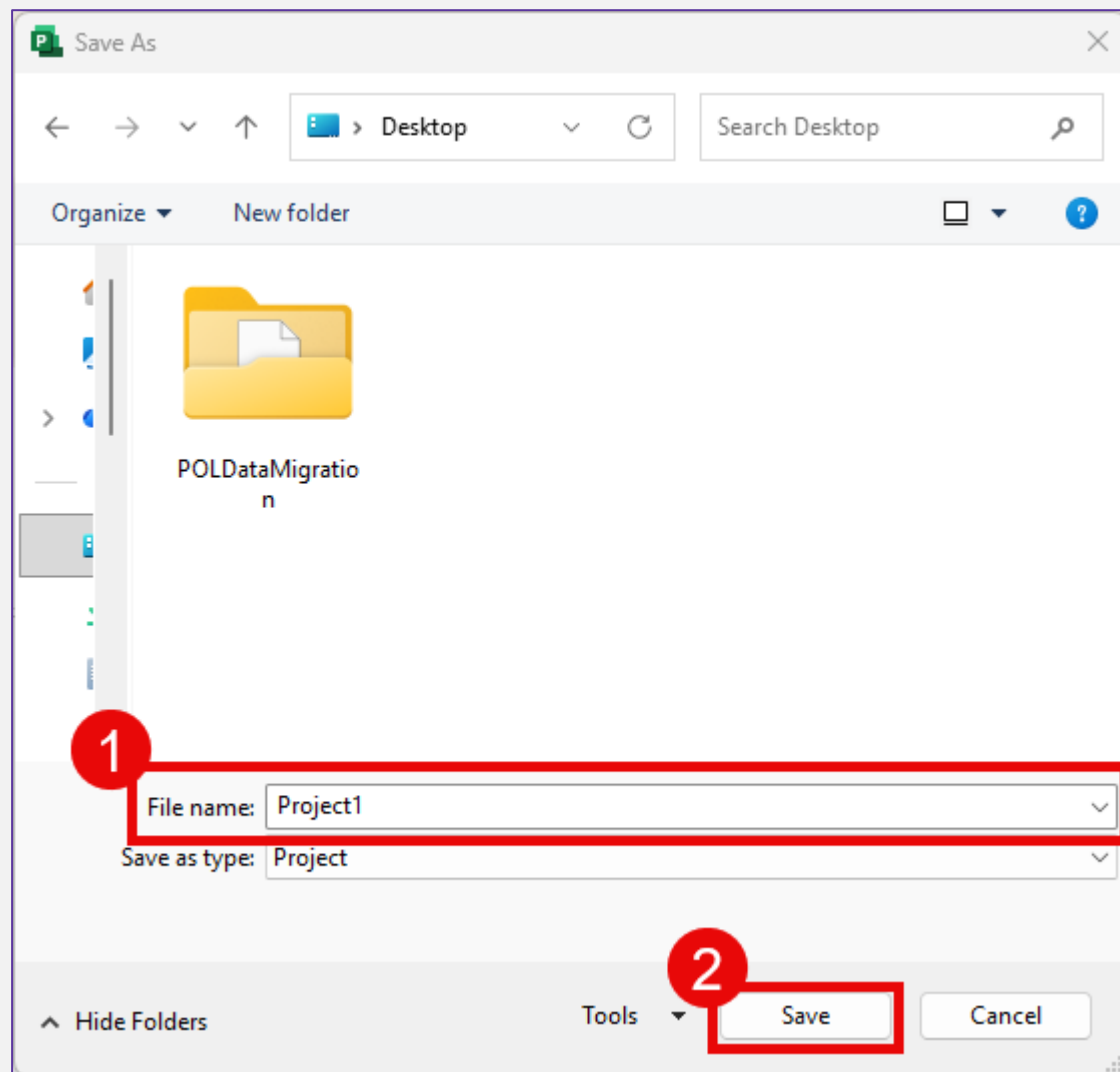
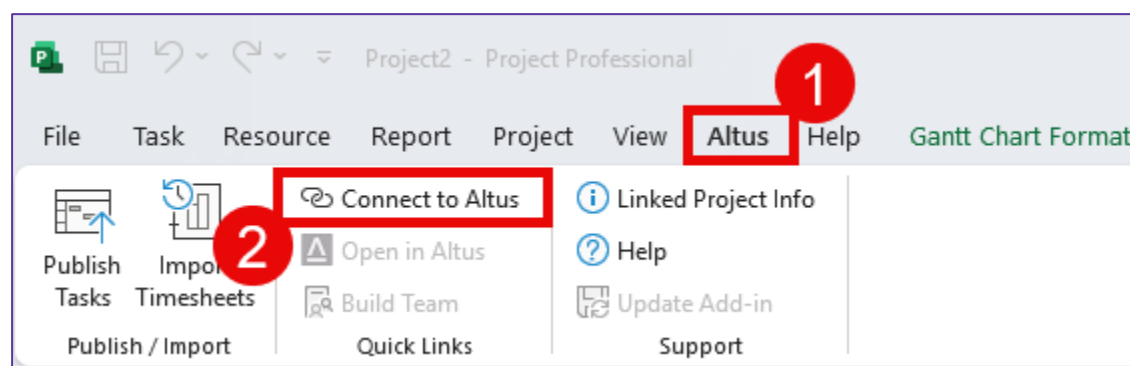
Video:

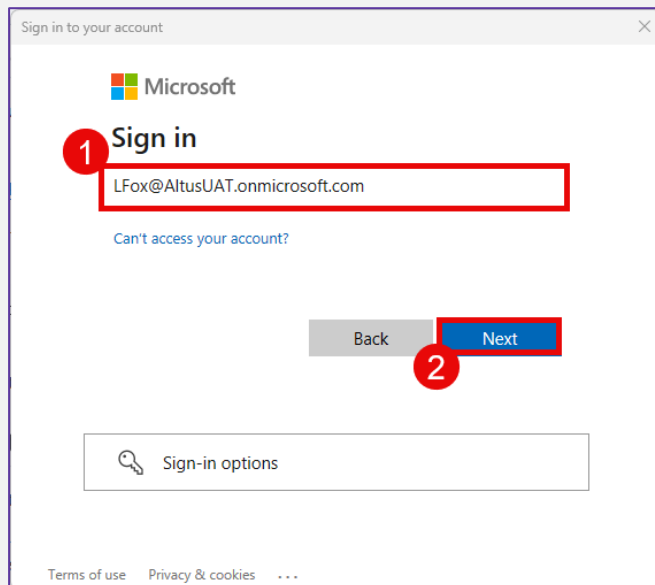
See link below to a video which covers the above steps with an example migration:

 [Project Online Data Migration - 8.1 Connect to MS Project to Altus.mp4](#)



1. Connect to Altus via Microsoft Project**2. Select Blank Project.**

1. Connect to Altus via Microsoft Project**3. Save the blank project.****4. Select the Altus Tab > Connect to Altus.**

1. Connect to Altus via Microsoft Project**5. Sign in using your Microsoft Account.**

Sign in to your account

Microsoft

1 Sign in

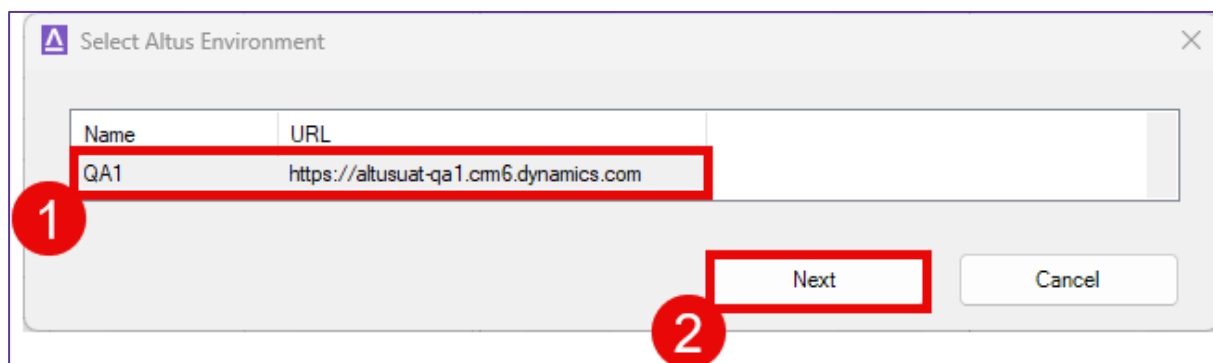
LFox@AltusUAT.onmicrosoft.com

Can't access your account?

Back **2** Next

Sign-in options

Terms of use Privacy & cookies ...

6. Select the Altus environment to connect to:

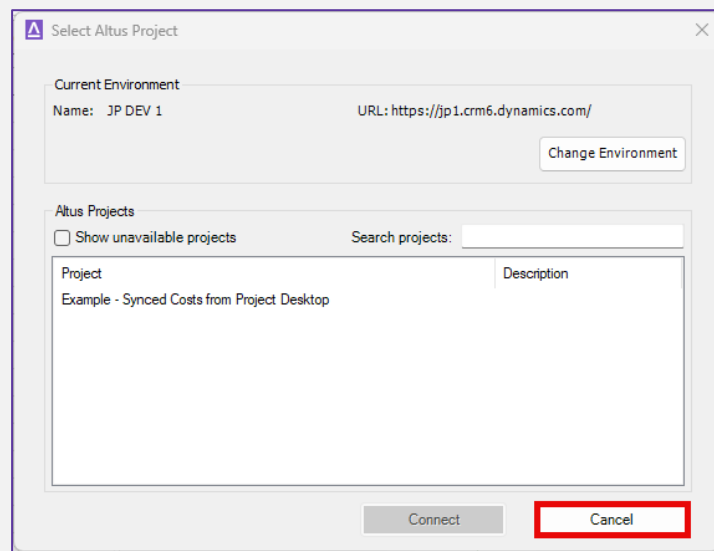
Select Altus Environment

Name	URL
1 QA1	https://altusuat-qa1.crm6.dynamics.com

2 Next Cancel

1. Connect to Altus via Microsoft Project

7. Once the list of projects load you can click **Cancel**.



2. Run the Schedule Publish script

The process below covers how to publish the schedules whilst maintaining the connection to Altus. Before running this process it is worth noting the following:

- This script will publish the .MPP files that were downloaded into this folder (in the [export project online data](#) process): ...**2.Project-and-Resource-Migration\Files**
- When files are successfully published, they will be moved to this folder: ...**2.Project-and-Resource-Migration\Files\Published**
- If you need to re-run the migration, you will need to move the .MPP files back to this location: ...**2.Project-and-Resource-Migration\Files**

Changing the Currency

The schedule migration scripts

Quick Summary:

1. Open **SharePoint Online Management Shell**.
2. Navigate to the location where this script is located: **.\Publish-MPPsToAltus.ps1**
3. Open **Microsoft Project** (If not already open).
4. Run the **script**.

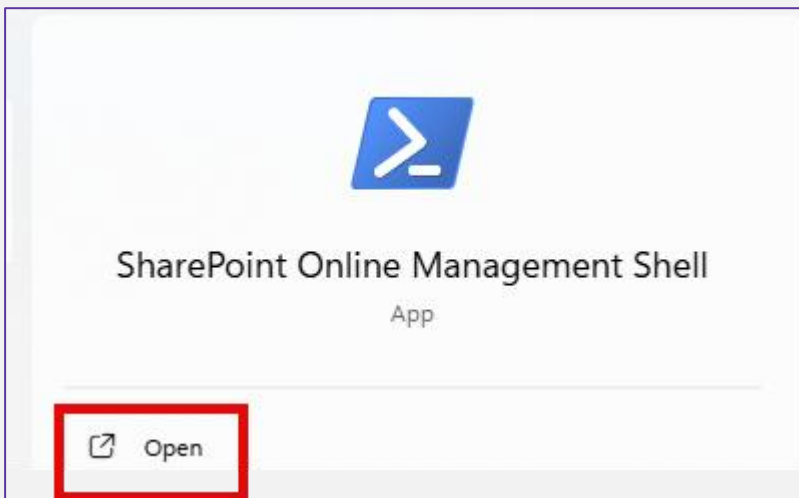
Video:

See link below to a video which covers the above steps with an example migration:

 [Project Online Data Migration - 8.2 Run Schedule Migration.mp4](#)

2. Run the Publish-MPPsToAltus.ps1 script

1. Open SharePoint Online Management Shell

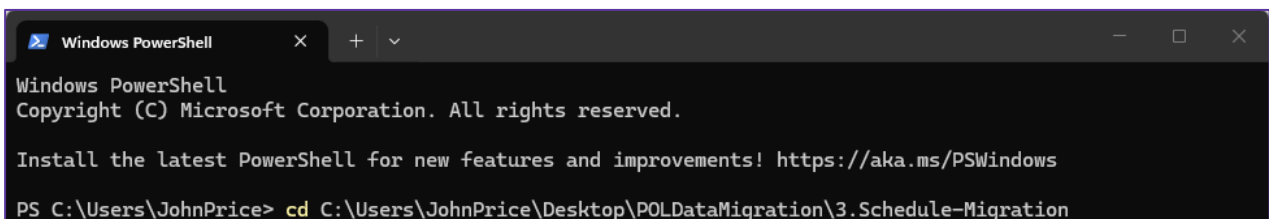


Note: Do not run as an Administrator as this will cause the script to error.

2. Navigate to the location where this script is located: `.\Publish-MPPsToAltus.ps1`

Example:

`cd C:\Users\JohnPrice\Desktop\POLDataMigration\3.Schedule-Migration`



3. Open Microsoft Project (If not already open).

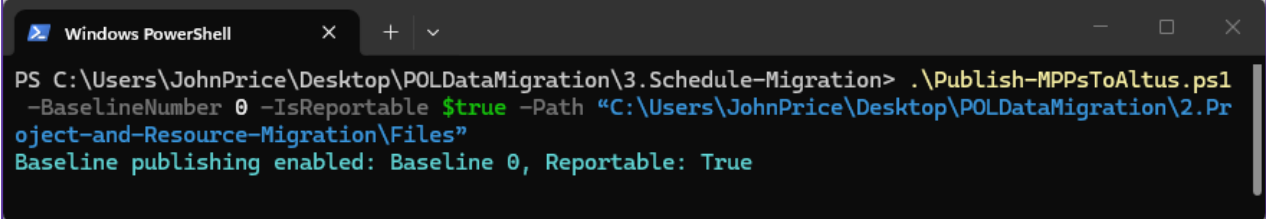
2. Run the Publish-MPPsToAltus.ps1 script

4. Run the **script**, including these extra parts:

Baseline to Migrate	BaselineNumber 0 -IsReportable \$true
Location of .MPP Files	...\2.Project-and-Resource-Migration\Files

Example:

```
.\Publish-MPPsToAltus.ps1 -BaselineNumber 0 -IsReportable $true -Path  
"C:\Users\JohnPrice\Desktop\POLDataMigration\2.Project-and-Resource-Migration\Files"
```



```
Windows PowerShell
PS C:\Users\JohnPrice\Desktop\POLDataMigration\3.Schedule-Migration> .\Publish-MPPsToAltus.ps1  
-BaselineNumber 0 -IsReportable $true -Path "C:\Users\JohnPrice\Desktop\POLDataMigration\2.Pr  
oject-and-Resource-Migration\Files"  
Baseline publishing enabled: Baseline 0, Reportable: True
```

Note: To migrate additional baselines, simply change BaselineNumber0 to the required baseline and re-run the script.

Example of a Published Schedule

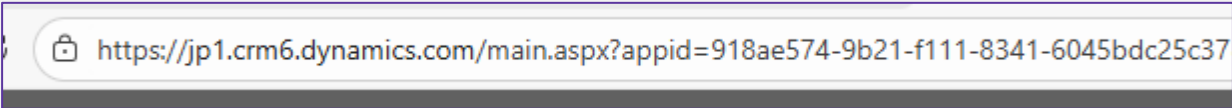
Once the schedule publish script has been completed you can continue to access your schedule via Microsoft Project. If you would like to access the data that has been migrated to your project, follow these steps:

Quick Summary:

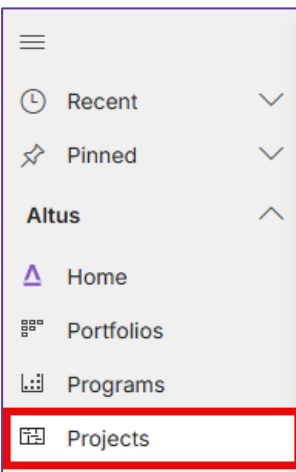
1. **Login** to Altus.
2. Open **Initiatives > Projects**.
3. Open **your project**.
4. Open the **Tasks** tab.

Access Schedule data in Altus

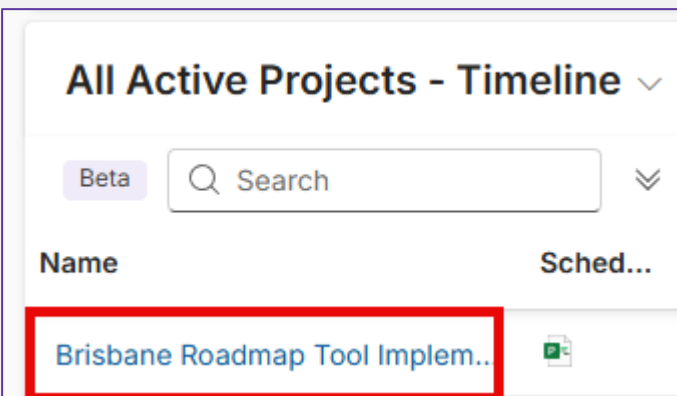
1. Login to Altus.



2. Open Initiatives > Projects.

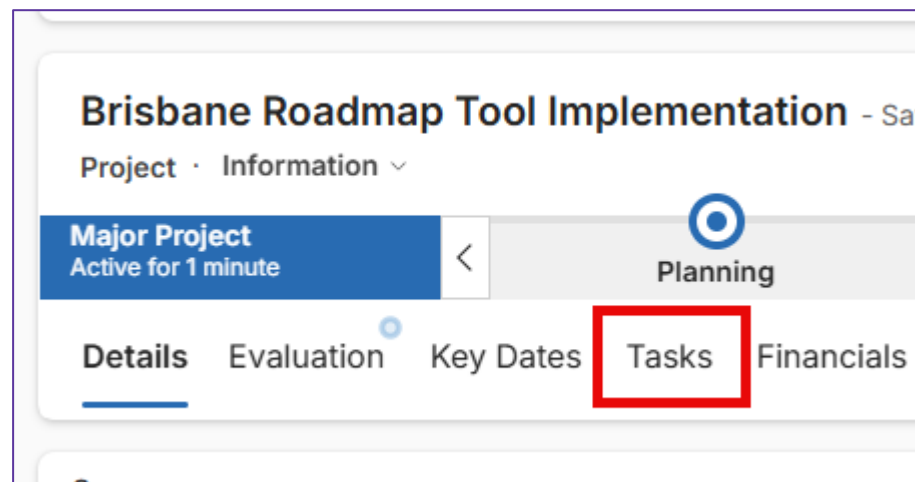


3. Open your project.

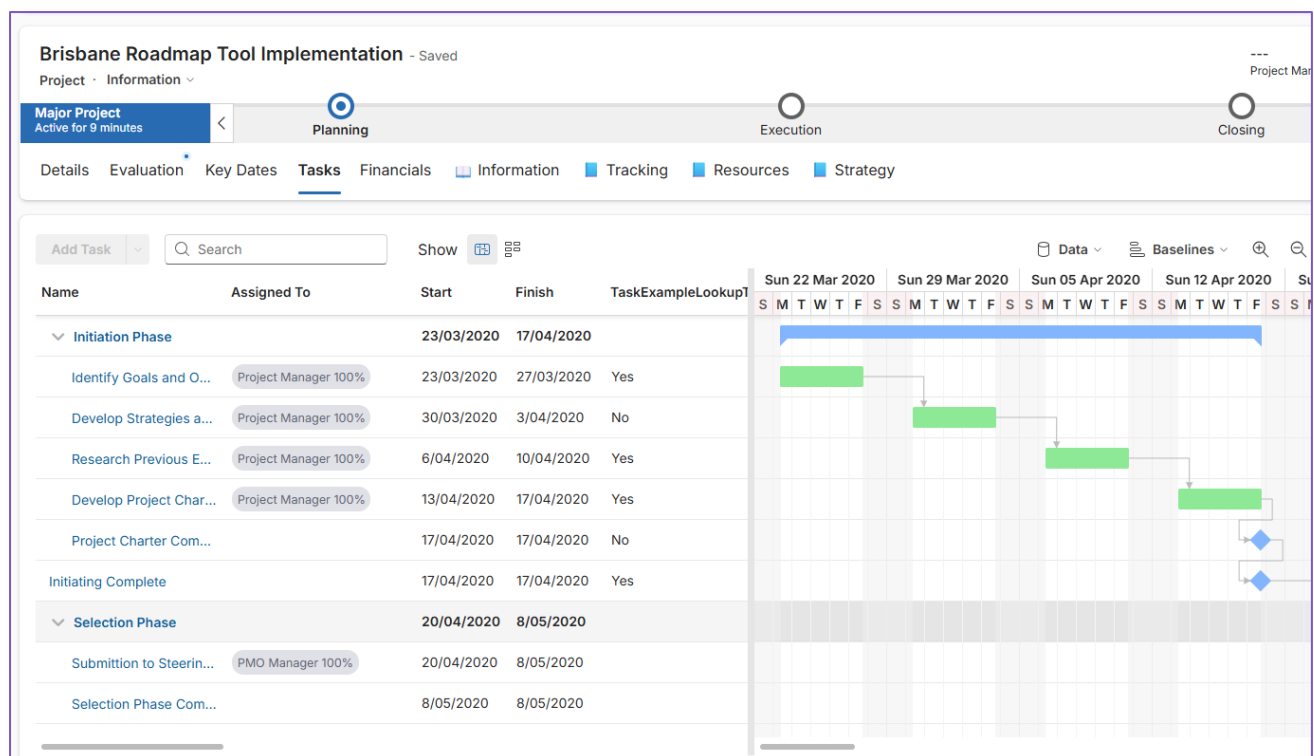


Access Schedule data in Altus

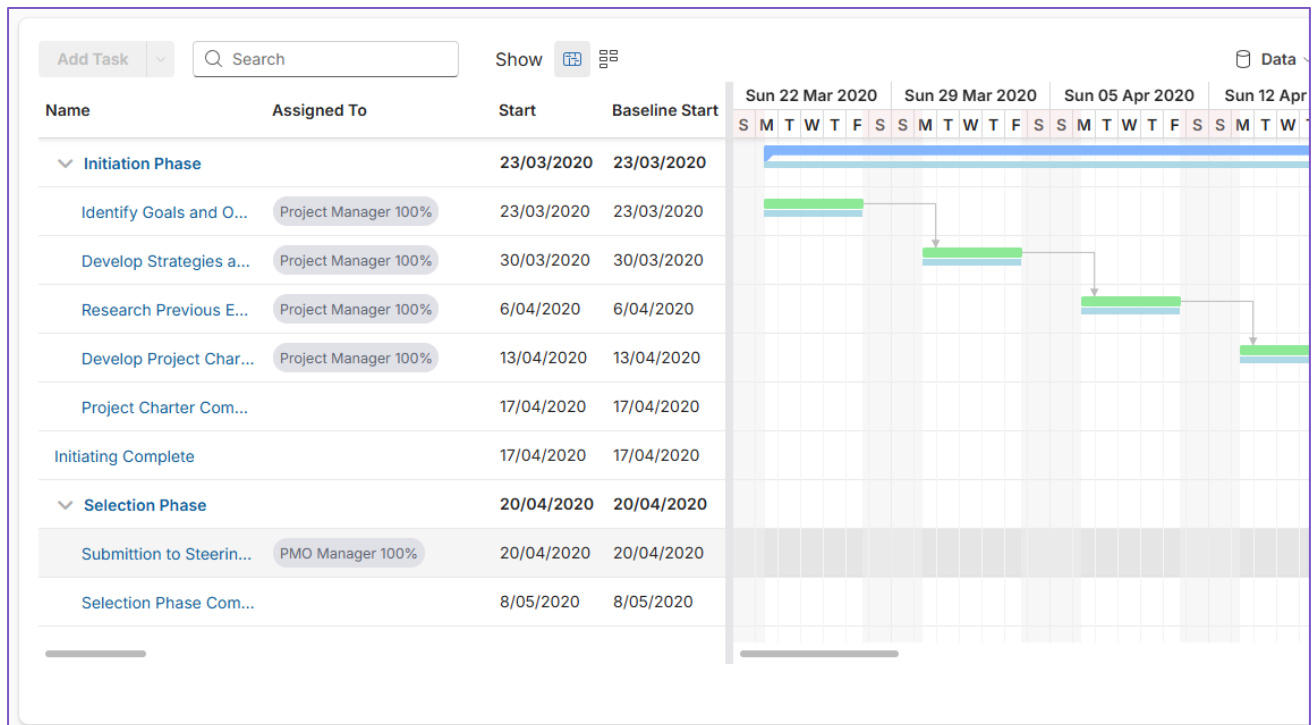
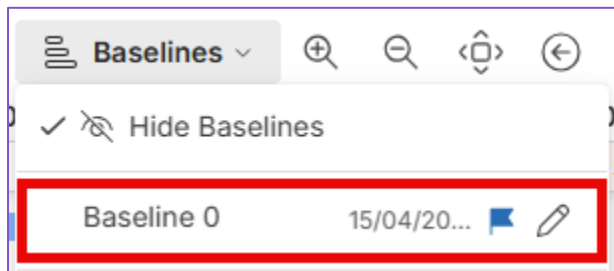
4. Open the **Tasks** tab.



This will show the schedule data that has been migrated into Altus.



You can view the baseline information by selecting this control.



Migration of Resource Rates

The schedule publish script also migrates resource rates from MS Project resources to Altus bookable resources. Details of this have been provided below.

Mapping between Systems

MS Project	Altus
Resource Sheet	Bookable Resources

Mapping of Rates

The resource rates are mapped as follows:

MS Project	Altus
Cost Rate Table A	Bookable Resource > Rates

The fields map as follows:

MS Project Field Name	Altus Field Display Name	Altus Field Internal Name
Standard Rate	Rate (Hourly)	sensei_rate
Effective Date	Effective Date	sensei_effectivedate

Time-phased mapping

Resource rates are migrated for each time period.

For example:

MS Project | Time-phased Rates

Resource Information

General Costs Notes Custom Fields

Resource Name: John Price

Cost rate tables

For rates, enter a value or a percentage increase or decrease from the previous rate. For instance, if a resource's Per Use Cost is reduced by 20%, type -20%.

A (Default)	B	C	D	E
Effective Date	Standard Rate	Overtime Rate	Per Use Cost	
--	\$100.00/h	\$0.00/h	\$0.00	
Tue 1/07/25	\$110.00/h	\$0.00/h	\$0.00	
Wed 1/07/26	\$120.00/h	\$0.00/h	\$0.00	

Cost accrual: Prorated

Help Details... OK Cancel

Altus | Time-phased Rates

John Price - Saved

Bookable Resource

General

Calendar Exceptions

Rates

Allocation

Cost Rates

+

New Cost Rate

↺

Refresh

📉

Flow

📊

Run Report

⋮

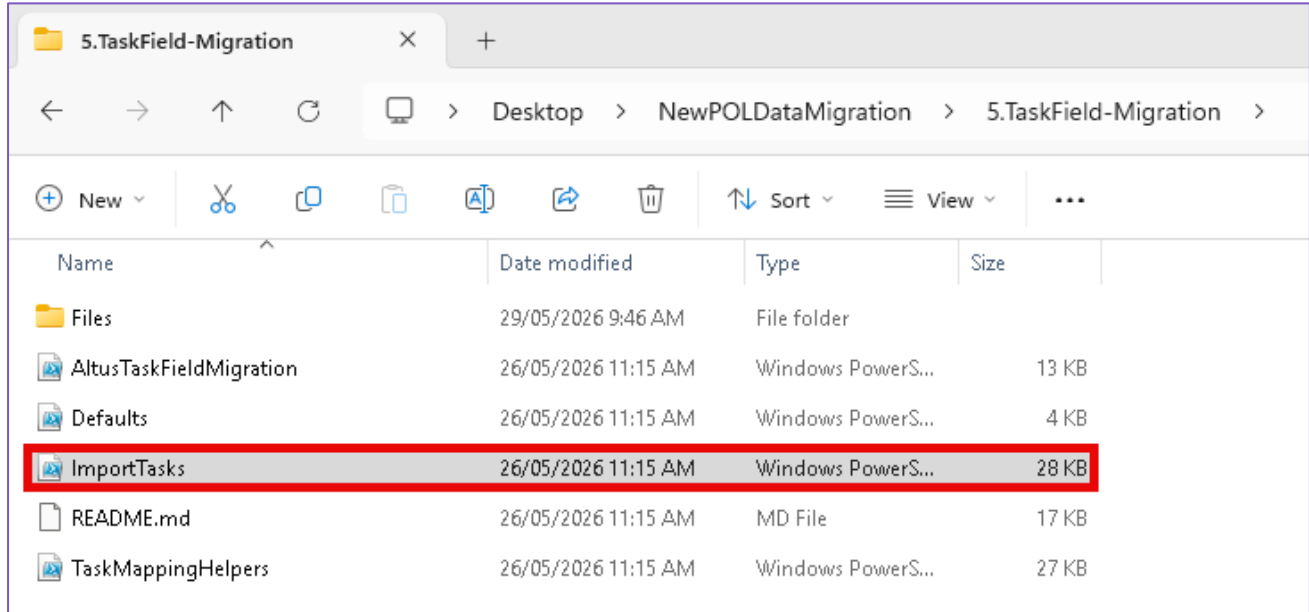
☐ Effective Date ↑	Rate (Hourly) ↓	☐
☐ 1/06/2025	\$110.00	☐
☐ 1/07/2026	\$120.00	☐

Rows: 2

Overview

This script can be extended to migrate custom fields:

ImportTasks.ps1



The script can be extended to migrate the following field types from Project Online to Altus:

Project Online Field Type	Dataverse Field Type	Dataverse Field Sub-type
Date (Date only)	Date and time	Date only
Duration	Whole Number	Duration
Flag	Choice	Yes/no
Number (Whole)	Whole Number	None
Number (Decimal)	Decimal	None
Text (Single line of Text)	Single line of text	Text
Lookup (Single Select)	Lookup	Text
Lookup (Multi Select)	Lookup	Text

1. Map Task Custom fields

The ImportTasks.ps1 script needs to be updated to map custom fields between systems.

The parts of the script that are highlighted below are as follows. Only these parts of the script block need to be updated.

Attribute	Details
Project Online Field	Display name for the field as it appears in Project Online.
Altus Field	This is the Logical Name for the Altus field.
Default.ps1 script reference	Reference to another Altus script, specifically the alias.
Additional Altus variables	Altus references to related tables.

Quick Summary:

1. Open this folder, then edit this script: **ImportTasks.ps1**.
2. Map all **Text** fields.
3. Map all **Date** fields.
4. Map all **Number** fields.
5. Map all **Duration** fields.
6. Map all **Flag** fields (Flag to Flag).
7. Map all **Flag** fields (Choice to Flag).
8. Map all **Lookup** fields (Lookup to Choice).
9. Map all **Lookup** fields (Single-select).

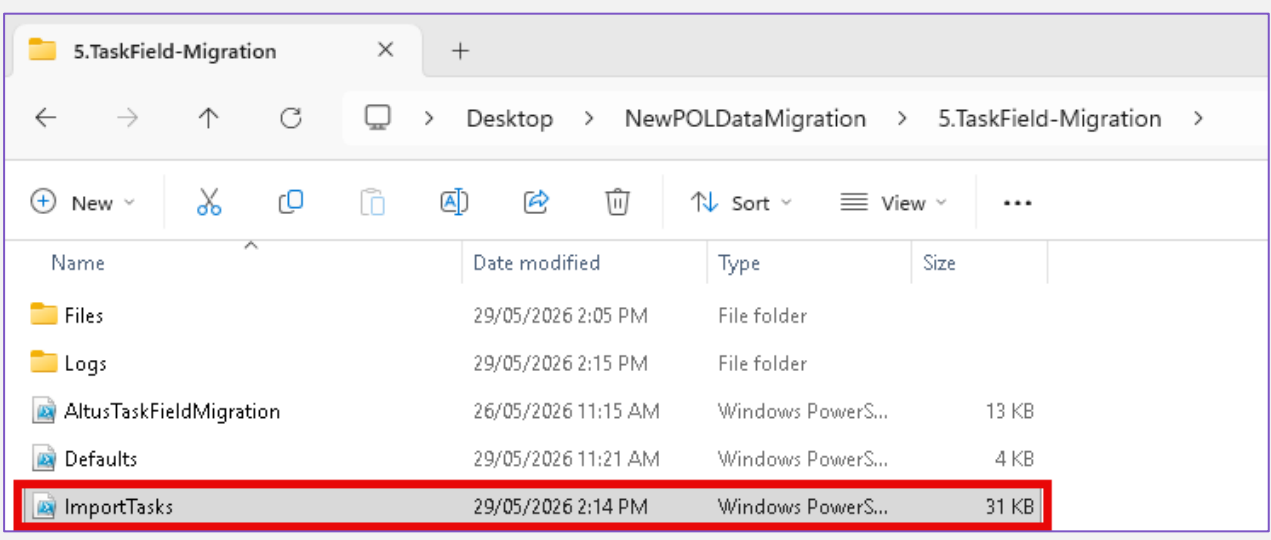
Video:

See link below to a video which covers the above steps with an example migration:

 [Data Migration - 9.1 Map Task Custom Fields.mp4](#)

1. Map Tasks Custom Fields

1. Open this folder, then edit this script: **ImportTasks.ps1**



1. Map Tasks Custom Fields

2. Map all **Text** fields.

```
$textValue = Get-ReportingTaskCustomFieldTextValue -ReportingTask $ReportingTask -
CustomFieldName 'Task Example Text Field'
if ($textValue) { $Task['ce_taskexampletextfield'] = [string]$textValue }
```

3. Map all **Date** fields.

```
$dateValue = Get-ReportingTaskCustomFieldTextValue -ReportingTask $ReportingTask -
CustomFieldName 'Task Example Date Field'
if ($dateValue) { $Task['ce_taskexampledatefield'] = ($dateValue -as [datetime]).ToString("yyyy-
MM-dd") }
```

4. Map all **Number** fields.

```
$numberValue = Get-ReportingTaskCustomFieldTextValue -ReportingTask $ReportingTask -
CustomFieldName 'Task Example Number Field'
if ($numberValue) { $Task['ce_taskexamplenumberfield'] = [int][double]$numberValue }
```

5. Map all **Duration** fields.

```
$durationValue = Get-ReportingTaskCustomFieldTextValue -ReportingTask $ReportingTask -
CustomFieldName 'Task Example Duration Field'
if ($durationValue) { $Task['ce_taskexampledurationfield'] = [int][double]$durationValue }
```

6. Map all **Flag** fields (Flag to Flag).

```
$cfValue = Get-ReportingTaskCustomFieldTextValue -ReportingTask $ReportingTask -
CustomFieldName 'Task Example Flag Field'
if ($cfValue) {
    if ($cfValue -eq 'true') { $ynValue = $true } else { $ynValue = $false } # must be set as True and
    False not 1 and 0.
    $Task['ce_taskexampleflagfield'] = $ynValue }
```

7. Map all **Flag** fields (Lookup to Flag).

```
$cfValue = Get-ReportingTaskCustomFieldTextValue -ReportingTask $ReportingTask -
CustomFieldName 'Task Example LookupToFlag Field'
if ($cfValue) {
    if ($cfValue -eq 'Yes') { $ynValue = $true } else { $ynValue = $true }
    $Task['ce_taskexamplelookuptoflagfield'] = $ynValue }
```

8. Map all **Lookup** fields (Lookup to Choice).

```
Set-TaskChoiceField -TaskBody $Task -ReportingTask $ReportingTask -POLCustomFieldName 'Task
Example LookupToChoice Field' -ChoiceName 'choice_YesNo' -DataverseFieldName
'ce_taskexamplelookuptochoicefield'
```

1. Map Tasks Custom Fields

9. Map all **Lookup** fields (Single-select).

```
Set-TaskLookupField -TaskBody $Task -ReportingTask $ReportingTask -POLCustomFieldName 'Task  
Example Lookup Field' -LookupName 'lookup_TeamName' -DataverseFieldName  
'ce_TaskExampleLookupField'
```

Note: The Dataverse field name in this example is case sensitive.

2. Migrate task custom field data

Once you have created the custom fields and mapped them in the script.

Quick Summary:

1. Open SharePoint Online Management Shell.
2. Navigate to the location of this script.
3. Run the following command: `.\AltusTaskFieldMigration.ps1`
4. Enter the Target Altus environment URL.
5. Select the Microsoft account to login with.
6. Select mode of operation (Execute or What-If).

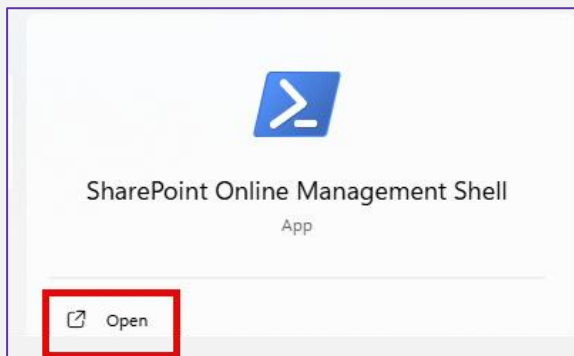
Video:

See link below to a video which covers the above steps with an example migration:

 [Data Migration - 9.2 Migrate task custom field data.mp4](#)

2. Migrate task custom field data

1. Open SharePoint Online Management Shell.

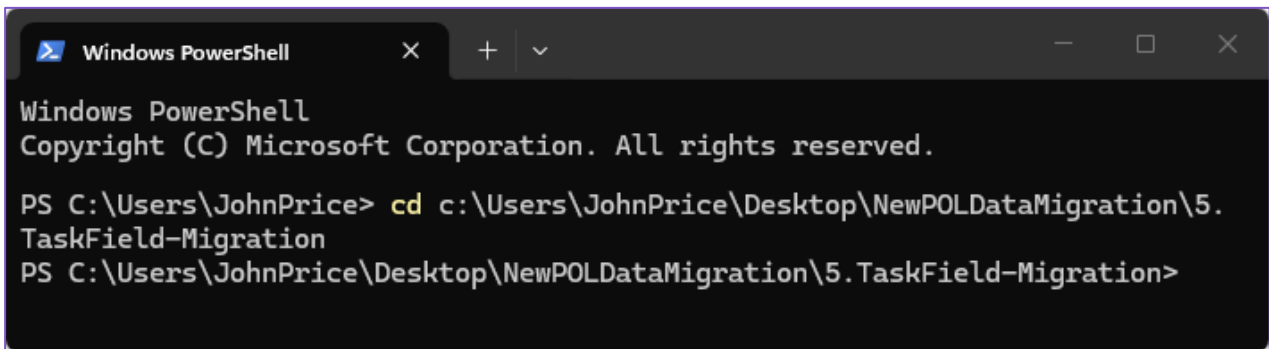
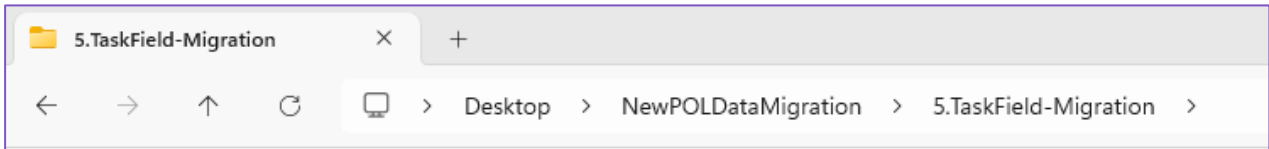


2. Migrate task custom field data

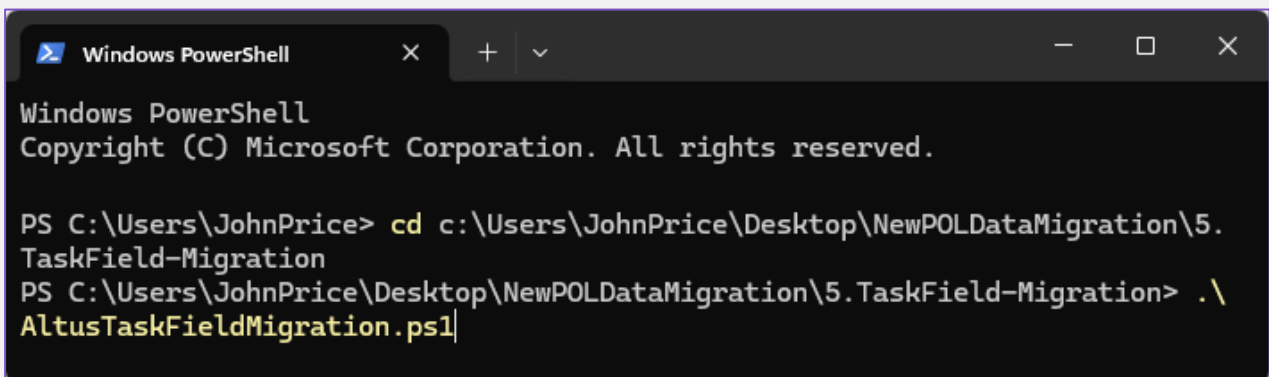
2. Navigate to the location of this script: `.\AltusTaskFieldMigration.ps1`

Example:

`cd c:\Users\JohnPrice\Desktop\NewPOLDataMigration\5.TaskField-Migration`



3. Run the following command: `.\AltusTaskFieldMigration.ps1`

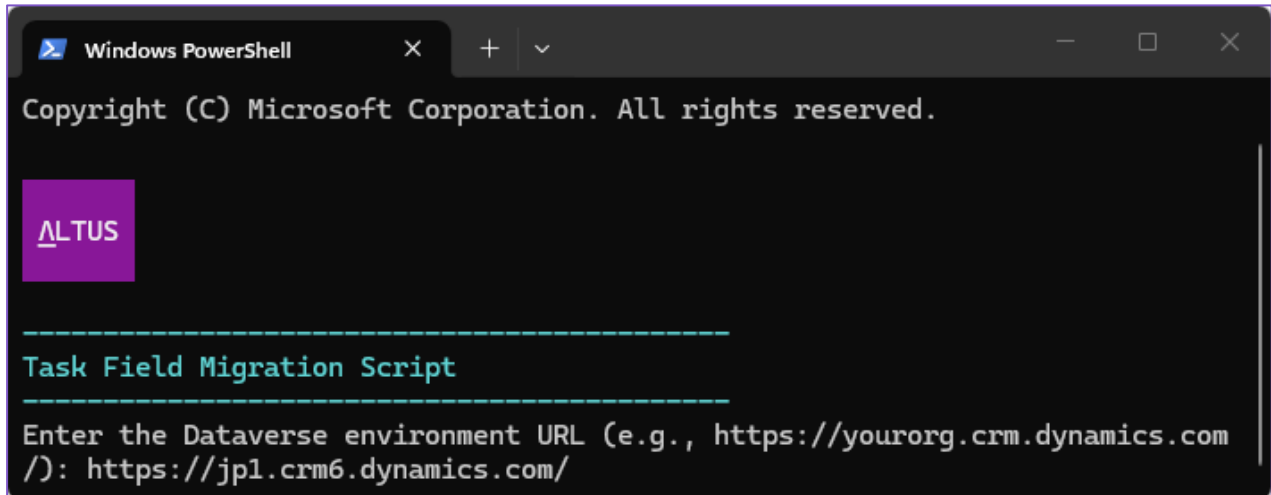


2. Migrate task custom field data

4. Enter the Target Altus environment URL.

Example:

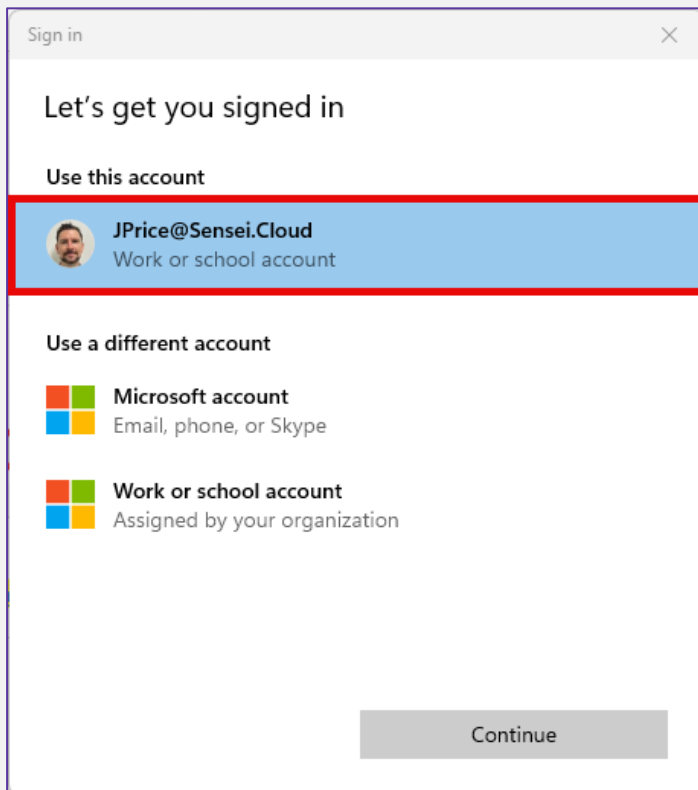
<https://jp1.crm6.dynamics.com/>



5. Select the Microsoft account to login with.

Example:

jprice@sensei.cloud

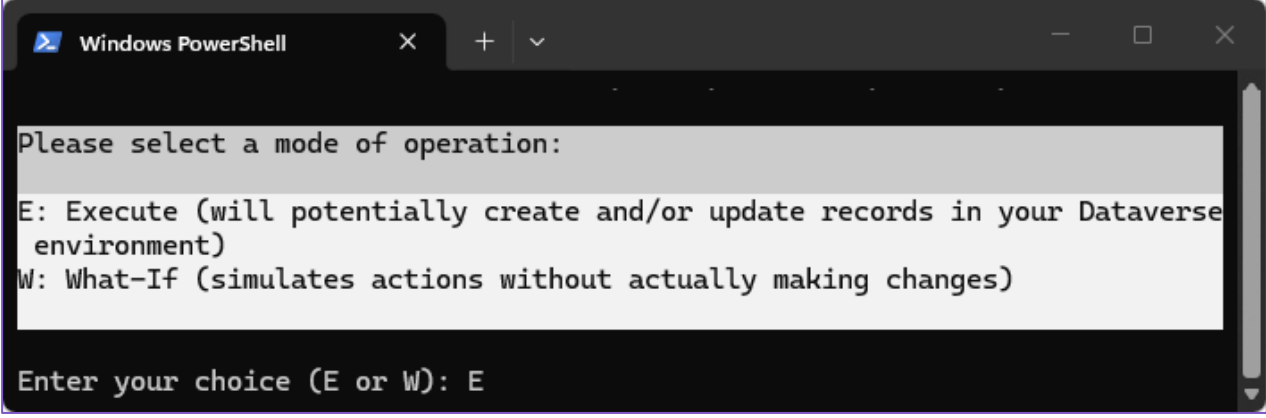


2. Migrate task custom field data

6. Select mode of operation:

E: Execute = Commit records to Dataverse

W: What-If = Simulate actions with creating records in Dataverse



```
Windows PowerShell
Please select a mode of operation:
E: Execute (will potentially create and/or update records in your Dataverse environment)
W: What-If (simulates actions without actually making changes)
Enter your choice (E or W): E
```

OOTB Mapping

The table below shows how the fields from the .MPP file are mapped to Altus.

.MPP Table	Altus Table
Task Entry (<i>Task Level 0</i>)	sensei_project
Resource Entry	sensei_bookableresource
Task Entry	sensei_task

The following OOTB fields are mapped as follows:

.MPP Field	Altus Field	Example Data
Task Name	Name	Identify Goals and Objectives
Resource Names	Assigned To	Project Manager
Start	Start	2-12-19
Finish	Finish	6-12-19
% Complete	Percent Complete	25%
Work	Effort (Hours)	40
Remaining Work	Effort Remaining (Hours)	30
Actual Work	Effort Completed (Hours)	10
Duration	Duration (Days)	5 days
Type	Task Type	Fixed Units
Task Mode	Manually Schedule	Yes
Notes	Notes	Goals were discussed
Deadline	Due Date	6-12-19
Summary	Summary	No
Milestone	Milestone	No
Locked	Locked	No

Example of migrated task custom field data

The screenshots below show examples of task custom field data that has been migrated to Altus.

Altus Tasks View

Name	TaskExampleDurationField	TaskExampleTextField	Task Example LookupToChoice ...	Task Example LookupToFlag F...
 Initiation Phase	0		No	No
 Identify Goals and O...	8	Identify Goals and Ob...	Yes	Yes
 Develop Strategies a...	16	Develop Strategies a...	No	Yes
 Research Previous Ex...	24	Research Previous Ex...	No	Yes
 Develop Project Char...	8	Identify Goals and Ob...	Yes	Yes
 Project Charter Comp...	16	Develop Strategies a...	No	No
 Initiating Complete	24	Research Previous Ex...	No	No
 Selection Phase	0		No	No
 Submission to Steerin...	0		Yes	Yes
 Selection Phase Com...	0		No	Yes

Name	TaskExampleDateField	TaskExampleFlagField	TaskExampleLookupField	TaskExampleNumberField
 Initiation Phase		No		0
 Identify Goals and O...	2/04/2026	Yes	Yes	123
 Develop Strategies a...	3/04/2026	No	No	456
 Research Previous Ex...	4/04/2026	Yes	Yes	789
 Develop Project Char...	2/04/2026	Yes	Yes	123
 Project Charter Comp...	3/04/2026	No	No	456
 Initiating Complete	4/04/2026	Yes	Yes	789
 Selection Phase		No		0
 Submission to Steerin...		No		0
 Selection Phase Com...		No		0

Altus Tasks Form

Task Details

General

Advanced

Notes

Summary

false

TaskExampleDurationF...

8

TaskExampleDateField

2/04/2026

TaskExampleTextField

Identify Goals and Objectives

TaskExampleFlagField

Yes

TaskExampleNumberFi...

123

Task Example Decimal ...

123

Task Example LookupT...

Yes

Task Example LookupT...

Yes

Bucket

Not Started

Save

Cancel

10. Migration calendars and exceptions

This is the final scripted part of the process. It enables you to do the following:

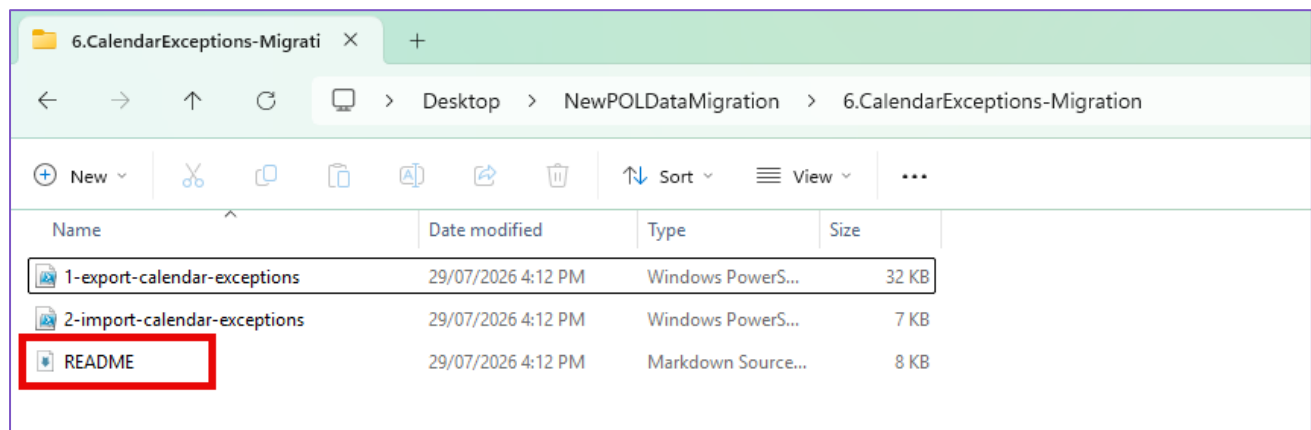
1. Migrate enterprise and project calendars from Project Online to Altus (including any exceptions within these calendars).
2. Migrate calendar exceptions for any enterprise resources matched to Altus (e.g., periods of leave or sickness).

There are two parts to this process:

Step	Details
1. Run the export script	This creates an XML file with calendars and resource exceptions.
2. Run the import script	This creates enterprise calendars in Altus and adds calendar exceptions for bookable resources.

This process reads all `\*\_published.xml` files in `InputFolder` as a single combined batch, extracts exceptions from `Calendar/Exceptions/Exception` nodes, and writes one CMT-compliant `Data.xml` to `OutputFolder`.

See the readme in the folder for further technical information where required:



1. Run the export script

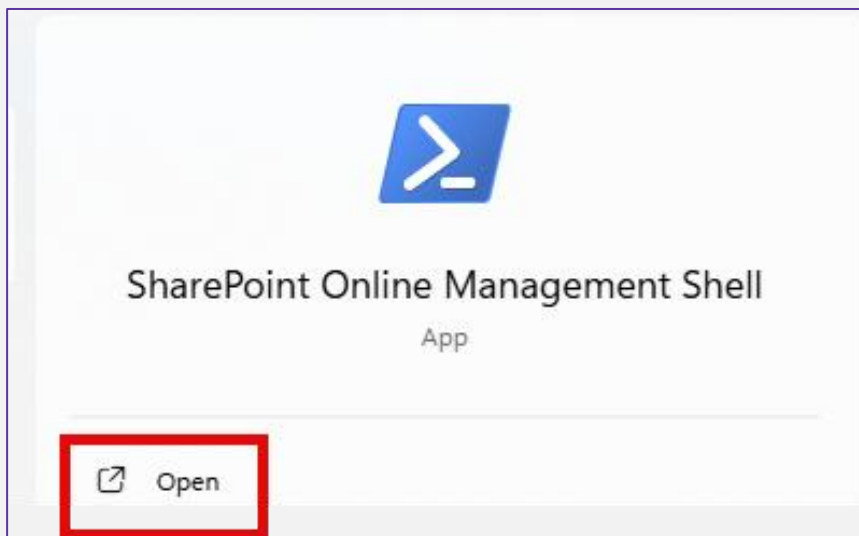
Quick Summary:

1. Open **SharePoint Online Management Shell**.
2. Connect to this folder: **\6.CalendarExceptions-Migration**
3. Run this script: **1-export-calendar-exceptions.ps1**

[10.1 Calendar and Resource Exceptions](#)

1. Run the export script

1. Open **SharePoint Online Management Shell**.



2. Connect to this folder: **\6.CalendarExceptions-Migration**

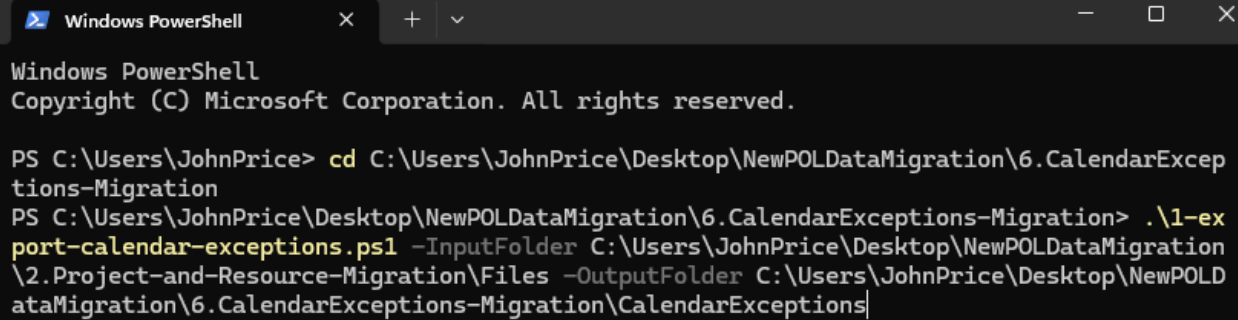


3. Run this script: **1-export-calendar-exceptions.ps1**

Parse in these parameters:

-InputFolder	Location where you exported the Project Online data to.
-OutputFolder	The location where you want the XML files to be output to – Note, you will need to create this location before running the script.

1. Run the export script

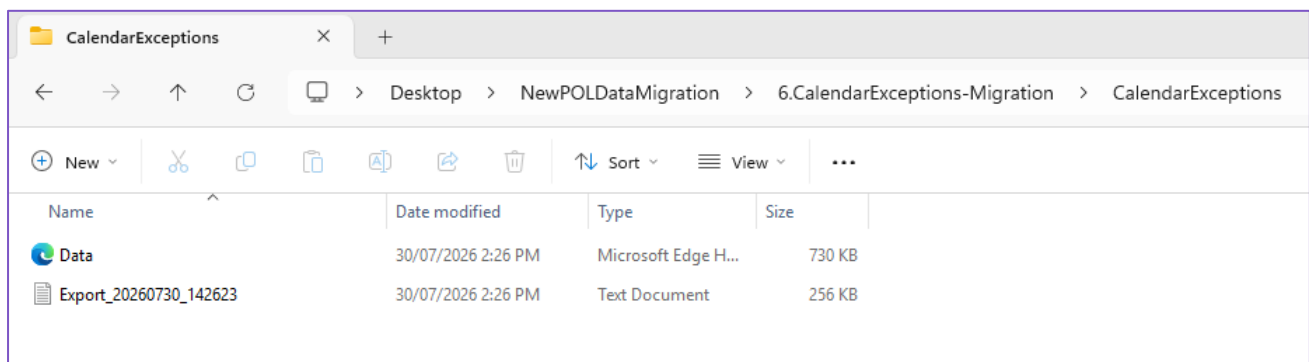


```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\JohnPrice> cd C:\Users\JohnPrice\Desktop\NewPOLDataMigration\6.CalendarExceptions-Migration
PS C:\Users\JohnPrice\Desktop\NewPOLDataMigration\6.CalendarExceptions-Migration> .\1-export-calendar-exceptions.ps1 -InputFolder C:\Users\JohnPrice\Desktop\NewPOLDataMigration\2.Project-and-Resource-Migration\Files -OutputFolder C:\Users\JohnPrice\Desktop\NewPOLDataMigration\6.CalendarExceptions-Migration\CalendarExceptions
```

Once the process has completed, the following information will be output.

Note: You don't need to do anything with the outputted content.



CalendarExceptions				
Desktop > NewPOLDataMigration > 6.CalendarExceptions-Migration > CalendarExceptions				
Name	Date modified	Type	Size	
Data	30/07/2026 2:26 PM	Microsoft Edge H...	730 KB	
Export_20260730_142623	30/07/2026 2:26 PM	Text Document	256 KB	

2. Run the import script

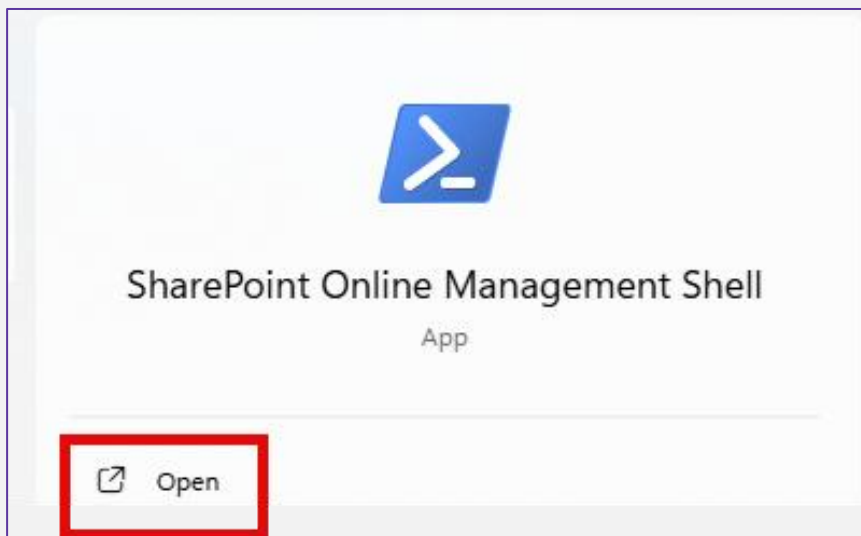
Quick Summary:

1. Open **SharePoint Online Management Shell**.
2. Connect to this folder: **\6.CalendarExceptions-Migration**
3. Run this script: **2-import-calendar-exceptions.ps1**

[10.2 Import Calendars and Resource Exceptions](#)

2. Run the import script

1. Open **SharePoint Online Management Shell**.



2. Connect to this folder: **\6.CalendarExceptions-Migration**



3. Run this script: **2-import-calendar-exceptions.ps1**

Parse in these parameters:

-D365Url	Address for the Altus environment you are connecting to.
-DataFolder	The location where the XML file was added to in the export process

2. Run the import script

```

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\JohnPrice> cd C:\Users\JohnPrice\Desktop\NewPOLDataMigration\
6.CalendarExceptions-Migration
PS C:\Users\JohnPrice\Desktop\NewPOLDataMigration\6.CalendarExceptions-Mi
gration> .\2-import-calendar-exceptions.ps1 -D365Url "https://jpl.crm6.dy
namics.com/" -DataFolder C:\Users\JohnPrice\Desktop\NewPOLDataMigration\6
.CalendarExceptions-Migration\CalendarExceptions|
  
```

Output

Project Online | Example of Resource Calendar Exceptions

Below is an example of how resource calendar exceptions are captured in Project Online:

Change Working Time

Resource calendar for 'Chee Ho Tai':

Base calendar: Standard

Legend:

- Working
- Nonworking
- 31** Edited working hours
- On this calendar:
- 31** Exception day
- 31** Nondefault work week

Click on a day to see its working times: Working times for 30 July 2026:

- 8:00 AM to 12:00 PM
- 1:00 PM to 5:00 PM

Based on: Default work week on calendar 'Standard'.

July 2026

M	T	W	Th	F	S	S
	1	2	3	4	5	
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

Exceptions

	Name	Start	Finish
1	2026 - Grand Final Eve	25/09/2026	25/09/2026
2	2026 - Melbourne Cup	3/11/2026	3/11/2026
3	2026 - Christmas Day	25/12/2026	25/12/2026
4	2026 - Boxing Day	28/12/2026	28/12/2026
5	2027 - New Year's Day	1/01/2027	1/01/2027

Details... Delete

Help OK Cancel

Altus | Example of Resource Calendar Exceptions

Below is an example of how calendar exceptions are captured in Altus:

11. Post Migration Steps

The following post-migration steps need to be completed to ensure a complete process:

1. Set Project Type
2. Set Project Ownership
3. Update Remaining Project Information
4. Set the Current Project Stage
5. Update Remaining Bookable Resource Information
6. Share the Altus Application with custom security roles
7. Remove the connection to Project Online

1. Set Project Type

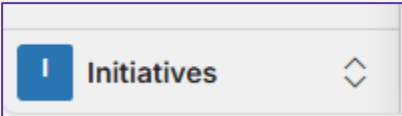
By default, all projects are set to Project Type = Major Project. There may be instances where you need to update this.

Quick Summary:

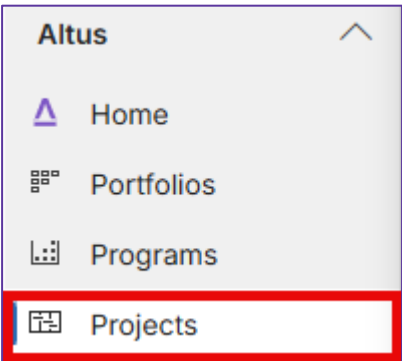
1. Select **Initiatives**.
2. Select **Projects**.
3. Select **Project** to update.
4. Remove the current project type and select a new one.
5. Select **Save & Close** to commit the change.

1. Set Project Type

1. Select **Initiatives**.



2. Select **Projects**.



3. Select **Project** to update.

1. Set Project Type

Name	Sched...
Brisbane Roadmap Tool Implem...	

4. Remove the current project type and select a new one.

Brisbane Roadmap Tool Implementation - Saved
Project · Information ▾

Major Project
Active for 5 days

<

 Planning (5 D)

Details Evaluation Key Dates Tasks Financials  Information  Trac

Investment Category

RunGrowTransform

Location

Project Type

 Major Project ×



5. Select **Save & Close** to commit the change.

<





 Save

 **Save & Close**

 Delete



Brisbane Roadmap Tool Implementation - Saved
Project · Information ▾

2. Set Project Ownership

All projects will need to be re-assigned to the project manager who is managing the project in Altus.

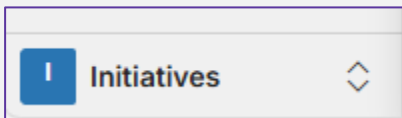
When the project is assigned to a new user, the security model will ensure that only the correct users can edit the project data.

Quick Summary:

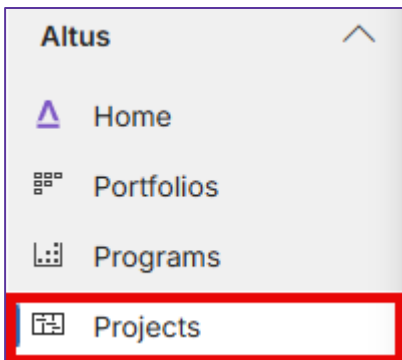
1. Select **Initiatives**.
2. Select **Projects**.
3. Select **Project** to update.
4. Select the **user** from the **Project Manager** lookup field who owns the project.
5. Select **Save & Close** to commit the change.

2. Set Project Ownership

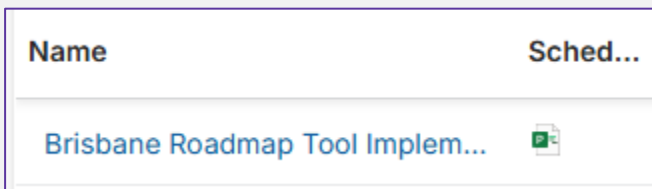
1. Select Initiatives.



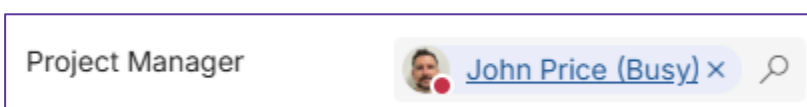
2. Select Projects.



3. Select Project to update.

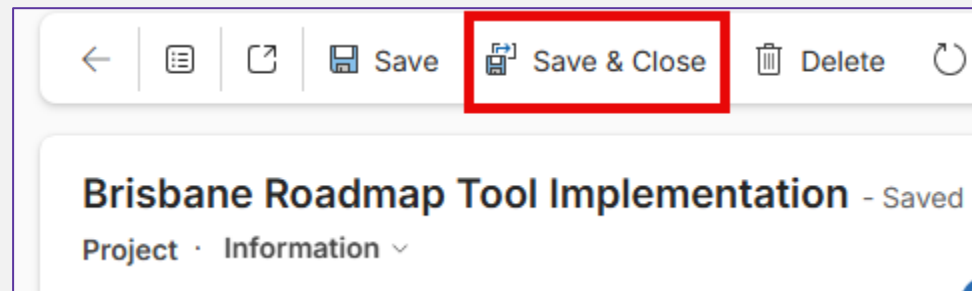


4. Select the user from the Project Manager lookup field who owns the project.



2. Set Project Ownership

5. Select **Save & Close** to commit the change.



3. Update Remaining Project information

The script only adds basic information to Projects within Altus.

Recommendation is to update the following information post migration, for example:

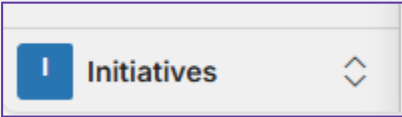
- Description
- Department
- Target Finish
- Investment Category
- Location
- Portfolio / Program
- Sponsor
- Status

Quick Summary:

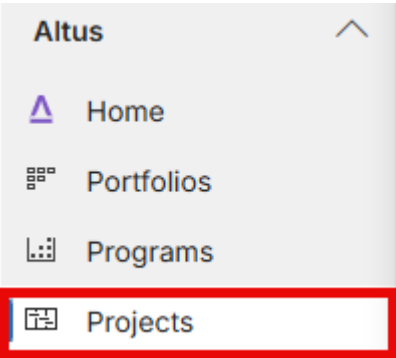
1. Select **Initiatives**.
2. Select **Projects**.
3. Select **Project** to update.
4. **Update** the **fields** as required.
5. Select **Save & Close** to commit the change.

3. Update Remaining Project information

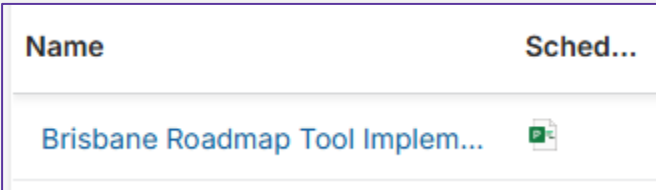
1. Select **Initiatives**.



2. Select **Projects**.



3. Select **Project** to update.



3. Update Remaining Project information**4. Update the fields as required.**

For Example:

Summary

Name * Brisbane Roadmap Tool Implementation

Description

Department

5. Select **Save & Close to commit the change.**

← | [Icon] | [Icon] | [Icon] Save | **Save & Close** | [Icon] Delete | [Icon]

Brisbane Roadmap Tool Implementation - Saved

Project · Information ▾

4. Set the Current Project Stage

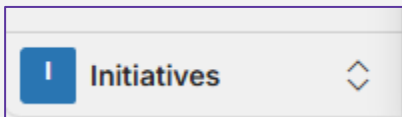
By default, each newly created project will start at the first stage in the project workflow (BPF). If the project has progressed beyond this stage, it will need to be moved to the correct stage:

Quick Summary:

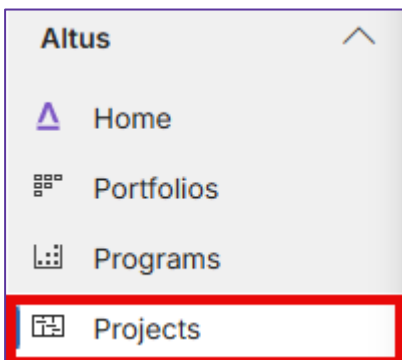
1. Select **Initiatives**.
2. Select **Projects**.
3. Select **Project** to update.
4. Select the **current stage** > **Update** the **checklist** items > **Next Stage**.
5. Select **Save & Close** to commit the change.

4. Set the Current Project Stage

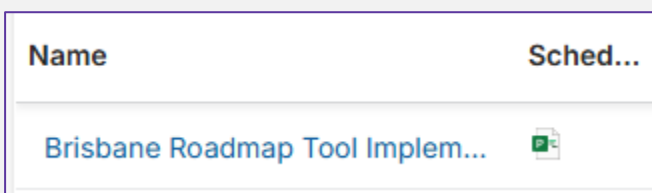
1. Select **Initiatives**.



2. Select **Projects**.



3. Select **Project** to update.



4. Set the Current Project Stage

4. Select the **current stage** > **Update** the **checklist** items > **Next Stage**.

Brisbane Roadmap Tool Implementation - Unsaved

Project · Information ▾

Major Project
Active for 5 days

Planning (5 D)

Active for 5 days

Scope and schedule * Incomplete ▾

Target Finish Date * ✓ 4/7/2026

Next Stage

Summary

Name * Brisbane Roadmap Tool Imple

Description ---

Project stage will move forward to the next stage in the lifecycle.

Brisbane Roadmap Tool Implementation - Saved

Project · Information ▾

Major Project
Active for 5 days

Planning

Execution (< 1 Min)

Details Evaluation Key Dates Tasks Financials Information Tracking Resources Strategy

5. Select **Save & Close** to commit the change.

Brisbane Roadmap Tool Implementation - Saved

Project · Information ▾

Save & Close

5. Update Remaining Bookable Resources information

The script only adds basic information to bookable resources within Altus.

Recommendation is to update the following information post migration:

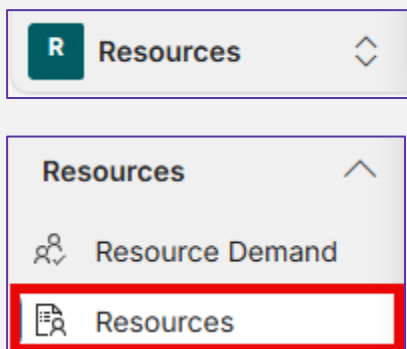
- Primary Role
- Enterprise Calendar
- Location
- Start/End Date
- Calendar Exceptions

Quick Summary:

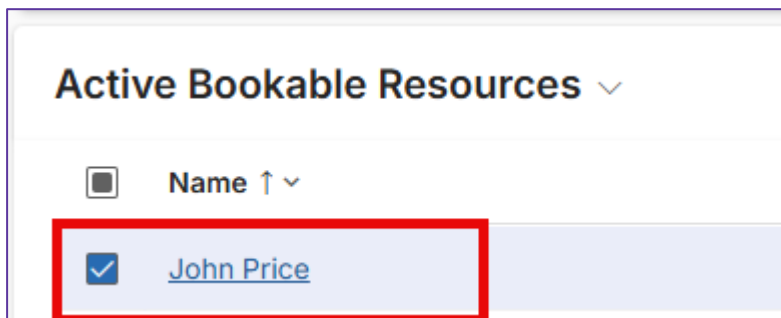
1. Select **Resources > Resources**.
2. Select the **resource** to update by clicking on the **name**.
3. Update **fields** as required.
4. Select **Save & Close** to commit the change.

5. Update Remaining Bookable Resources information

1. Select Resources > Resources.



2. Select the resource to update by clicking on the name.



5. Update Remaining Bookable Resources information**3. Update fields as required.**

John Price - Saved
Bookable Resource

General Calendar Exceptions Rates Allocation Related ▾

Resource Type *

Named ▾

User *

 John Price (Available) × 🔍

Name *

John Price

Department

--- ▾

Primary Role *

 Unclassified × 🔍

4. Select **Save & Close to commit the change.**

←

☰

↗

 Save

 **Save & Close**

John Price - Saved
Bookable Resource

6. Share the Altus application with custom security roles

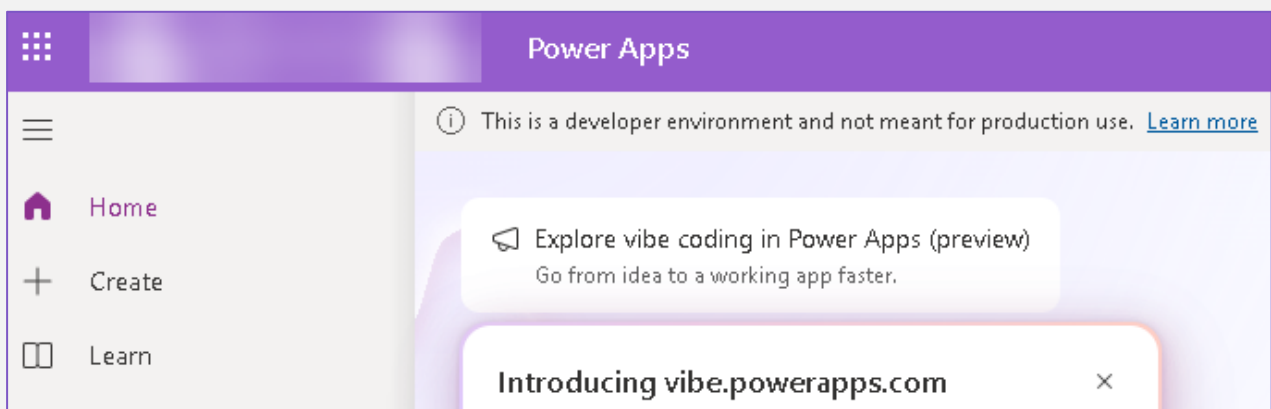
The script only adds basic information to bookable resources within Altus.

Quick Summary:

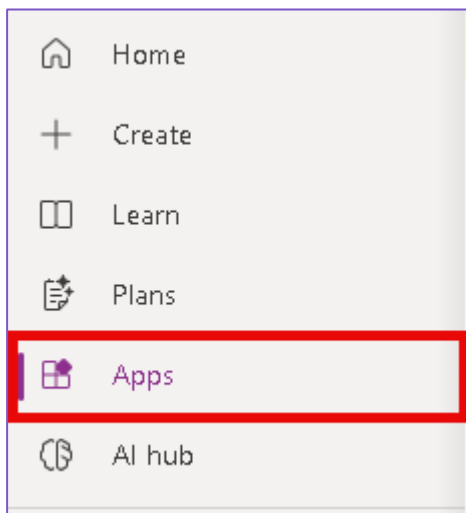
1. Open make.powerapps.com
2. Select Apps
3. Select Altus > Share
4. Select App > Altus
5. From Dataverse, select your custom security role > Share

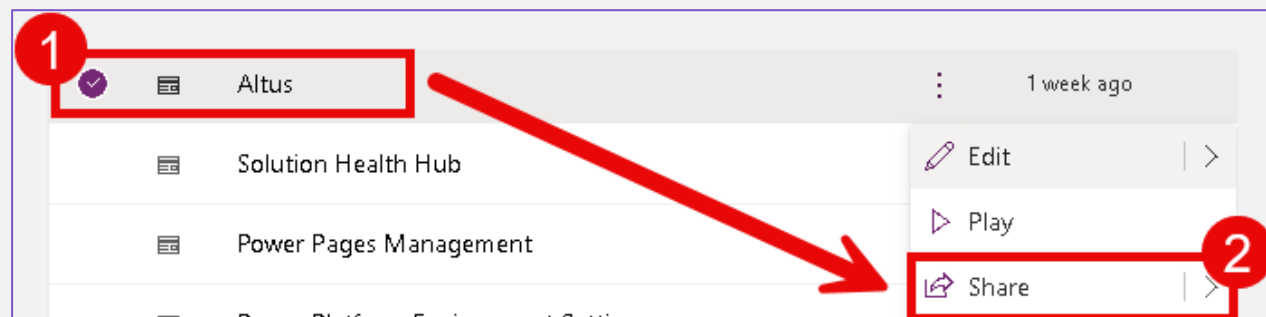
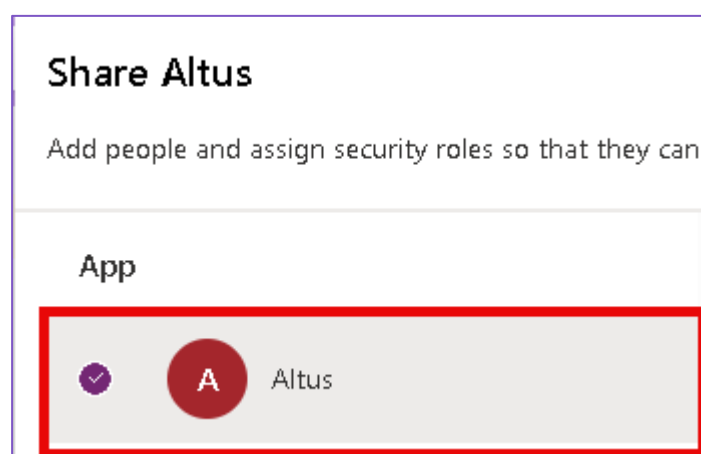
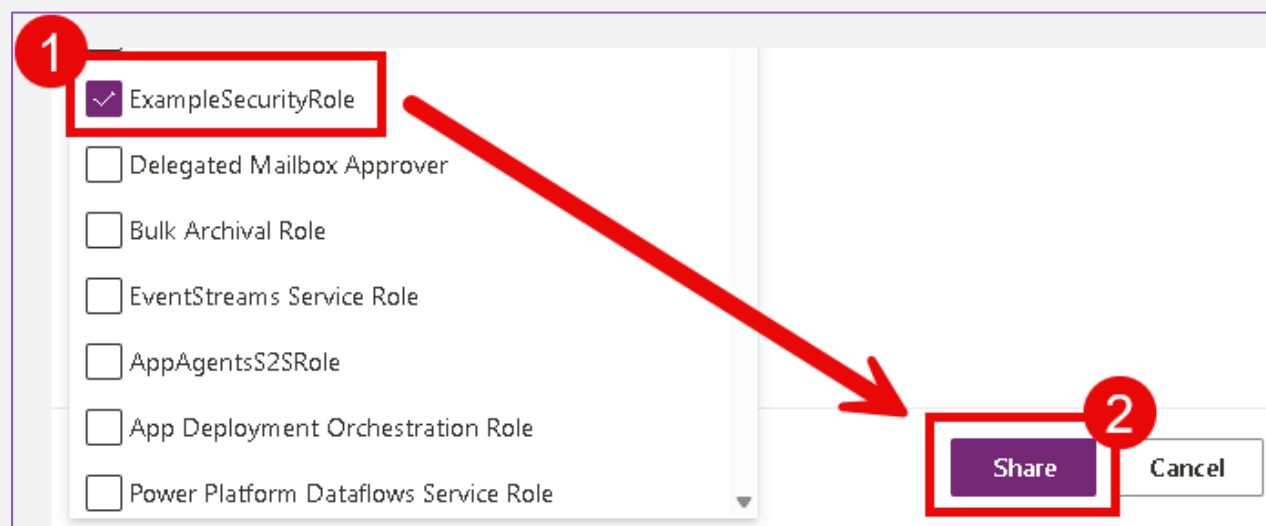
6. Share the Altus application with custom security roles

1. Open make.powerapps.com



2. Select Apps



6. Share the Altus application with custom security roles**3. Select Altus > Share****4. Select App > Altus****5. From Dataverse, select your custom security role > Share**

7. Remove the connection to Project Online

Once the migration to Altus has been completed, the connection between Microsoft Project and Project Online is no longer required and should be removed.

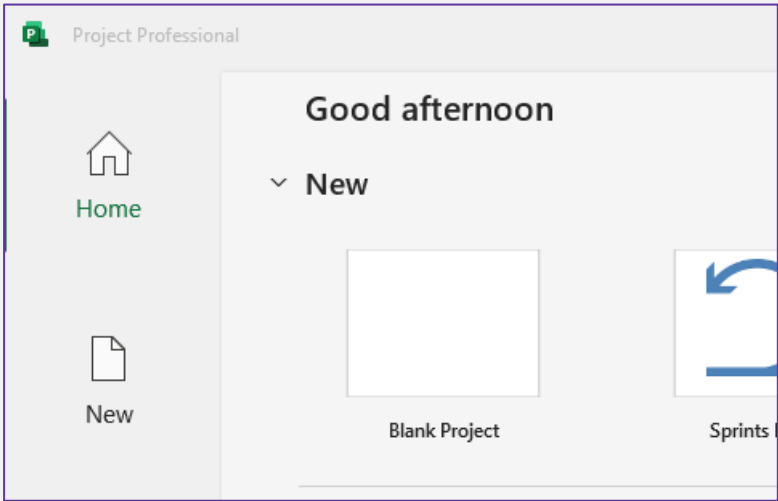
This will need to be done by each individual user – There is no global process to set and remove this.

Quick Summary:

1.

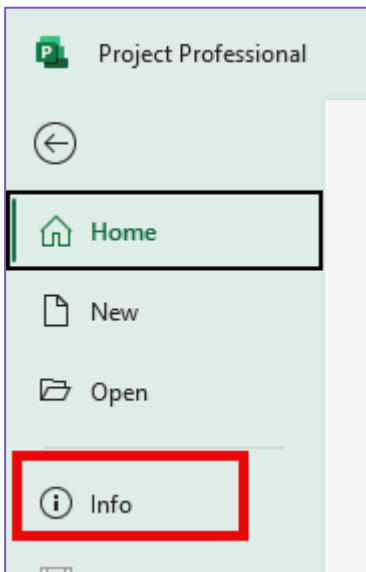
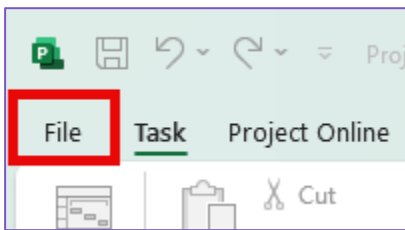
7. Remove the connection to Project Online

1. Open Microsoft Project.

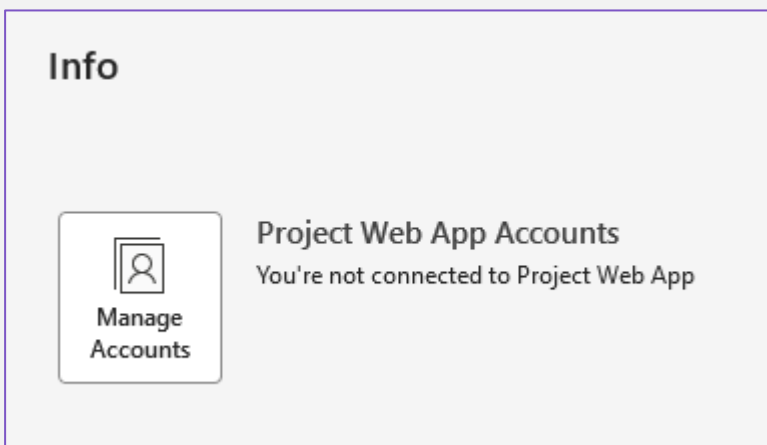


7. Remove the connection to Project Online

2. Select File > Info

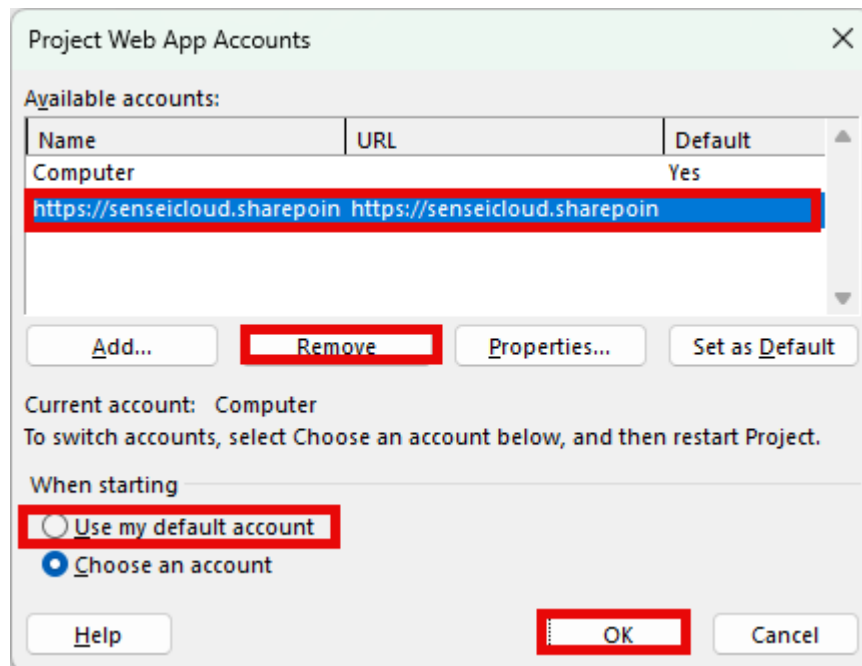


3. Select Manage Accounts



7. Remove the connection to Project Online

4. Remove all entries (Computer is default and cannot be removed) > Set When Starting = Use my default account.

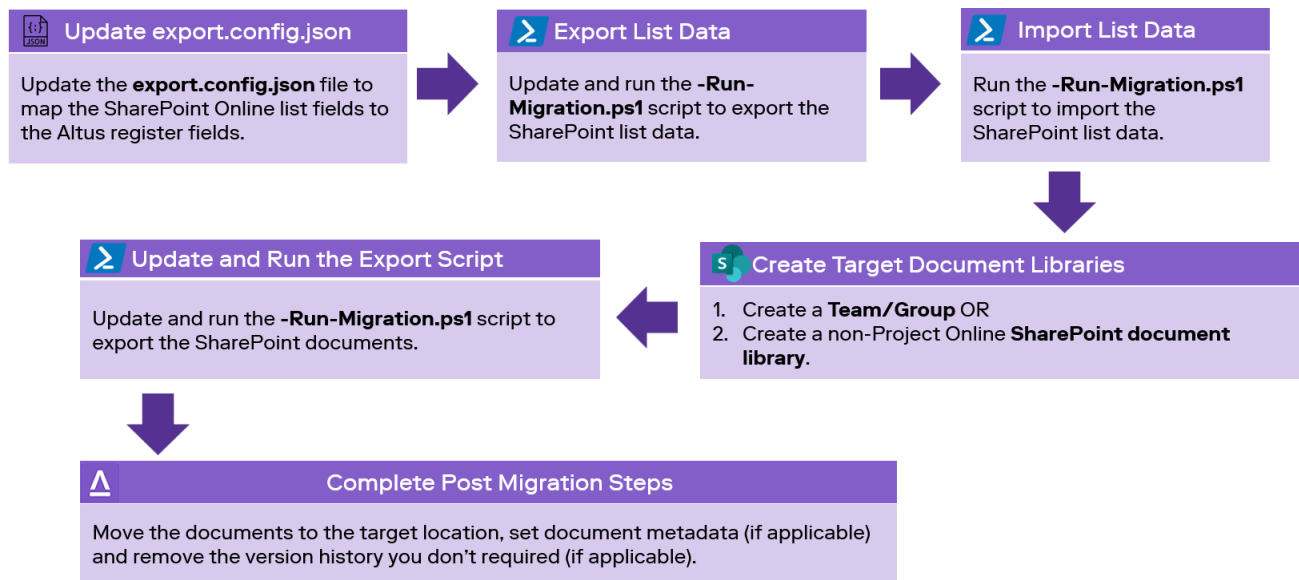


SharePoint Data Migration

Summary

The table below articulates the end-to-end process for migrating SharePoint Online data:

1. SharePoint List Data
2. SharePoint Documents



Lists

Step	Details
1. Update export.config.json	With the mapping for the SharePoint custom fields to Altus fields.
2. Run the export script	Run <code>.\1-export-lists.ps1 -SiteCollectionUrl "<pol url>" -MappingJsonPath ".\export.config.json" -OutputFolder ".\Output" -POLExportPath "path to project Files"</code>
3. Run the import script	Standalone script: Run <code>.\2-import-lists.ps1 -D365Url "altus url" -DataFolder ".\Output" -POLExportPath "path to project Files"</code>

Documents

Step	Details
1. Create/identify target library	Either: 1. Create a Teams channel for the Altus project OR 2. Use an existing non-Project Online SharePoint document library OR 1. Create a non-Project Online SharePoint document library.
2. Update and run the export script	Update the export script parameters and run for the select projects you wish to export documentation for.
3. Complete Post-Migration steps	Complete the post migration steps documented.

Before you start this process, you will need to have:

- Complete the [\[POL\] Prerequisites](#)
- Create the relevant tables and custom fields.
- Add the fields to the relevant forms.

SharePoint List Data

Before running this process, the target projects in Altus must already exist. If not, please run the Project Online process first then come back to these steps.

Step	Details
1. Update export.config.json	With the mapping for the SharePoint custom fields to Altus fields.
2. Run the export script	Run <code>.\1-export-lists.ps1 -SiteCollectionUrl "<pol url>" -MappingJsonPath ".\export.config.json" -OutputFolder ".\Output" -POLExportPath "path to project Files"</code>
3. Run the import script	Standalone script: Run <code>.\2-import-lists.ps1 -D365Url "altus url" -DataFolder ".\Output" -POLExportPath "path to project Files"</code>

The SharePoint List data migration script supports UPSERTS. This means it can be run multiple times and will write updated data into Altus.

Note: The UPSERT process doesn't support deletions. If you delete the list item from SharePoint, you will need to manually delete the corresponding item from the Altus register.

1. Update export.config.json

Default Mapping

By default, the scripts map the following lists and fields between SharePoint Online and Altus.

SharePoint List	Altus Table
Risks	sensei_risk
Issues	sensei_issue

Risks

SharePoint Field	Altus Field	Example Data
Title	sensei_name	Requirements Not Clearly Defined
Description	sensei_description	Project requirements are incomplete or ambiguous, leading to rework and delays.
DueDate	sensei_duedate	30/04/2026 0:00
Status	statuscode	(1) Active
Category	sensei_category	(2) Category2
Probability	sensei_likelihood	50%
Impact	sensei_consequence	5
Exposure	sensei_riskscore	2.5
Cost	sensei_cost	\$10,000.00
Mitigation_x0020_plan	sensei_mitigationplan	Conduct workshops with stakeholders, document requirements, and obtain formal sign-off.
Contingency_x0020_plan	sensei_contingencyplan	Rebaseline scope and timeline if major requirement changes emerge.
Trigger_x0020_Description	sensei_triggerdescription	When the due date is lapsed, a decision will be required.
Trigger	sensei_trigger	Date
Owner	sensei_riskowner	John Price
AssignedTo	sensei_assignedto	Marc Soester

Issues

SharePoint Field	Altus Field	Example Data
Title	sensei_name	Resource Overallocation
Discussion	sensei_description	Key developer assigned to multiple projects causing delays.
DueDate	sensei_duedate	24/04/2026 00:00
Status	statuscode	(1) Active
Category	sensei_category	(3) Category3
Priority	sensei_priority	(2) Medium
AssignedTo	sensei_assignedto	John Price
Cost_x0020_Impact	sensei_costimpact	\$10,000.00
Resolution	sensei_resolution	Rebalance workload and secure additional support.

If you haven't created any custom fields, you can skip this step and proceed to 4. Run the export script.

If you have created additional custom fields, you will need update this script.

Example Mapping for Issues Custom fields

In this example, the following custom fields have been added to the Issues register:

SharePoint Display Name	SharePoint Internal Name	SharePoint Field Type
Example Boolean Field	ExampleBooleanField	Yes/No (Checkbox)
Example Date and Time Field	ExampleDateAndTimeField	Date & Time
Example Date Only Field	ExampleDateOnlyField	Date Only
Example Decimal Field	ExampleDecimalField	Number (2 decimal places)
Example Lookup Field	ExampleLookupField	Lookup
Example Money Field	ExampleMoneyField	Currency
Example Number Field	ExampleNumberField	Number (0 decimal places)
Example Option Set Field	ExampleOptionSetField	Choice
Example Text Field	ExampleTextField	Single line of text

The SharePoint Internal name is set when the field is created. This can be obtained by opening the field in edit mode and getting the name from the URL:



If the field was created with spaces in, these spaces will be replaced with **_x0020_**

For example, "Cost Impact" (with a space between Cost and Impact) would have an internal name = "Cost_x0020_Impact"

Altus Display Name	Altus Internal Name	Altus Field Type
Example SP Boolean Field	ce_examplespbooleanfield	Yes/no
Example SP DateAndTime Field	ce_examplespdateandtimefield	Date and time
Example SP DateOnly Field	ce_examplespdateonlyfield	Date only
Example SP Decimal Field	ce_examplespdecimalfield	Decimal
Example SP Lookup Decision Category	ce_examplesplookupdecisioncategory	Lookup
Example SP Money Field	ce_examplespmoneyfield	Currency
Example SP Number Field	ce_examplespnumberfield	Whole number
Example SP OptionSet Field	ce_examplespoptionsetfield	Choice
Example SP Text Field	ce_examplesptextfield	Single line of text

Update the Script

The parts highlighted below are as follows:

Attribute	Details
SharePoint Online Field	Internal name of the SharePoint field. This is set when the field is created in SharePoint.
Altus Field	This is the logical name for the field in the Altus Table.
SharePoint Online Field Format (type)	This is the SharePoint Online field type such as Text, Boolean, DateTime, etc.
SharePoint Online Value	This is the choice field value selected in SharePoint Online.
Altus Value	This is the Value field in Altus e.g. 955000000. This is the identifier for the label.

Quick Summary:

1. Open **4.SharePoint-Migration > export.config.json** to edit it.
2. Map **Text** fields.
3. Map **Boolean** fields.
4. Map **Date and Time** fields.
5. Map **Date Only** fields.
6. Map **Decimal** fields.
7. Map **Lookup** fields.
8. Map **Money** fields.
9. Map **Whole Number** fields.
10. Map **Option** Set fields.

Video:

See link below to a video which covers the above steps with an example migration:

 [SharePoint Online List Migration - 1 Update export.config.json.mp4](#)

Update the Script

1. Open 4.SharePoint-Migration > export.config.json to edit it.

2. Map **Text** fields.

Example:

```
{ "_fielddetails": "This is a custom Single line of Text field.",  
  "spFieldInternalName": "ExampleTextField",  
  "entityAttribute": "ce_examplesptextfield",  
  "type": "Text"  
}
```

3. Map **Boolean** fields.

Example:

```
{ "_fielddetails": "This is a custom Flag (Boolean) field.",  
  "spFieldInternalName": "ExampleBooleanField",  
  "entityAttribute": "ce_examplespbooleanfield",  
  "type": "Bool"  
}
```

4. Map **Date and Time** fields.

Example:

```
{ "_fielddetails": "This is a custom Date and Time field.",  
  "spFieldInternalName": "ExampleDateAndTimeField",  
  "entityAttribute": "ce_examplespdateandtimefield",  
  "type": "DateTime"  
}
```

5. Map **Date Only** fields.

Example:

```
{  "_fielddetails": "This is a custom Date Only field.",  
    "spFieldInternalName": "ExampleDateOnlyField",  
    "entityAttribute": "ce_examplespdateonlyfield",  
    "type": "DateOnly"  
}
```

Update the Script**6. Map Decimal fields.**Example:

```
{  "_fielddetails": "This is a custom Decimal field.",
    "spFieldInternalName": "ExampleDecimalField",
    "entityAttribute": "ce_examplespdecimalfield",
    "type": "Decimal",
    "divider": 2
}
```

7. Map Lookup fields.Example:

```
{  "_fielddetails": "This is a custom Lookup field.",
    "spFieldInternalName": "ExampleLookupField",
    "entityAttribute": "ce_examplesplookupdecisioncategory",
    "type": "Lookup",
    "lookupEntity": "DecisionCategory"
}
```

8. Map Money fields.Example:

```
{  "_fielddetails": "This is a custom Money field.",
    "spFieldInternalName": "ExampleMoneyField",
    "entityAttribute": "ce_examplespmoneyfield",
    "type": "Money"
}
```

9. Map Whole Number fields.Example:

```
{  "_fielddetails": "This is a custom Whole Number field.",
    "spFieldInternalName": "ExampleNumberField",
    "entityAttribute": "ce_examplespnumberfield",
    "type": "Number"
}
```

Update the Script**10. Map Option Set fields.**

Example:

```
{  "_fielddetails": "This is a custom OptionSet field.",
    "spFieldInternalName": "ExampleOptionSetField",
    "entityAttribute": "ce_examplesoptionsetfield",
    "type": "OptionSet",
    "choiceMap": {
      "Resource": 955000000,
      "Scope": 955000001,
      "Schedule": 955000002,
      "Technology": 955000003,
      "Other": 955000004,
      "Business Process": 955000005,
      "Vendor": 955000006,
      "Management Escalation": 955000007
    }
}
```

Attached below is an example of the script:



export.config.zip

2. Update and run the export script

Quick Summary:

1. Open this script: **.\-Run-Migration.ps1** and make the following changes.
2. Open **SharePoint Online Management Shell**.
3. Connect to the folder where this script is located: **.\-Run-Migration.ps1**
4. Run this command: **.\-Run-Migration.ps1**
5. Select **1** to Export the lists.
6. Select the **account** you use to access **Project Online**.
7. Once the export process has completed, you can close SharePoint Online Management Shell.

Video:

See link below to a video which covers the above steps with an example migration:

 [SharePoint Online List Migration - 2 Update and run the export script.mp4](#)

4. Update and run the export script

1. Open this script: `.\Run-Migration.ps1` and make the following changes.

Variable	Details
Url	Set this to the source Project Online environment.
FolderName	Specify a folder name where you want the export files to be output to.
ProjectFilter	Specify the names of projects to export from Project Online.
\$D365Url	Set this to the target Altus environment.
\$PolExportRoot	Specify the location of the files exported during this step.
\$ClientId	Specify the ClientId set in the prerequisites process.
\$CmtSchemaPath	Remove this line from the script.
-CmtSchemaPath \$CmtSchemaPath '	Remove this line from the script.

Examples:

```
@{ Url      = "https://senseicloud.sharepoint.com/sites/vNext"
  FolderName = "vNext"    # Folder name for output
  ProjectFilter = @("Brisbane Roadmap Tool Implementation", "D365 Deployment")
}
```

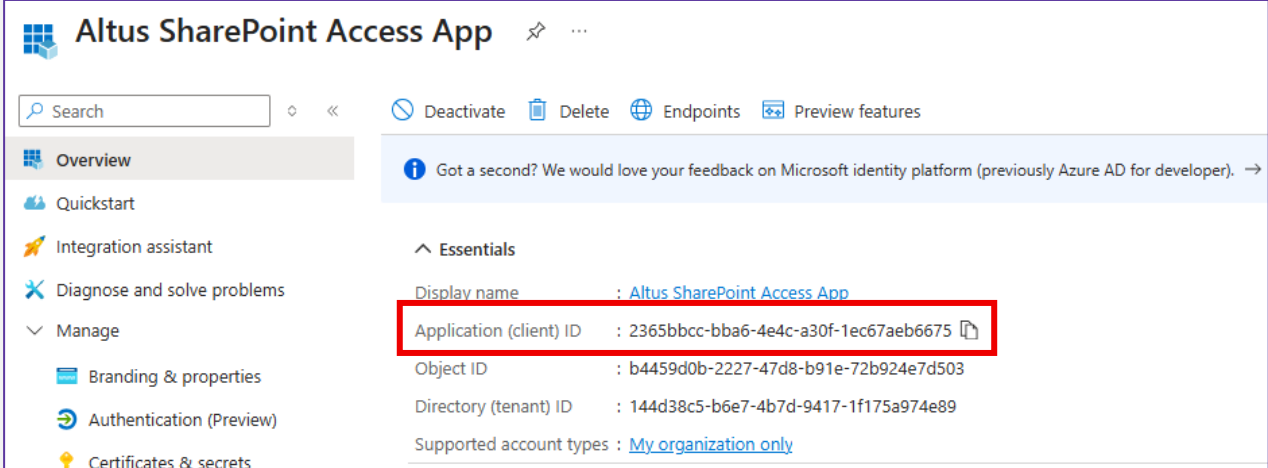
\$D365Url = <https://jp1.crm6.dynamics.com/>

\$PolExportRoot = "C:\Users\JohnPrice\Desktop\POLDataMigration\2.Project-and-Resource-Migration\Files"

\$ClientId = "30b4ad0b-d939-4cd4-bb6d-fa2d39fb4694"

Note: If you aren't using the Sensei app ClientID, you will need to grab this from the application you have configuration in [this document](#).

This is where you source the client ID from:

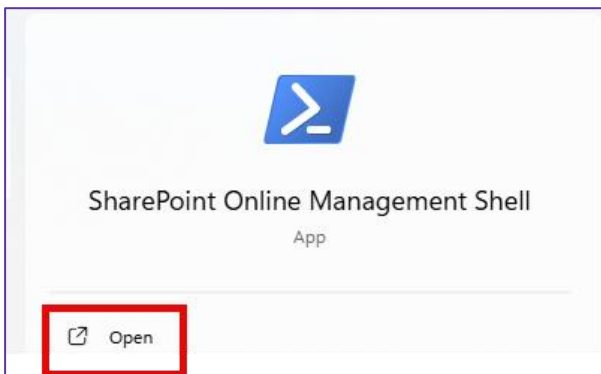


The screenshot shows the 'Altus SharePoint Access App' configuration page. The 'Essentials' section is expanded, displaying the following information:

- Display name: Altus SharePoint Access App
- Application (client) ID: 2365bbcc-bba6-4e4c-a30f-1ec67aeb6675 (highlighted with a red box)
- Object ID: b4459d0b-2227-47d8-b91e-72b924e7d503
- Directory (tenant) ID: 144d38c5-b6e7-4b7d-9417-1f175a974e89
- Supported account types: My organization only

4. Update and run the export script

2. Open SharePoint Online Management Shell.



3. Connect to the folder where this script is located: `.\-Run-Migration.ps1`

Example

```
cd C:\Users\JohnPrice\Desktop\POLDataMigration\4.SharePoint-Migration
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\JohnPrice> cd C:\Users\JohnPrice\Desktop\POLDataMigration\4.SharePoint-Migration
PS C:\Users\JohnPrice\Desktop\POLDataMigration\4.SharePoint-Migration>
```

4. Run this command: `.\-Run-Migration.ps1`

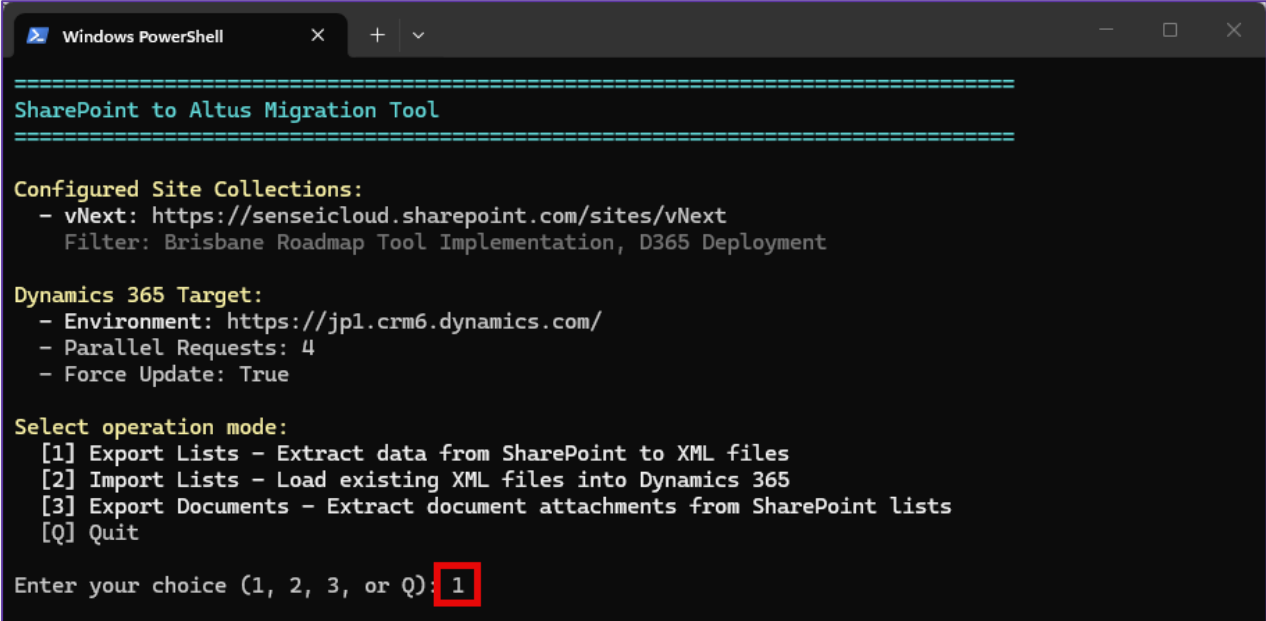
```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\JohnPrice> cd C:\Users\JohnPrice\Desktop\POLDataMigration\4.SharePoint-Migration
PS C:\Users\JohnPrice\Desktop\POLDataMigration\4.SharePoint-Migration> .\-Run-Migration.ps1
```

4. Update and run the export script

5. Select 1 to Export the lists.



```
Windows PowerShell

=====
SharePoint to Altus Migration Tool
=====

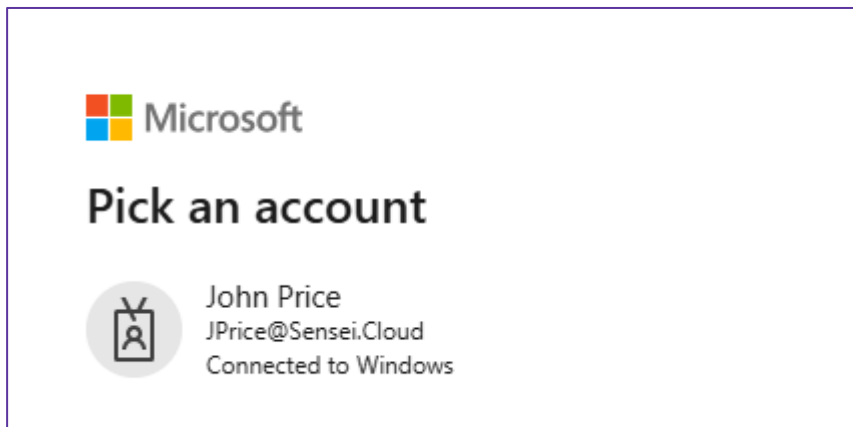
Configured Site Collections:
- vNext: https://senseicloud.sharepoint.com/sites/vNext
  Filter: Brisbane Roadmap Tool Implementation, D365 Deployment

Dynamics 365 Target:
- Environment: https://jpl.crm6.dynamics.com/
- Parallel Requests: 4
- Force Update: True

Select operation mode:
[1] Export Lists - Extract data from SharePoint to XML files
[2] Import Lists - Load existing XML files into Dynamics 365
[3] Export Documents - Extract document attachments from SharePoint lists
[Q] Quit

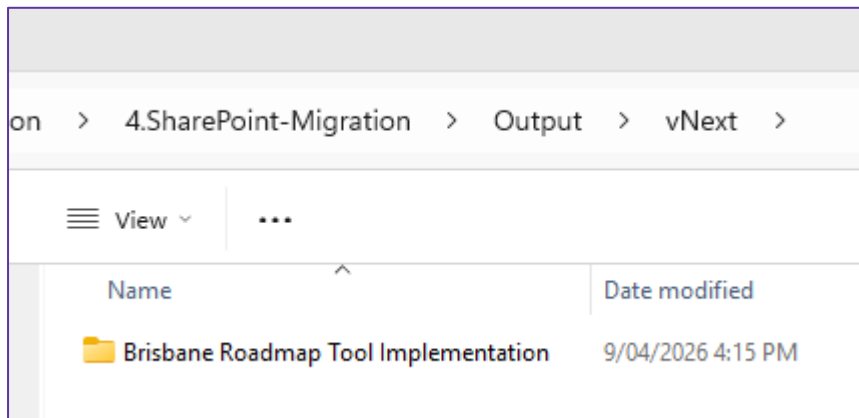
Enter your choice (1, 2, 3, or Q): 1
```

6. Select the **account** you use to access **Project Online**.

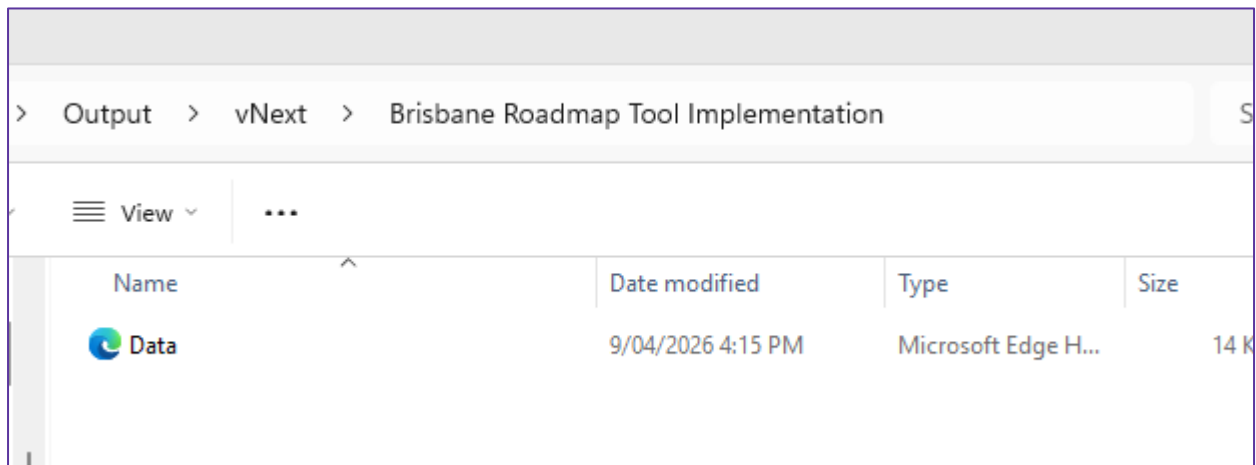


7. Once the export process has completed, you can close SharePoint Online Management Shell.

Once the process has been completed, a folder will be created for each project:



Within each folder is an XML file which contains the exported data:



3. Run the import script

Quick Summary:

1. Open **SharePoint Online Management Shell**.
2. Connect to the **folder** where this script is located: **.\-Run-Migration.ps1**
3. Run this command: **.\-Run-Migration.ps1**
4. Select **2** to **Import** the lists.
5. Once the export process has completed, you can close SharePoint Online Management Shell.

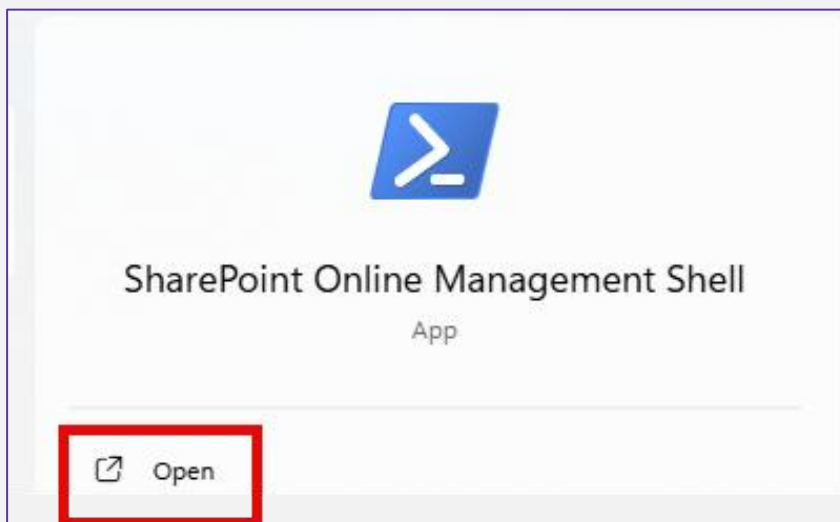
Video:

See link below to a video which covers the above steps with an example migration:

 [SharePoint Online List Migration - 3 Run the import script.mp4](#)

5. Run the import script

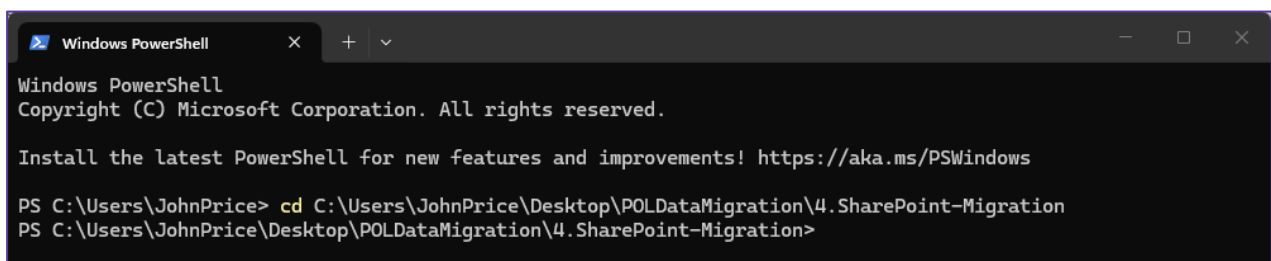
1. Open SharePoint Online Management Shell



2. Connect to the **folder** where this script is located: **.\-Run-Migration.ps1**

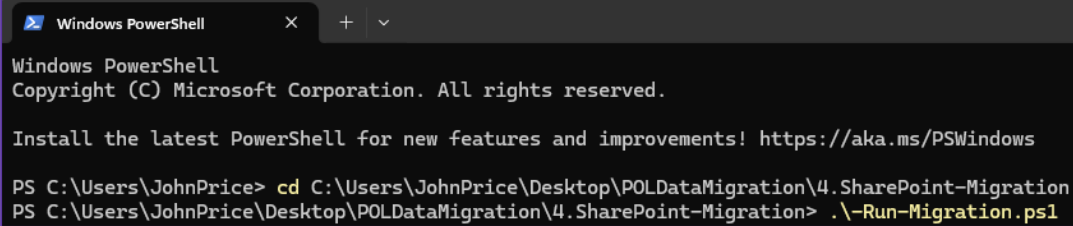
Example

```
cd C:\Users\JohnPrice\Desktop\POLDataMigration\4.SharePoint-Migration
```



3. Run this command: **.\-Run-Migration.ps1**

5. Run the import script

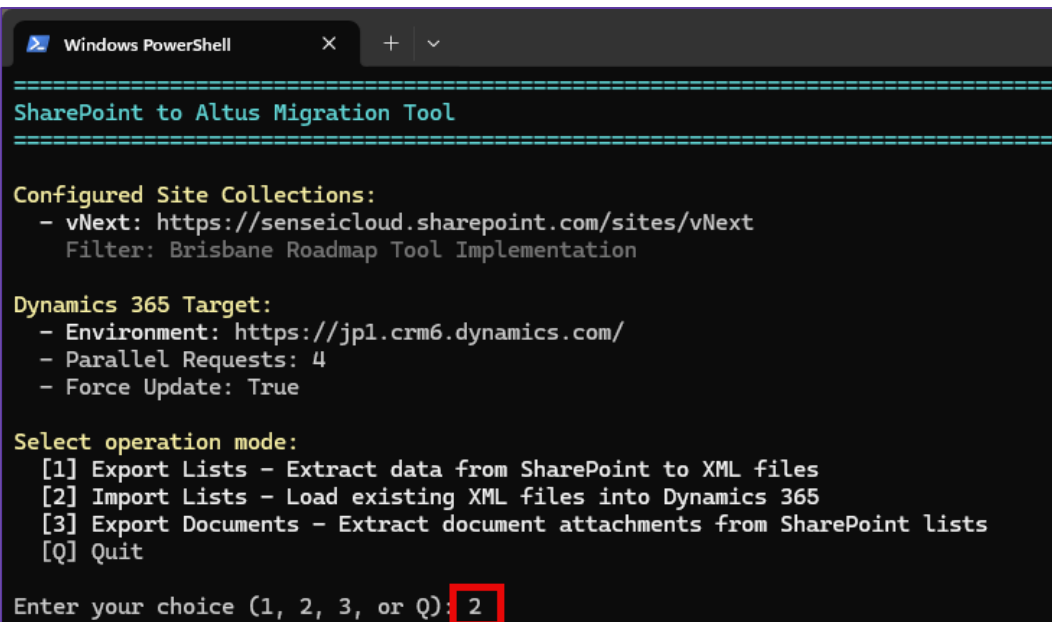


```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\JohnPrice> cd C:\Users\JohnPrice\Desktop\POLDataMigration\4.SharePoint-Migration
PS C:\Users\JohnPrice\Desktop\POLDataMigration\4.SharePoint-Migration> .\Run-Migration.ps1
```

4. Select 2 to Import the lists.



```
=====
SharePoint to Altus Migration Tool
=====

Configured Site Collections:
- vNext: https://senseicloud.sharepoint.com/sites/vNext
  Filter: Brisbane Roadmap Tool Implementation

Dynamics 365 Target:
- Environment: https://jpl.crm6.dynamics.com/
- Parallel Requests: 4
- Force Update: True

Select operation mode:
[1] Export Lists - Extract data from SharePoint to XML files
[2] Import Lists - Load existing XML files into Dynamics 365
[3] Export Documents - Extract document attachments from SharePoint lists
[Q] Quit

Enter your choice (1, 2, 3, or Q): 2
```

5. Once the export process has completed, you can close SharePoint Online Management Shell.

Example of Imported Risks

Project Online:

Home

EDIT LINKS

Risks ⓘ

+ new item

 or edit this list

All Items

All Active Risks

All Closed Risks

...

Find an item

✓	📎	ID	Title	Assigned To	Status	Exposure	Category	Due Date
Count= 4								
		1	Requirements Not Clearly Defined	...	☐ John Price (1) Active	2.5	(2) Category2	30/04/2026 0:00
		2	Stakeholder Misalignment	...	☐ John Price (1) Active	2	(1) Category1	15/04/2026 0:00
		3	Key Resource Unavailability	...	☐ John Price (3) Closed	0.5	(3) Category3	4/03/2026 0:00
		4	Budget Overrun	...	☐ John Price (1) Active	2.4	(1) Category1	30/06/2026 0:00

Altus PPM:

Brisbane Roadmap Tool Implementation - Saved

Project · Tracking ▾

Major Project

Active for 8 days

<

Planning

Execution (52 Hrs)

Deliverables

Status Updates

Dependencies

Issues

Risks

Decisions

Changes

Lessons

Information

Tracking

Resources

Strategy

Risks

Clear filter Refresh

Consequence

Likelihood

Risks for Project

☐ Name ▾	Assigned To ▾	Status Reason ▾	Category ▾
☐ Key Resource Unavailability	 John Price (Ava...	Closed	Other
☐ Stakeholder Misalignment		Active	Other
☐ Requirements Not Clearly Defined		Active	Other
☐ Budget Overrun	 John Price (Ava...	Active	Other

Rows: 4

An Example of a single risk's details is below:

Stakeholder Misalignment - Saved

Risk

Details Related ▾

Name	* Stakeholder Misalignment	Description	Key stakeholders have differing expectations about project outcomes.
Assigned To	 John Price (Available) ×		
Risk Owner	 John Price (Available) ×	Mitigation Plan	Establish governance meetings and maintain a stakeholder communication plan.
ID	🔒 1018	Contingency Plan	Escalate to steering committee for decision-making.
Status	Active ▾		
Category	Other ▾		
Due Date	15/04/2026 📅		
Likelihood	* Unlikely ▾	Trigger Description	The trigger for this risk is the date.
Consequence	* Major ▾		
Cost	\$5,000.00	Trigger	Date ▾

All OOTB fields are populated ready for use.

Example of Imported Issues

Project Online:

Home

EDIT LINKS

Issues ⓘ

+ new item

or edit this list

All Items

All Active Issues

All Closed Issues

...

Find an item

✓		ID	Title	Assigned To	Status	Priority	Category	Due Date
Count= 4								
		1	Resource Overallocation	... ☐ John Price	(1) Active	(2) Medium	(3) Category3	24/04/2026 0:00
		2	Incomplete Data Migration	... ☐ John Price	(1) Active	(1) High	(1) Category1	22/04/2026 0:00
		3	Stakeholder Feedback Outstanding	... ☐ Marc Soester	(1) Active	(3) Low	(3) Category3	7/05/2026 0:00
		4	Security Review Delayed	... ☐ John Price	(1) Active	(3) Low	(3) Category3	11/06/2026 0:00

Altus PPM:

Brisbane Roadmap Tool Implementation - Saved

Project · Tracking ▾

Major Project
Active for 8 days

<

Planning

Execution (52 Hrs)

Deliverables

Status Updates

Dependencies

Issues

Risks

Decisions

Changes

Lessons

Information

Tracking

Issues

0
Overdue
Issues

4
Active
Issues

0
Due Date
missing

Issues for Project

☐ Name ▾	Assigned To ▾
☐ Incomplete Data Migration	 John Price (Ava...
☐ Resource Overallocation	 John Price (Ava...
☐ Stakeholder Feedback Outstanding	
☐ Security Review Delayed	 John Price (Ava...

Rows: 4

An Example of a single issue details is below which shows the migrated custom field data:

Resource Overallocation - Saved

Issue

Program --- | Project Brisbane Roadmap Tool Implementation

Details

Related

Name

Resource Overallocation

ID

1326

Assigned To

John Price (Available)

Status

Active

Category

Other

Priority

High

Medium

Low

Linked to Risk

Created On

9/04/2026

7:32 AM

Due Date

24/04/2026

SP Custom Fields

Example SP Boolean Field

Yes

Example SP DateAndTime Field

2/04/2026

2:30 AM

Example SP Lookup DecisionCategory

Business Process

Example SP Decimal Field

61.73

Example SP DateOnly Field

2/04/2026

Example SP OptionSet Field

Resource

Example SP Text Field

Example Text 1

Example SP Number Field

123

Example SP Money Field

\$111.22

Description

Key developer assigned to multiple projects causing delays.

Resolution

Rebalance workload and secure additional support.

All OOTB fields and custom fields are populated ready for use.

Migration of Custom Lists / Example

The scripts that have been created by Altus can be modified to migrate custom lists. This section of the document covers an example of a custom list migration.

SharePoint Online source list

In this example, a custom list (Actions) is going to be migrated to Altus. Below is a screenshot of this list in a SharePoint Online sub-site for a Project Online project.

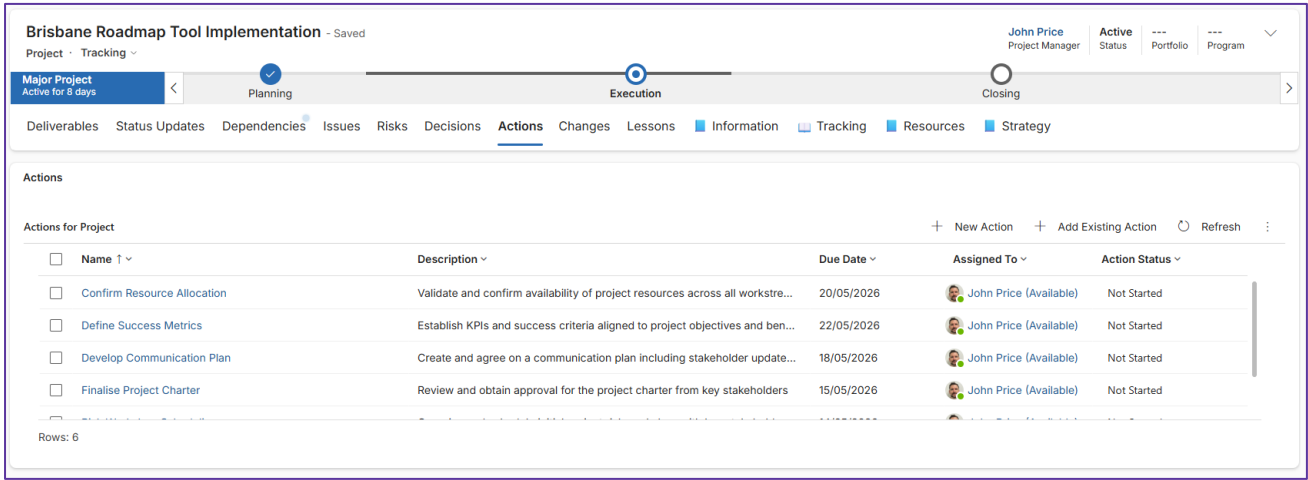
Brisbane Roadmap Tool Implementation					
+ Add new item Edit in grid view Undo Share Copy link Export Forms Workflows Integrate					
Actions ☆ ☺					
Title	Description	Due Date	Assigned To	Action Status	Completion Date
Finalise Project Charter	Review and obtain approval for the project charter from key stakeholders	15/05/2026	John Price	In Progress	
Confirm Resource Allocation	Validate and confirm availability of project resources across all workstreams	20/05/2026	John Price	Not Started	
Set Up Project Workspace	Configure SharePoint site, Teams channel, and document repository for the project	12/05/2026	John Price	Complete	3/04/2026
Develop Communication Plan	Create and agree on a communication plan including stakeholder updates and reporting cadence	18/05/2026	John Price	In Progress	
☐ Risk Workshop Scheduling	Organise and schedule initial project risk workshop with key stakeholders	14/05/2026	John Price	Impeded	
Define Success Metrics	Establish KPIs and success criteria aligned to project objectives and benefits realisation	22/05/2026		Not Started	

This list has the following columns:

SharePoint Display Name	SharePoint Internal Name	SharePoint Field Type
Title	Title	Single line of text
Description	Description	Multiple lines of text
Due Date	DueDate	Date and time
Assigned To	AssignedTo	Person or Group
Action Status	ActionStatus	Choice
Completion Date	CompletionDate	Date and time

Altus target register

Below is a screenshot of the target register in Altus:



This register has the following columns. **ce_** denotes the solution layer that the fields exist in:

Altus Table Name	Altus Internal Name
Action	ce_action

Altus Field Display Name	Altus Field Internal Name	Altus Field Type
Name	ce_name	Single line of text
Description	ce_description	Multiple lines of text
Due Date	ce_duedate	Date and time
Assigned To	ce_assignedto	Person or Group
Action Status	ce_actionstatus	Choice
Completion Date	ce_completiondate	Date and time

Example JSON file

Below is an example of the JSON used to extract the XML and data from the SharePoint list and map it to the corresponding Altus fields:

Note: When you create the many to one relationship between the custom register and Portfolio, Programs, and Projects, you will need to update the following additional parts of the script:

- projectLookupAttribute
- projectLookupEntityLogicalName

These will need to include the specific names in your solution layer.

```
{
  "version": "1.0",
  "lists": [
    {
      "spListTitle": "Actions",
      "entityLogicalName": "ce_actions",
      "projectLookupAttribute": "ce_sensei_project",
      "projectLookupEntityLogicalName": "ce_sensei_project",
      "projectLookupTextSource": "Web.Title",
      "backlinkAttribute": "sensei_sourceitemurl",
      "columnMap": [
        {
          "spFieldInternalName": "Title",
          "entityAttribute": "ce_name",
          "type": "Text"
        },
        {
          "spFieldInternalName": "Description",
          "entityAttribute": "ce_description",
          "type": "Text"
        },
        {
          "spFieldInternalName": "DueDate",
```

```
"entityAttribute": "ce_duedate",
"type": "DateOnly"
},
{
"spFieldInternalName": "AssignedTo",
"entityAttribute": "ce_assignedto",
"type": "Lookup",
"lookupEntity": "systemuser"
},
{
"spFieldInternalName": "Status",
"entityAttribute": "ce_actionstatus",
"type": "Status",
"choiceMap": {
"Not Started": 121460000,
"In Progress": 121460001,
"Complete": 121460002,
"Impeded": 121460003
}
},
{
"spFieldInternalName": "CompletionDate",
"entityAttribute": "ce_completiondate",
"type": "DateOnly"
}
]
}
]
```

Migration Process

To recap, you will need to:

1. Open the solution layer.
2. Create the register (as a new table).
3. Create the custom fields.
4. Add many-to-one relationships to Portfolio, Program, and Project tables.
5. Update the script (as per the example above).
6. Run the export script.
7. Run the import script.

Project Documents

The Altus scripts enable you to export all the documentation from the associated SharePoint Online project sub-sites.

Step	Details
1. Create/identify target library	Either: 1. Create a Teams channel for the Altus project OR 2. Use an existing non-Project Online SharePoint document library OR 3. Create a non-Project Online SharePoint document library.
2. Update and run the export script	Update the export script parameters and run for the select projects you wish to export documentation for.
3. Complete Post-Migration steps	Complete the post migration steps documented.

What is migrated?

Data	Details
Documentation	All files in the Project document library.
Folder Structure	All folders and their structure within the document library.
Metadata	Associated document metadata, including custom attributes.
Version History	All previous versions of the document that have been created.

The script will migrate documents for all document libraries in the source SharePoint Online sub-sites. This part of the script grabs documents from any folder with a type like Document.

```
# Get all lists
$lists = Get-PnPList
$documentLists = $lists | Where-Object {
    $_.BaseTemplate -eq 101 -or $_.EntityType -like "*Document*"
}
You, 2 months ago • restructure and copy over
Write-LogMessage " Found $($documentLists.Count) document libraries"
```

If you want to omit certain libraries from being exported, you can use this part of the script to omit these libraries from being exported.

```
[Parameter(Mandatory = $false)]
[string] $ExcludeLibraries = "Style Library,Preservation Hold Library,Form Templates,Recycle Bin,Site Assets"
```

What isn't migrated?

Data	Details
Permissions	Access controls regarding who can access the site and to what level.
Site Structure	How the site sits within a broader SharePoint hierarchy.
Content Types	Reusable definitions of content across the sub-site.
Workflows	Automated processes such as document approval workflows.
Retention Policies	How long content is retained in the SharePoint document library.
Site Columns	Reusable metadata fields that apply across multiple sites.

If you require a complete migration of project documentation (including what isn't migrated), Altus recommends that the user purchase [ShareGate](#).

1. Create/identify target library

Within this part of the process, you have three options:

1. Create a Teams channel for the Altus project **OR**
2. Use an existing non-Project Online SharePoint document library **OR**
3. Create a non-Project Online SharePoint document library.

1. Create a Teams channel for the Altus project

Quick Summary

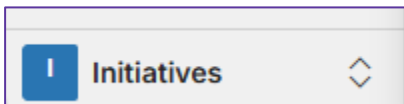
1. Login to the Altus environment:
2. Open **Initiatives**.
3. Open **Projects**.
4. Select the project you wish to create the Team/Group for.
5. Select **Project Group > No Group**.
6. Select **Create Group > Create**.
7. Once created, close the dialog box.
8. Once done, select **Open in SharePoint** in the ribbon.
9. Select **Documents**.

1. Create a Teams channel for the Altus project

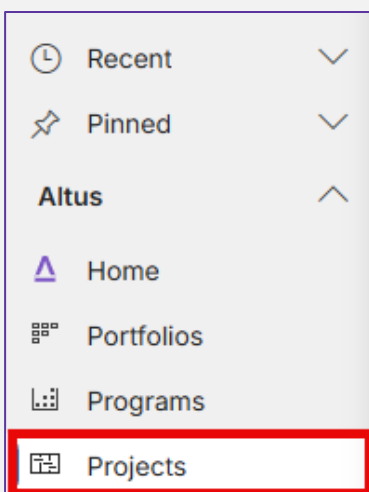
1. Login to the Altus environment:

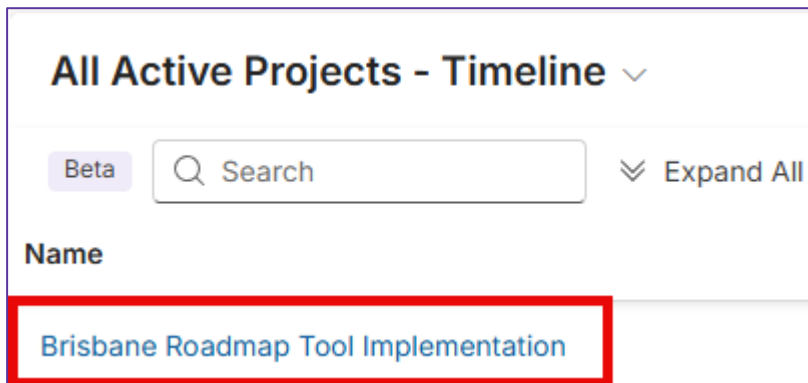
Example: <https://jp1.crm6.dynamics.com/main.aspx?appid=918ae574-9b21-f111-8341-6045bdc25c37>

2. Open **Initiatives**.



3. Open **Projects**.



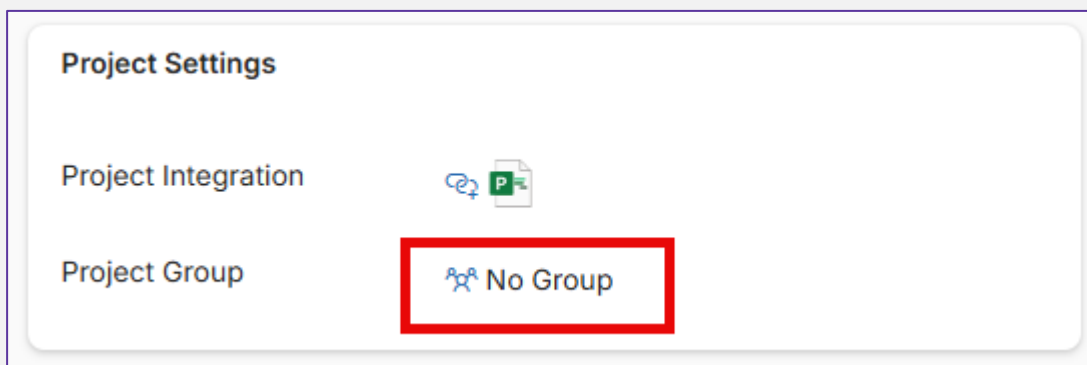
1. Create a Teams channel for the Altus project**4. Select the project you wish to create the Team/Group for.**

All Active Projects - Timeline ▾

Beta ▾ Expand All

Name

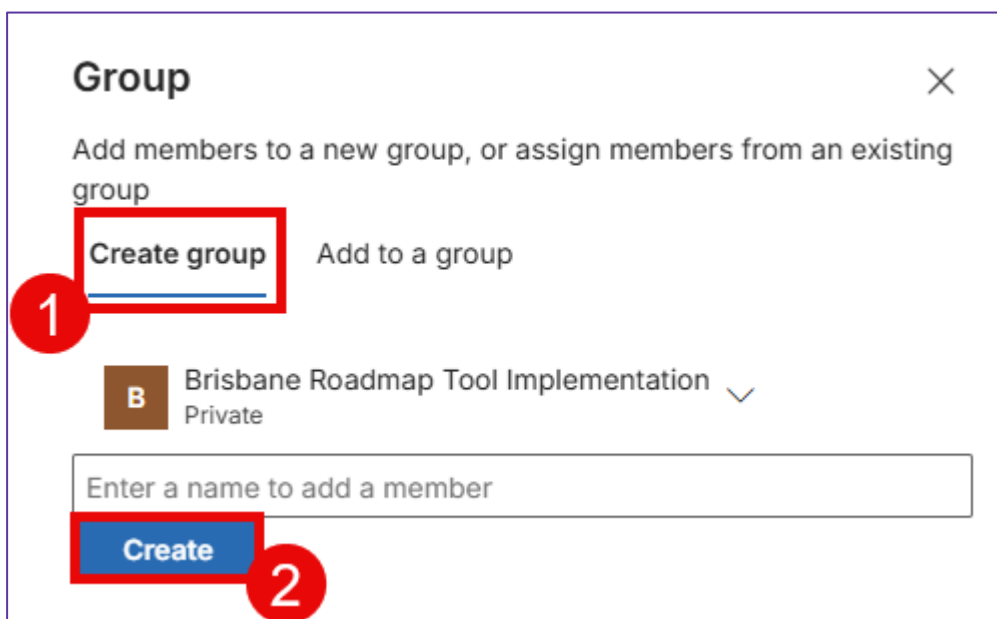
Brisbane Roadmap Tool Implementation

5. Select Project Group > No Group.

Project Settings

Project Integration 

Project Group **No Group**

6. Select Create Group > Create.

Group ×

Add members to a new group, or assign members from an existing group

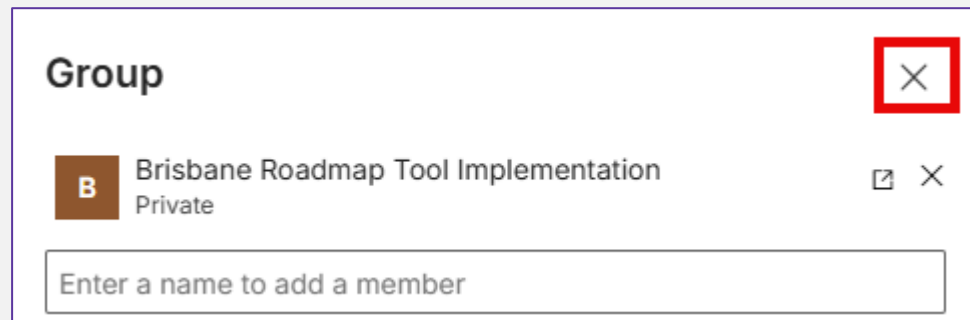
Create group Add to a group

B Brisbane Roadmap Tool Implementation ▾
Private

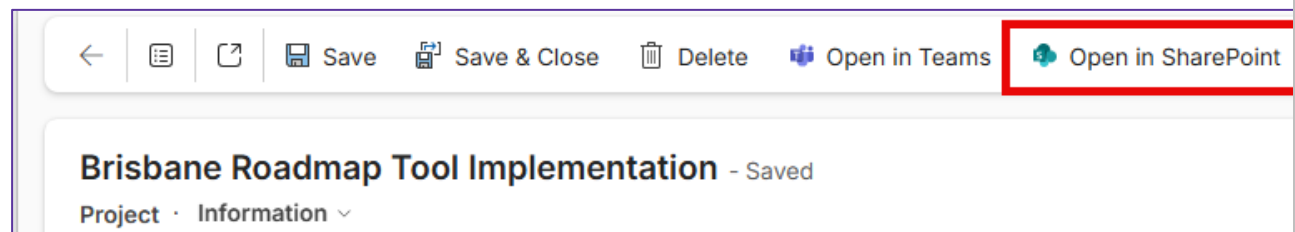
Create

1. Create a Teams channel for the Altus project

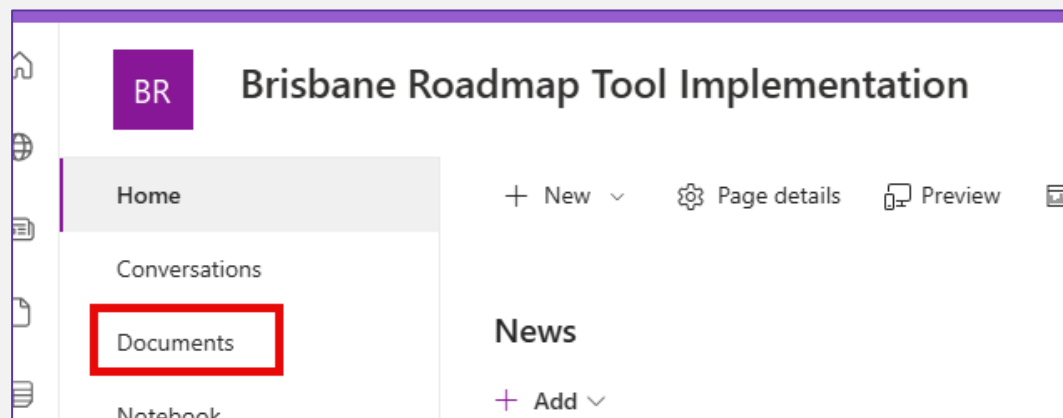
7. Once created, close the dialog box.



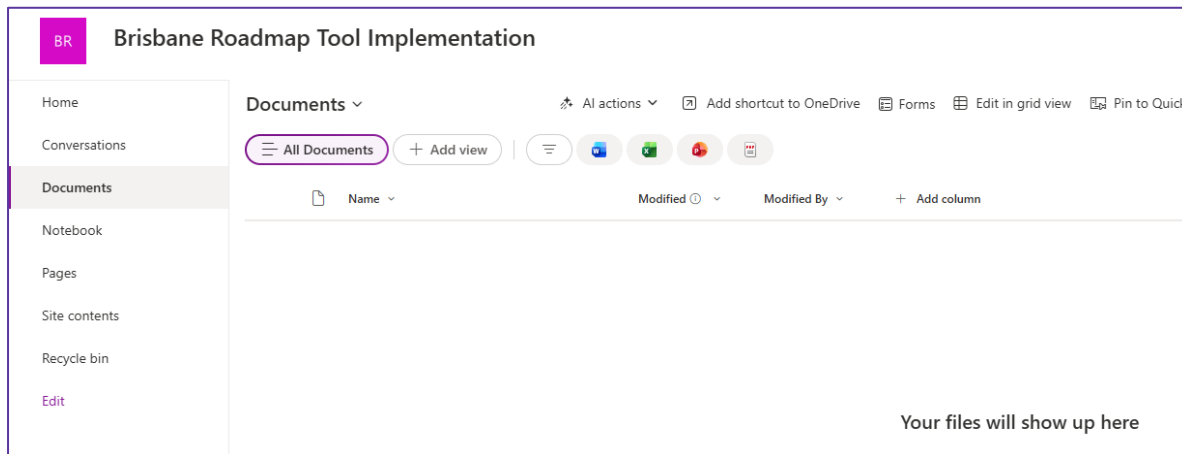
8. Once done, select **Open in SharePoint** in the ribbon.



9. Select **Documents**.



This is the location to migrate document to for this project.



2. Use an existing non-Project Online SharePoint document library

If you don't want to create a Team for the project, you can use an existing non-Project Online SharePoint site for the documentation. This would be a good solution for closed projects that are being migrated.

Quick Summary:

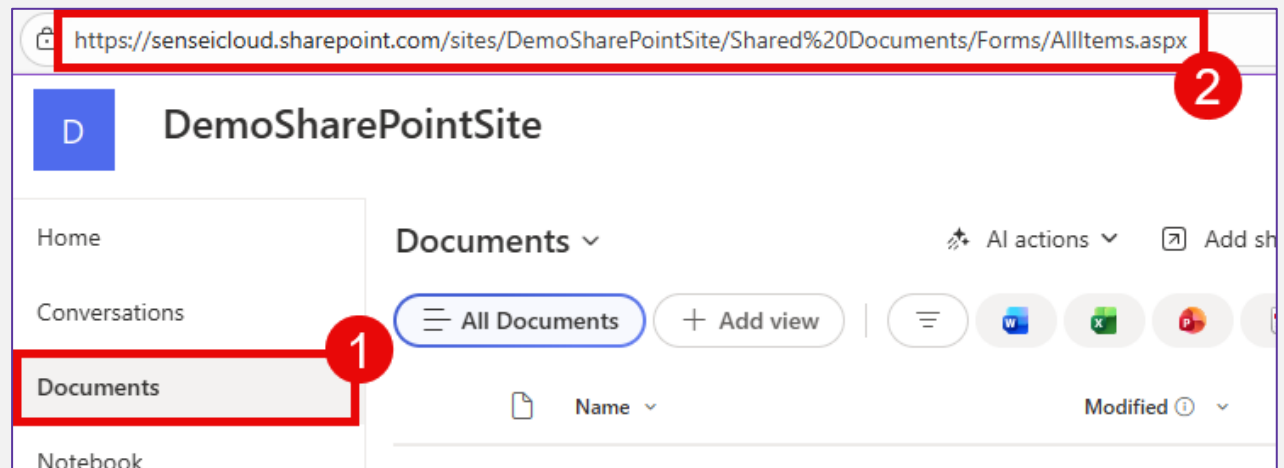
1. Open the SharePoint site document library > Copy the URL.
2. Open this link: <https://make.powerapps.com/>
3. Open **Solutions** > Solution Layer that you have already created:
4. Open the **Projects** table.
5. Select **Edit**.
6. Select **+000 more** > Search **document** > Select **Documents Location** > **Save**.
7. Paste the SharePoint document library URL into this field for the selected project.
8. Open the **Project** in **Altus** > **Documents**.
9. Documents are shown for the select SharePoint document library.

2. Use an existing non-Project Online SharePoint document library

1. Open the SharePoint site document library > Copy the URL.

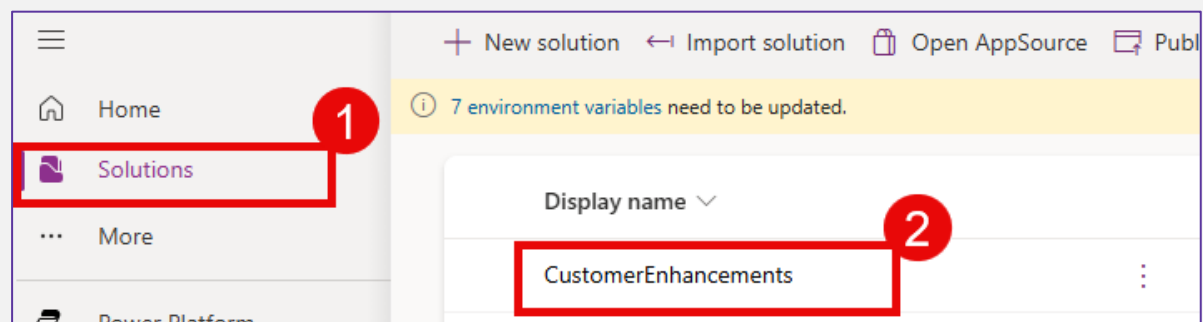
Example:

<https://senseicloud.sharepoint.com/sites/DemoSharePointSite/Shared%20Documents/Forms/AllItems.aspx>

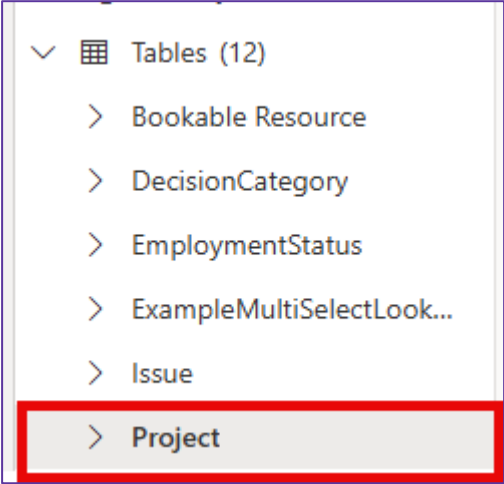


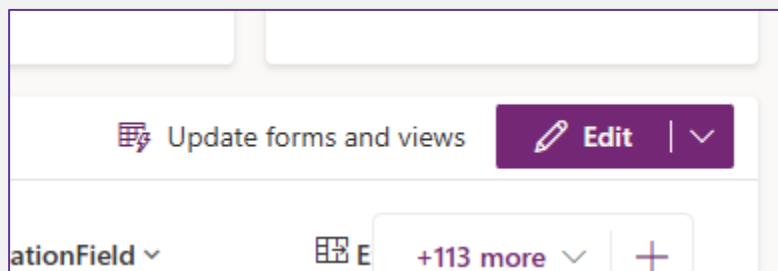
2. Open this link: <https://make.powerapps.com/>

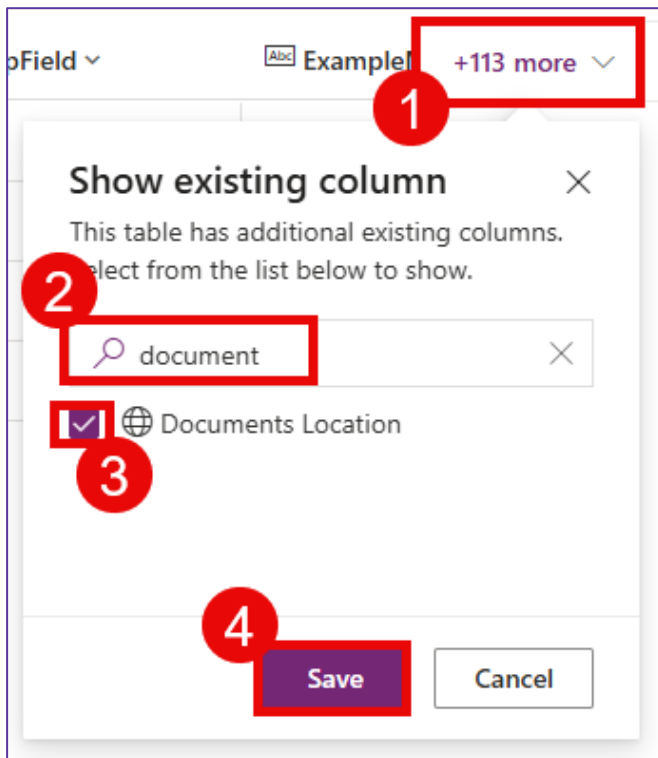
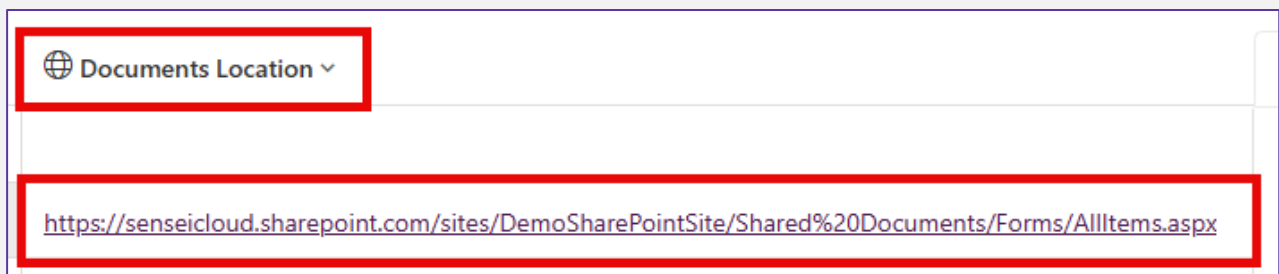
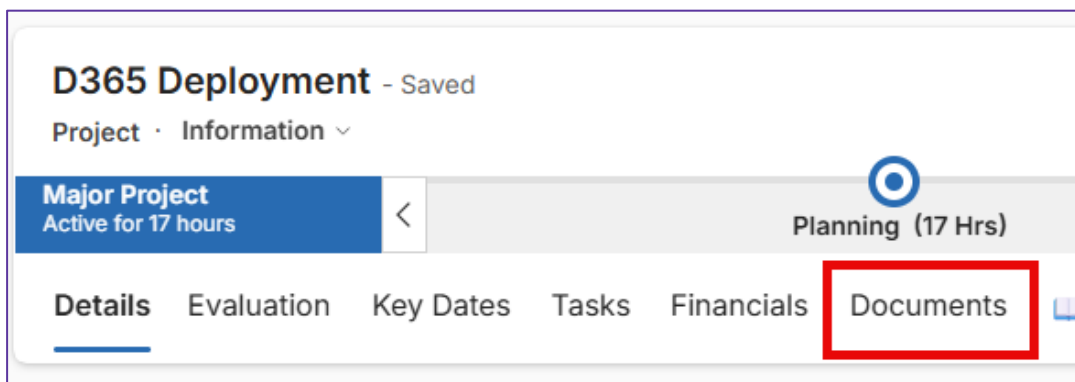
3. Open **Solutions** > **Solution Layer** that you have already created:



2. Use an existing non-Project Online SharePoint document library**4. Open the **Projects** table.**

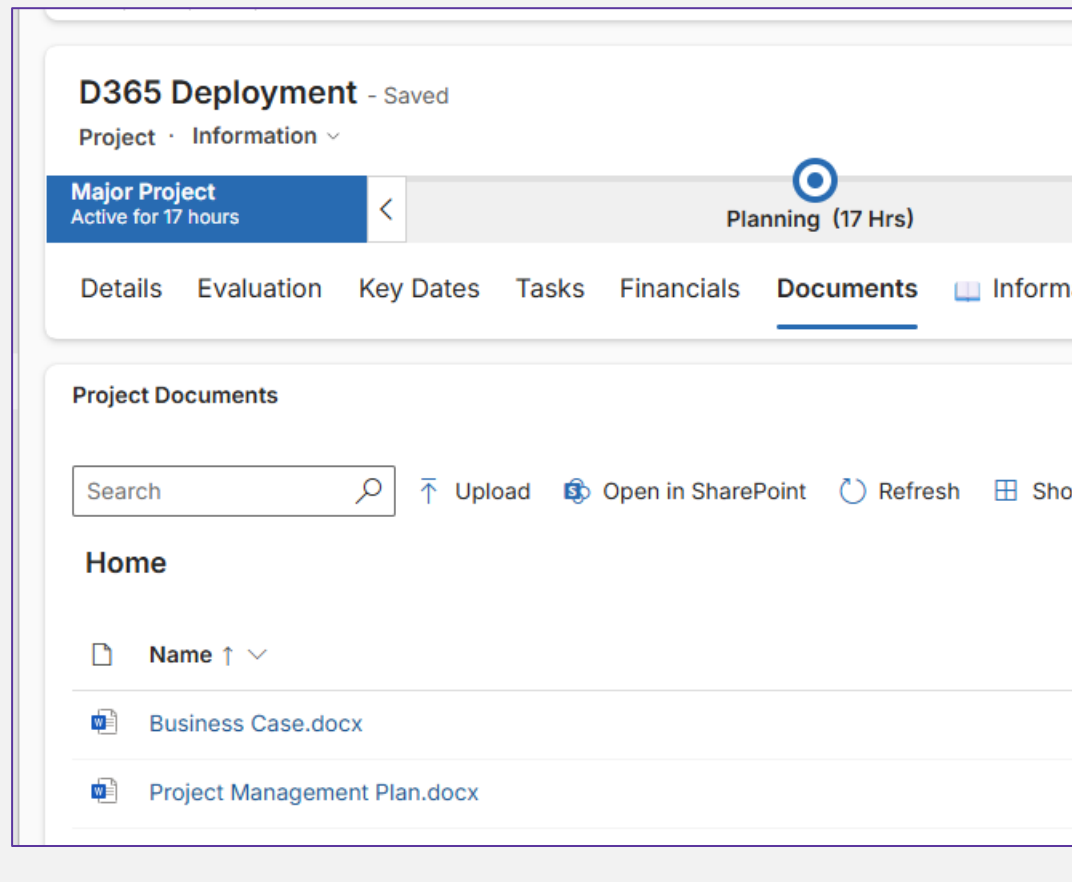
- 
- A screenshot of a SharePoint interface showing a list of tables. The list is titled 'Tables (12)' and includes several items: 'Bookable Resource', 'DecisionCategory', 'EmploymentStatus', 'ExampleMultiSelectLook...', 'Issue', and 'Project'. The 'Project' item is highlighted with a red rectangular border.
- ▼  Tables (12)
 - > Bookable Resource
 - > DecisionCategory
 - > EmploymentStatus
 - > ExampleMultiSelectLook...
 - > Issue
 - > **Project**

5. Select **Edit.**

2. Use an existing non-Project Online SharePoint document library**6. Select +000 more > Search document > Select Documents Location > Save.****7. Paste the SharePoint document library URL into this field for the selected project.****8. Open the Project in Altus > Documents.**

2. Use an existing non-Project Online SharePoint document library

9. Documents are shown for the select SharePoint document library.



3. Create a non-Project Online SharePoint document library

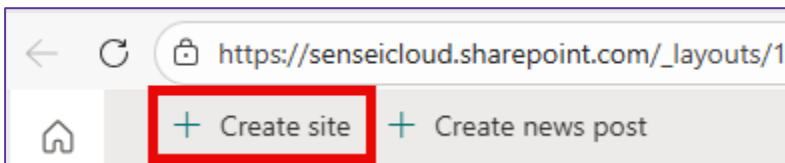
With this step, you will need to ensure you don't create this SharePoint document library in the same site collection as Project Online, otherwise it will be removed in October this year.

Quick Summary:

1. Open **SharePoint**.
2. Select **+ Create site**.
3. Select **Team site**.
4. Select the required template > **Use Template**.
5. Provide a **Site name** > **Next**.
6. **Add members** > **Finish**.

3. Create a non-Project Online SharePoint document library

1. Open **SharePoint**.

3. Create a non-Project Online SharePoint document library**2. Select + Create site.****3. Select Team site.****Create a site: Select the site type****Team site**

Create a private space to collaborate with your team.

-  Track and stay updated on project status
-  Share team resources and co-author content
-  All site owners and members publish site content
-  Can connect to other Microsoft 365 products

3. Create a non-Project Online SharePoint document library**4. Select the required template > Use Template.**

Select a template

From Microsoft From your organization



Standard team
Manage projects, share content, and stay connected with your team.

Back

Use template

5. Provide a Site name > Next.

Give your site a name

Decide on a unique name that follows your organization's naming standards. The description is optional, but useful for people to understand what your site is for.

Your team site will be connected to a Microsoft 365 group, which gives your site a shared OneNote notebook, group email address, and team calendar.



Standard team  Change template

1

Site name *

Demo SP Site

The site name is available.

Site description

Tell people the purpose of your site

Group email address *

DemoSPSite

The group alias is available.

Site address *

https://senseicloud.sharepoint.com/sites/ DemoSPSite

The site address is available.

2

Back

Next

ALTUS

Version 18

3. Create a non-Project Online SharePoint document library

6. Add members > Finish.

Add site owners and members

Consider assigning one additional site owner or a group to help manage site content. [Learn more about permissions](#)

Site owners have full control of site content, theme, permissions, hub associations, and other site settings.

Site members can edit and view site content, including files, pages, lists, and navigation.



Standard team

Add members

Start typing a name

Finish

2. Update and run the export script

Quick Summary:

1. Open this script in edit mode: **3-export-documents.ps1**
2. Set these variables in the script.
3. Open **SharePoint Online Management Shell**.
4. Connect to the folder where this script is located: `.\-Run-Migration.ps1`
5. Run the command: **.\-Run-Migration.ps1**
6. Select **3** to export the documents.
7. Select **Y** to proceed.
8. Select the account you use to access Project Online.
9. All documents will be output to the selected location.

Video:

See link below to a video which covers the above steps with an example migration:

 [SharePoint Online Document migration - 2. Update and run the export script.mp4](#)

Update and run the export script

1. Open this script in edit mode: **3-export-documents.ps1**

2. Set these variables in the script:

Variable	Details
\$SiteCollectionUrl	Set this to the source Project Online environment.
\$OutputFolder	Specify a folder name where you want the documents to be output to.
\$ProjectFilter	Specify the names of projects to export from Project Online.
\$ClientId	Specify the ClientId set in this prerequisite step .
\$ExcludeVersionHistory	If you wish to include version history, set this to = \$false
\$DetailedMetadata	If you wish to include the detailed metadata, set this to = \$true
\$ExcludeLibraries	By default, only the documents library will be output.
\$ExcludeFilePatterns	You can use this parameter to exclude certain file types.

Example:

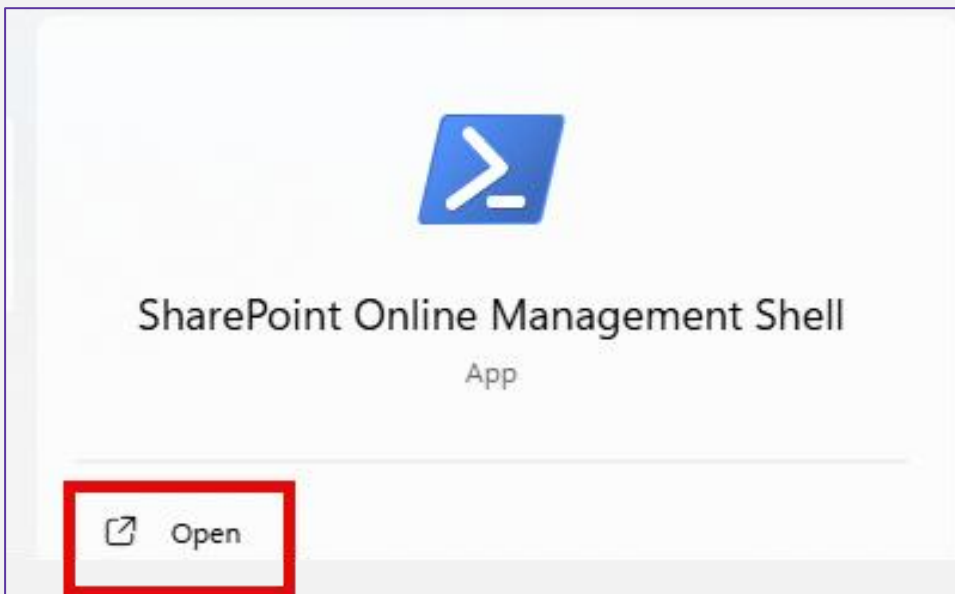
```
[Parameter(Mandatory = $false)]
[string] $SiteCollectionUrl = "https://sensei.sharepoint.com/sites/vNext",

[Parameter(Mandatory = $false)]
[string[]] $ProjectFilter = @("Brisbane Roadmap Tool Implementation", "D365 Deployment") ,

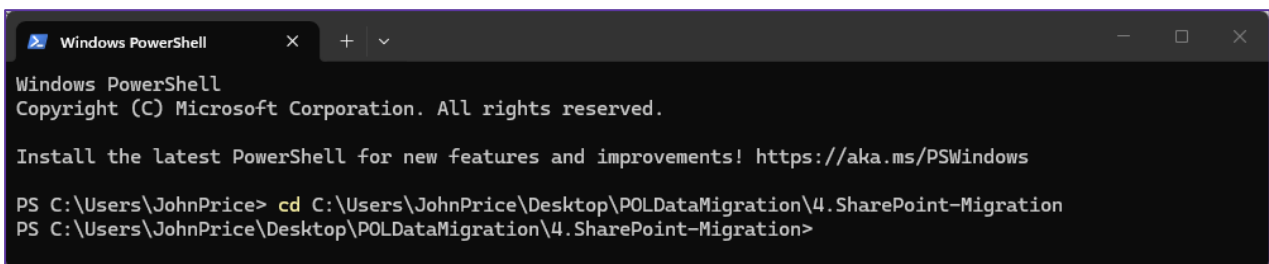
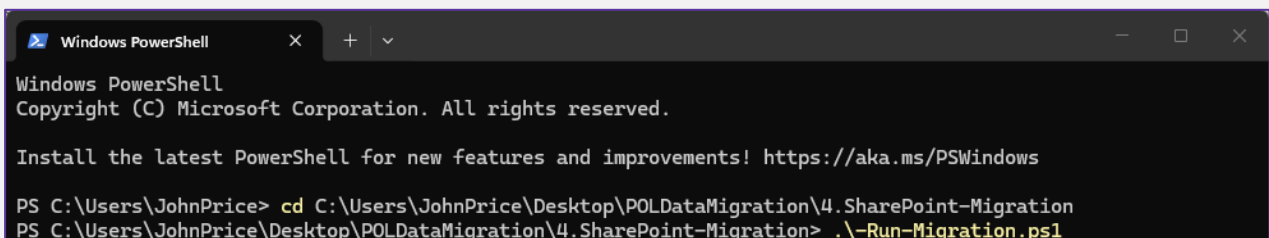
[Parameter(Mandatory = $false)]
[string] $ClientId = "30b4ad0b-d939-4cd4-bb6d-fa2d39fb4694",

[Parameter(Mandatory = $false)]
[bool] $ExcludeVersionHistory = $false,

[Parameter(Mandatory = $false)]
[bool] $DetailedMetadata = $true
```

Update and run the export script**3. Open SharePoint Online Management Shell****4. Connect to the folder where this script is located: .\Run-Migration.ps1****Example:**

`cd C:\Users\JohnPrice\Desktop\POLDataMigration\4.SharePoint-Migration`

**5. Run this command: .\Run-Migration.ps1**

Update and run the export script

6. Select 3 to export the documents.

```
Windows PowerShell
Summary exported to: C:\Users\JohnPrice\Desktop\POLDataMigration\4.SharePoint-Migration\Output\migration-summary.json
PS C:\Users\JohnPrice\Desktop\POLDataMigration\4.SharePoint-Migration> .\Run-Migration.ps1

=====
SharePoint to Altus Migration Tool
=====

Configured Site Collections:
- vNext: https://senseicloud.sharepoint.com/sites/vNext
  Filter: Brisbane Roadmap Tool Implementation

Dynamics 365 Target:
- Environment: https://jpl.crm6.dynamics.com/
- Parallel Requests: 4
- Force Update: True

Select operation mode:
[1] Export Lists - Extract data from SharePoint to XML files
[2] Import Lists - Load existing XML files into Dynamics 365
[3] Export Documents - Extract document attachments from SharePoint lists
[Q] Quit

Enter your choice (1, 2, 3, or Q): 3
```

7. Select Y to proceed.

```
Windows PowerShell
POLExportPath: C:\Users\JohnPrice\Desktop\POLDataMigration\2.Project-and-Resource-Migration\Files

=====
WARNING: DOCUMENT EXPORT SCOPE AND LIMITATIONS
=====

This tool extracts DOCUMENTS ONLY from SharePoint project sites.

What IS exported:
[OK] Documents and folder hierarchy from document libraries
[OK] Basic file metadata (Name, Path, Size, Created, Modified, etc.)
[OK] Version history for each document

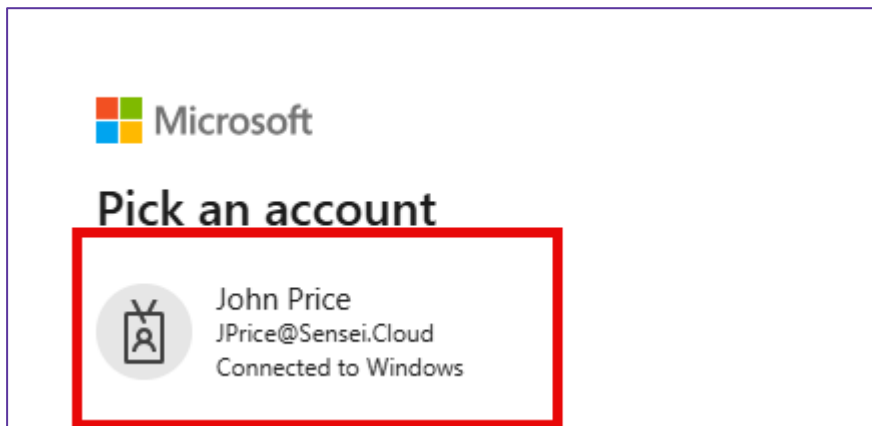
What IS NOT exported:
[NOT EXPORTED] Lists and list items (Risks, Issues, Tasks, etc.)
[NOT EXPORTED] Web part pages, modern site pages, wiki pages
[NOT EXPORTED] Picture libraries
[NOT EXPORTED] Custom lists
[NOT EXPORTED] Workflows and workflow approval history
[NOT EXPORTED] Site settings, permissions, and configurations
[NOT EXPORTED] Versions of pages
[NOT EXPORTED] Custom metadata beyond basic file properties

NOTE: For complete SharePoint migration including lists, pages, workflows,
      and full site structure, use dedicated migration tools (e.g., ShareGate).

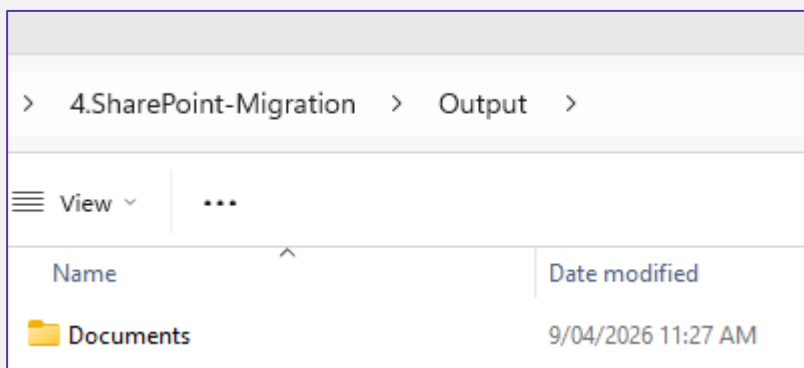
=====
Do you understand the scope and wish to continue? (Y/N): Y
```

Update and run the export script

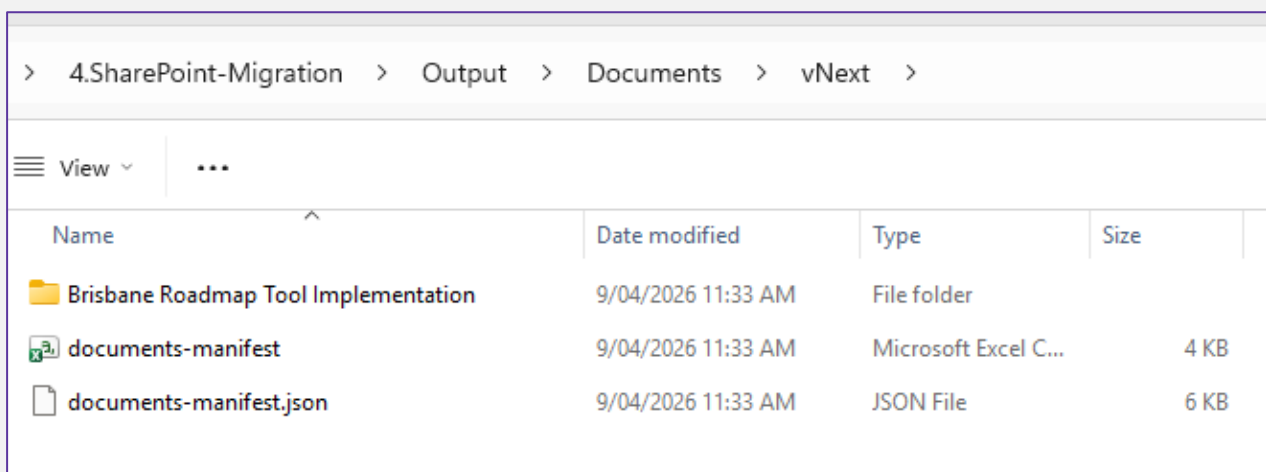
8. Select the account you use to access Project Online.



9. All documents will be output to this location.



One folder will be created per project site:



3. Complete Post-Migration steps

Once you have completed the documents export process there are several post-migration steps you will need to complete:

1. Move the documents to the target location.
2. Set the Document Metadata (where applicable).
3. Remove the version history (where applicable).

1. Move the documents to the target location.

Once you have exported all the documents, you will need to move them to the required Altus location.

The example below shows this process for the Teams/Groups channel.

Quick Summary:

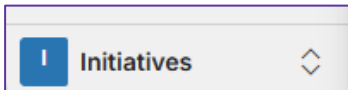
1. Login to the Altus environment:
2. Open **Initiatives**.
3. Open **Projects**.
4. Select the project you wish to create the Team/Group for.
5. Select **Open in SharePoint**.
6. Select **Documents**.
7. Select the exported documents folder and drill down to the documents folder.
8. Drag all contents except the manifest to SharePoint.

1. Move the documents to the target location.

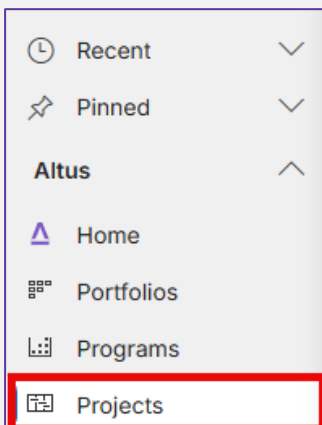
1. Login to the Altus environment:

Example: <https://jp1.crm6.dynamics.com/main.aspx?appid=918ae574-9b21-f111-8341-6045bdc25c37>

2. Open **Initiatives**.



3. Open **Projects**.



1. Move the documents to the target location.**4. Select the project you wish to create the Team/Group for.**

All Active Projects - Timeline ▾

Beta ▾ Expand All

Name

Brisbane Roadmap Tool Implementation

5. Select Open in SharePoint.

← Save Save & Close Delete Open in Teams

Brisbane Roadmap Tool Implementation - Saved
Project · Tracking ▾

6. Select Documents.

BR **Brisbane Roadmap Tool Implementation**

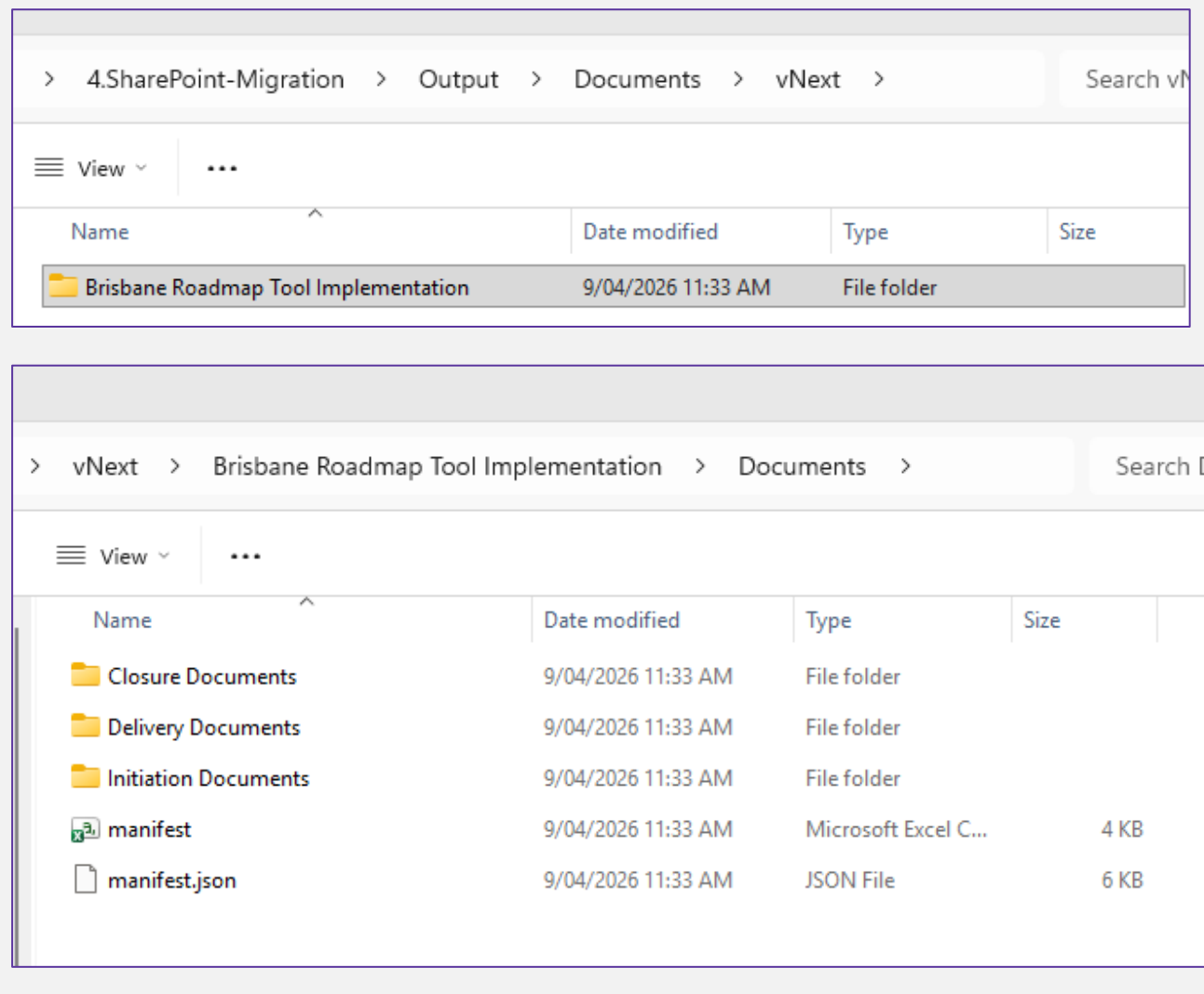
Home
Conversations
Documents
Notebook
Pages
Site contents
Recycle bin
Edit

Documents ▾
All Documents + Add view

AI actions ▾ Add shortcut to OneDrive Forms Edit in grid view Pin to Quick ad

Name ▾ Modified ⓘ ▾ Modified By ▾ + Add column

Your files will show up here

1. Move the documents to the target location.**7. Select the exported documents folder and drill down to the documents folder.**

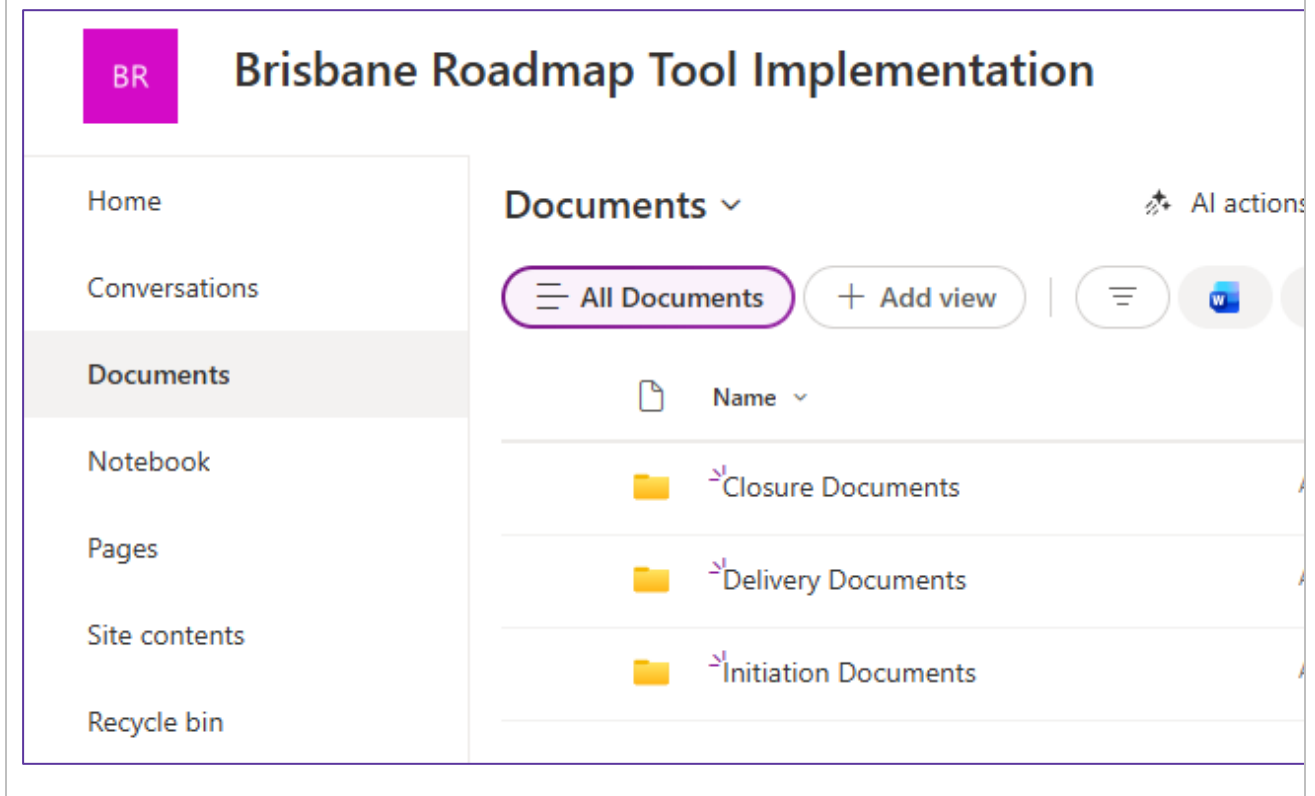
The image displays two screenshots of a file explorer interface, likely from a web application.

Top Screenshot: The breadcrumb path is `> 4.SharePoint-Migration > Output > Documents > vNext >`. The search bar contains "Search vN". The view is set to "View". A table lists the contents of the current directory:

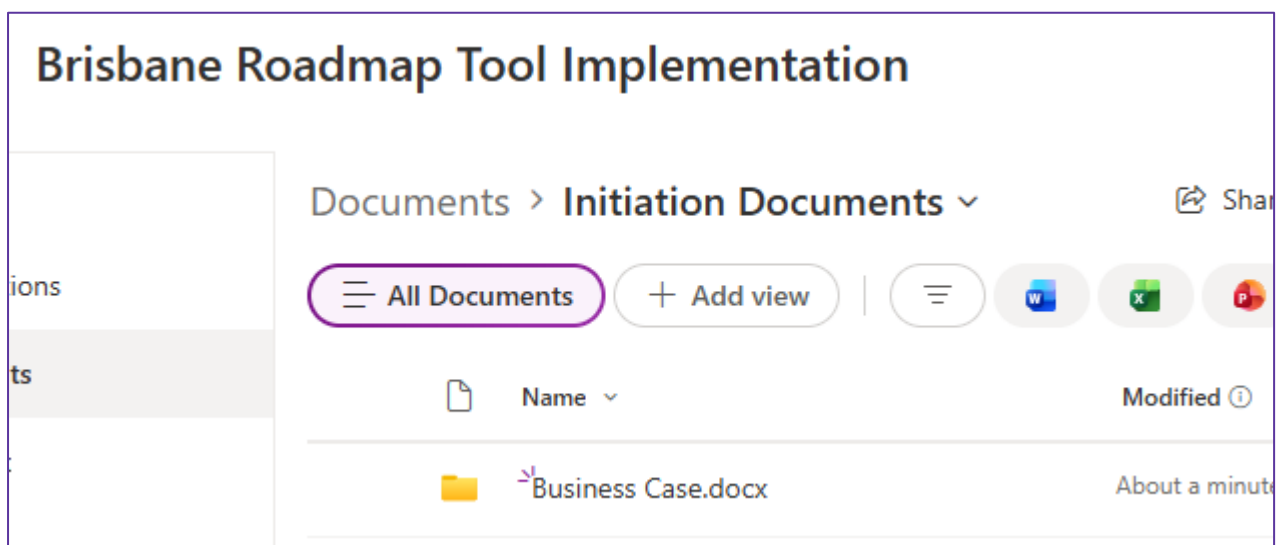
Name	Date modified	Type	Size
 Brisbane Roadmap Tool Implementation	9/04/2026 11:33 AM	File folder	

Bottom Screenshot: The breadcrumb path is `> vNext > Brisbane Roadmap Tool Implementation > Documents >`. The search bar contains "Search D". The view is set to "View". A table lists the contents of the 'Documents' folder:

Name	Date modified	Type	Size
 Closure Documents	9/04/2026 11:33 AM	File folder	
 Delivery Documents	9/04/2026 11:33 AM	File folder	
 Initiation Documents	9/04/2026 11:33 AM	File folder	
 manifest	9/04/2026 11:33 AM	Microsoft Excel C...	4 KB
 manifest.json	9/04/2026 11:33 AM	JSON File	6 KB

1. Move the documents to the target location.**8. Drag all contents except the manifest to SharePoint.**

Note: if you select to include Version History and/or Detailed Metadata, a folder will be created for each document. This will contain the metadata and each version of the document, see below for an example:



Brisbane Roadmap Tool Implementation

Documents > Initiation Documents > Business Case.docx

All Documents

+ Add view



	Name	Modified	Modified By	
	Business Case.docx	2 minutes ago	John Price	
	Business Case_v1.0.docx	2 minutes ago	John Price	
	metadata.json	2 minutes ago	John Price	

2. Move the .MPP to the project document library

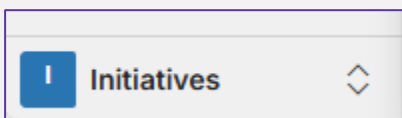
Once the schedule migration has been completed, we recommend that you move the .MPP files either to a central location or to each project document library. The steps below cover moving the schedule to the project document library.

Quick Summary:

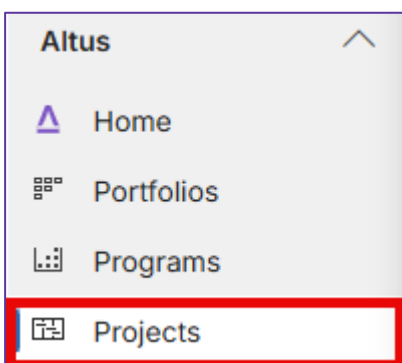
1. Select **Initiatives**.
2. Select **Projects**.
3. Select **Project** to update.
4. Open the **Documents** tab > **Open in SharePoint**.
5. Open this **location** where the .MPP files are located.
6. Drag the **published version** of the schedule to document library.

1. Set Project Type

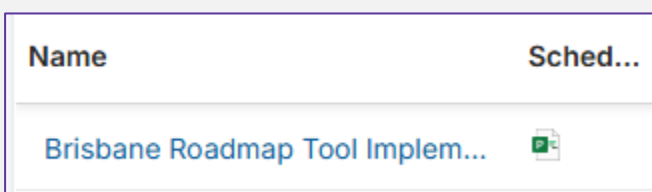
1. Select **Initiatives**.



2. Select **Projects**.

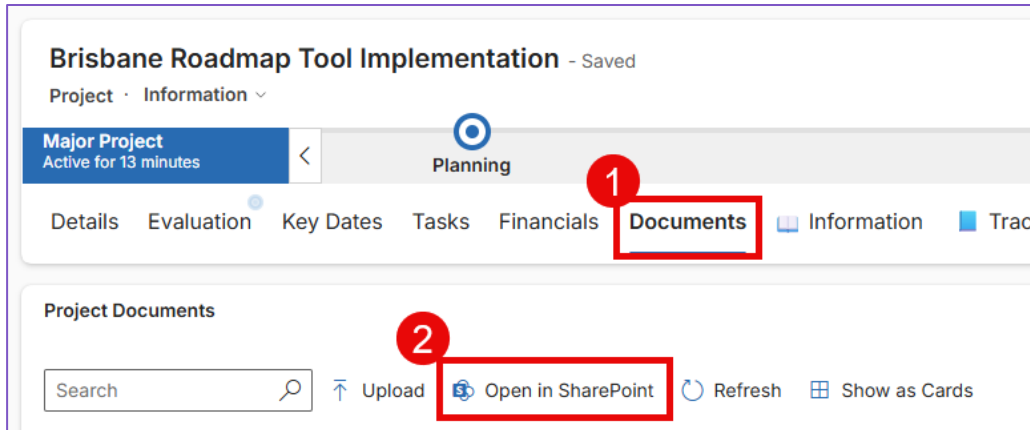


3. Select **Project** to update.



1. Set Project Type

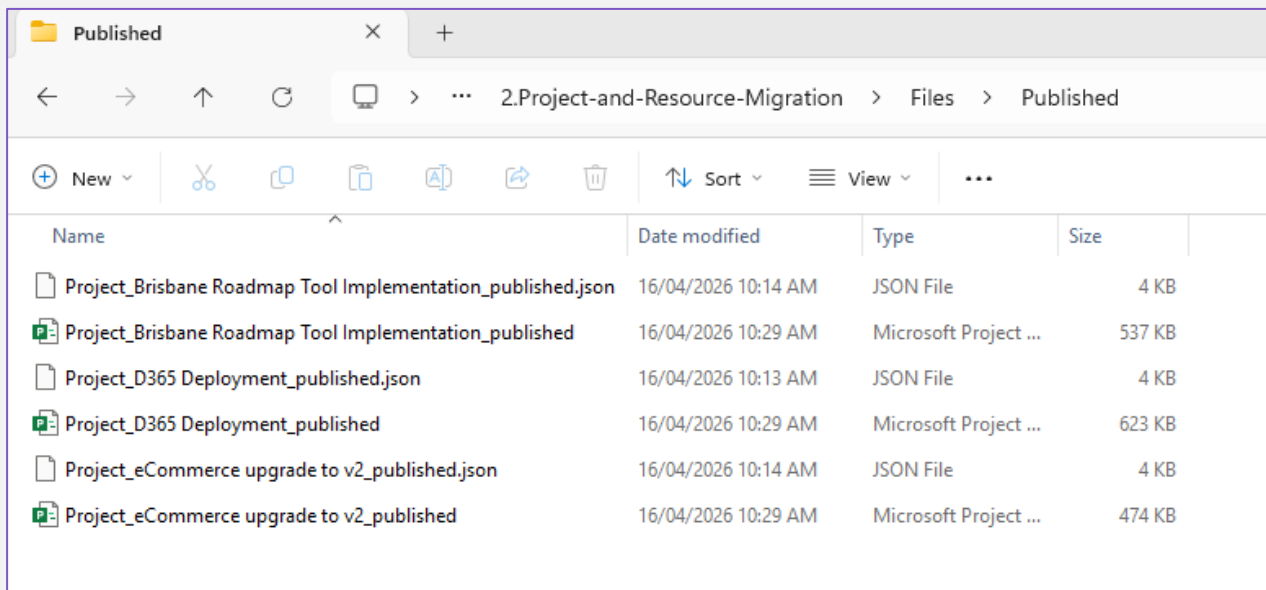
4. Open the **Documents** tab > **Open in SharePoint**.



5. Open this location where the .MPP files are located.

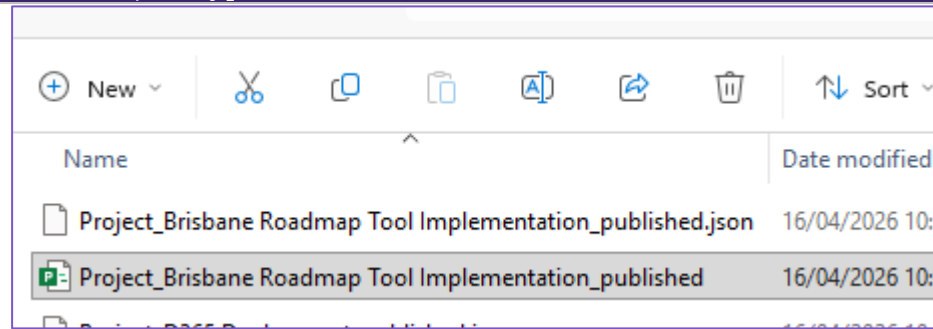
Example:

POLDataMigration\2.Project-and-Resource-Migration\Files\Published



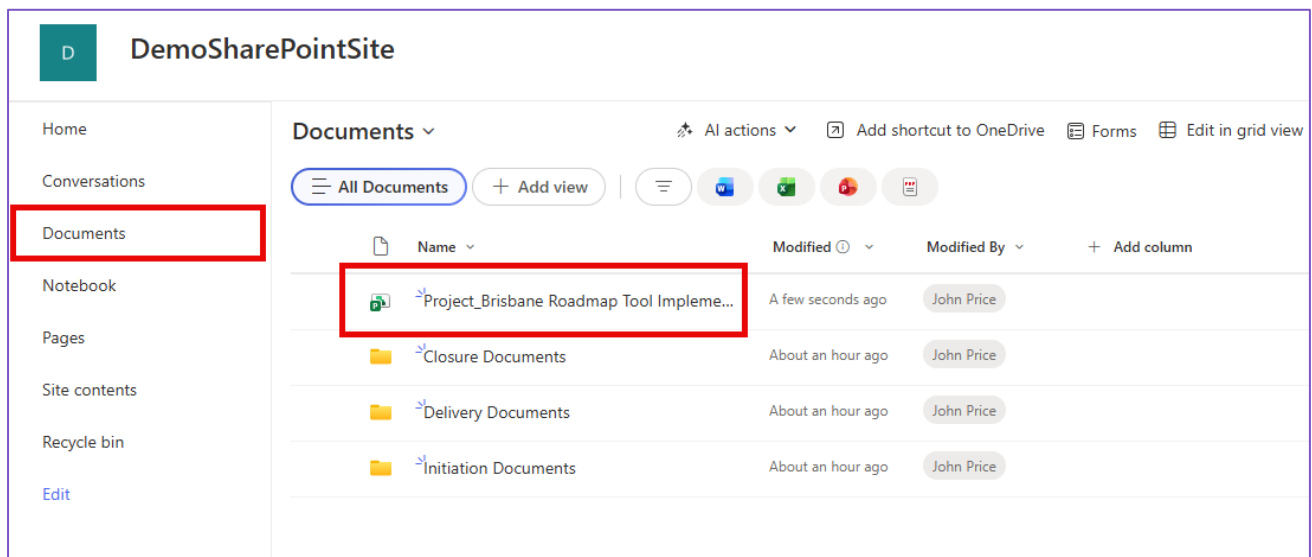
6. Drag the **published version** of the schedule to document library.

1. Set Project Type



Name	Date modified
Project_Brisbane Roadmap Tool Implementation_published.json	16/04/2026 10:
Project_Brisbane Roadmap Tool Implementation_published	16/04/2026 10:

Once done the .MPP will appear in the document library. The connection to the Altus project will be retained.



DemoSharePointSite

Home
Conversations
Documents
Notebook
Pages
Site contents
Recycle bin
Edit

Documents AI actions Add shortcut to OneDrive Forms Edit in grid view

All Documents + Add view

Name	Modified	Modified By	+ Add column
Project_Brisbane Roadmap Tool Impleme...	A few seconds ago	John Price	
Closure Documents	About an hour ago	John Price	
Delivery Documents	About an hour ago	John Price	
Initiation Documents	About an hour ago	John Price	

Troubleshooting

This section covers off some of the errors that might occur whilst running the migration scripts and how to resolve these issues.

Troubleshooting details can be accessed in this document: [Troubleshooting.pdf](#)

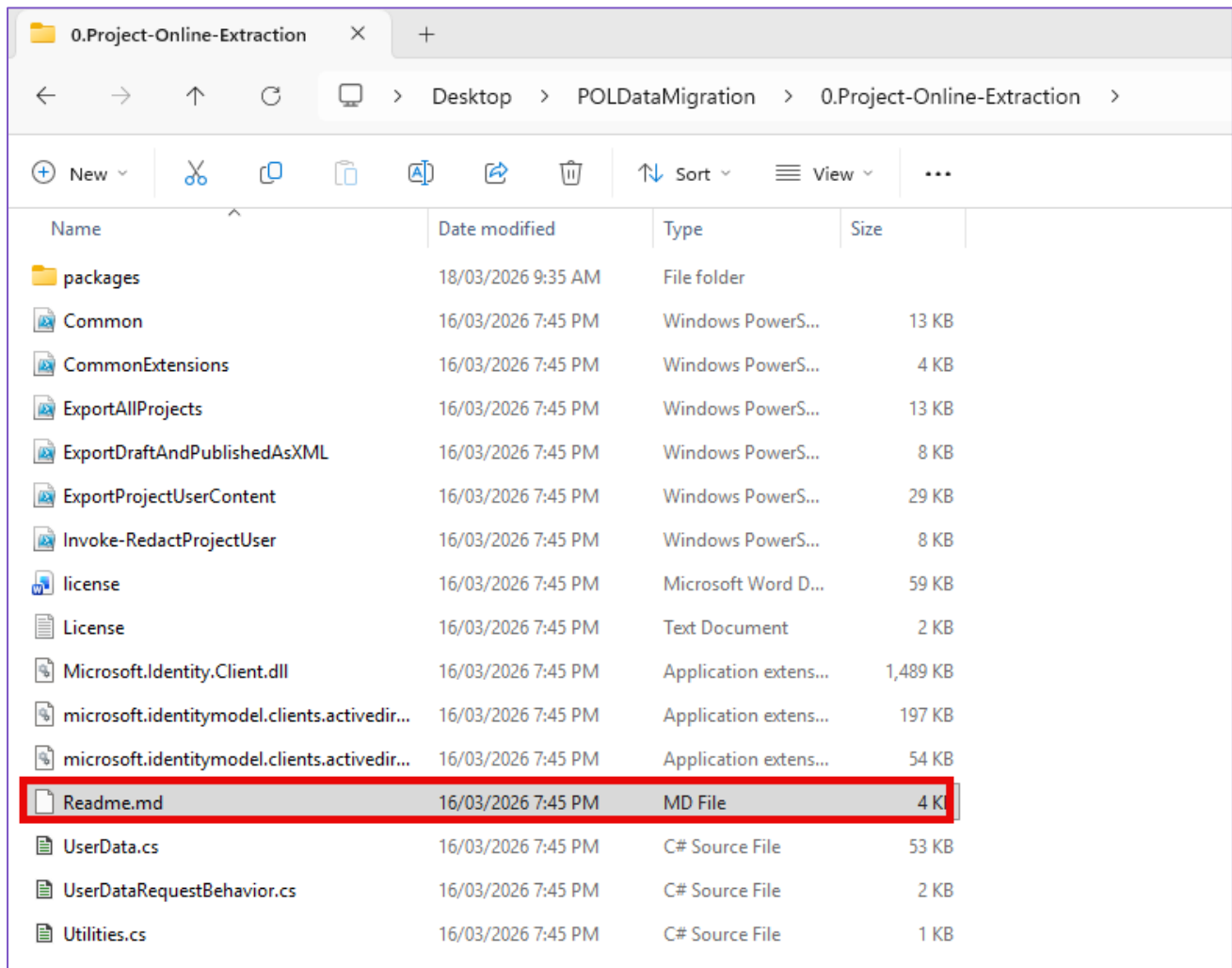
Additional Information

In the script folders that you downloaded, there are readme files that provide additional information about the migration scripts.

These can be accessed as follows:

0.Project-Online-Extraction

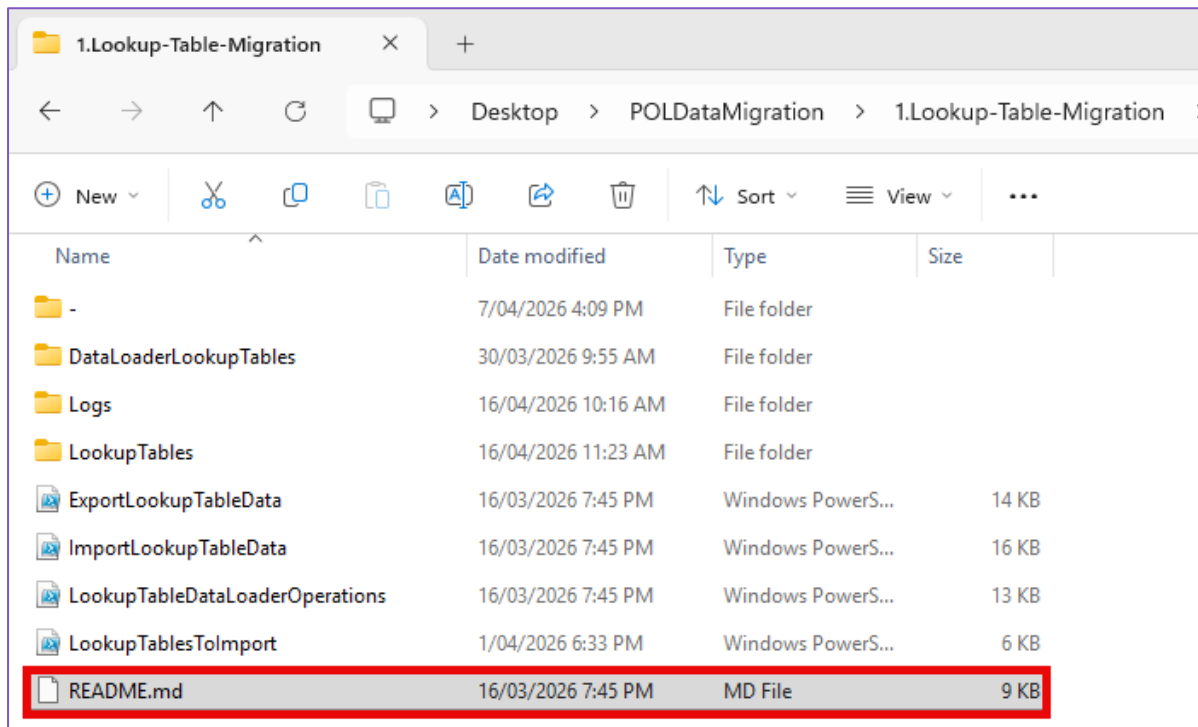
This readme provides additional information regarding the project extraction process:



Name	Date modified	Type	Size
packages	18/03/2026 9:35 AM	File folder	
Common	16/03/2026 7:45 PM	Windows PowerS...	13 KB
CommonExtensions	16/03/2026 7:45 PM	Windows PowerS...	4 KB
ExportAllProjects	16/03/2026 7:45 PM	Windows PowerS...	13 KB
ExportDraftAndPublishedAsXML	16/03/2026 7:45 PM	Windows PowerS...	8 KB
ExportProjectUserContent	16/03/2026 7:45 PM	Windows PowerS...	29 KB
Invoke-RedactProjectUser	16/03/2026 7:45 PM	Windows PowerS...	8 KB
license	16/03/2026 7:45 PM	Microsoft Word D...	59 KB
License	16/03/2026 7:45 PM	Text Document	2 KB
Microsoft.Identity.Client.dll	16/03/2026 7:45 PM	Application extens...	1,489 KB
microsoft.identitymodel.clients.activedir...	16/03/2026 7:45 PM	Application extens...	197 KB
microsoft.identitymodel.clients.activedir...	16/03/2026 7:45 PM	Application extens...	54 KB
Readme.md	16/03/2026 7:45 PM	MD File	4 K
UserData.cs	16/03/2026 7:45 PM	C# Source File	53 KB
UserDataRequestBehavior.cs	16/03/2026 7:45 PM	C# Source File	2 KB
Utilities.cs	16/03/2026 7:45 PM	C# Source File	1 KB

1.Lookup-Table-Migration

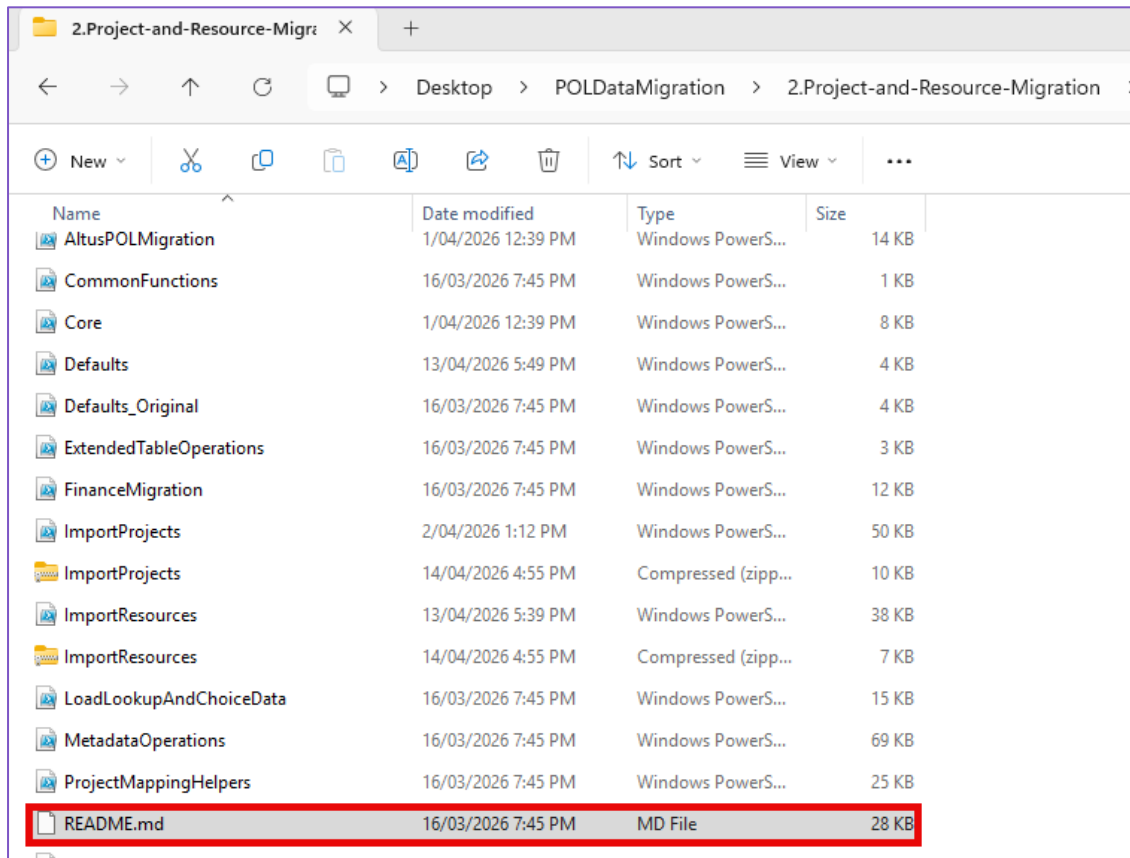
This readme provides additional information regarding the lookup tables migration:



Name	Date modified	Type	Size
-	7/04/2026 4:09 PM	File folder	
DataLoaderLookupTables	30/03/2026 9:55 AM	File folder	
Logs	16/04/2026 10:16 AM	File folder	
LookupTables	16/04/2026 11:23 AM	File folder	
ExportLookupTableData	16/03/2026 7:45 PM	Windows PowerS...	14 KB
ImportLookupTableData	16/03/2026 7:45 PM	Windows PowerS...	16 KB
LookupTableDataLoaderOperations	16/03/2026 7:45 PM	Windows PowerS...	13 KB
LookupTablesToImport	1/04/2026 6:33 PM	Windows PowerS...	6 KB
README.md	16/03/2026 7:45 PM	MD File	9 KB

2.Project-and-Resource-Migration

This readme provides additional information regarding the project and resource migration process:



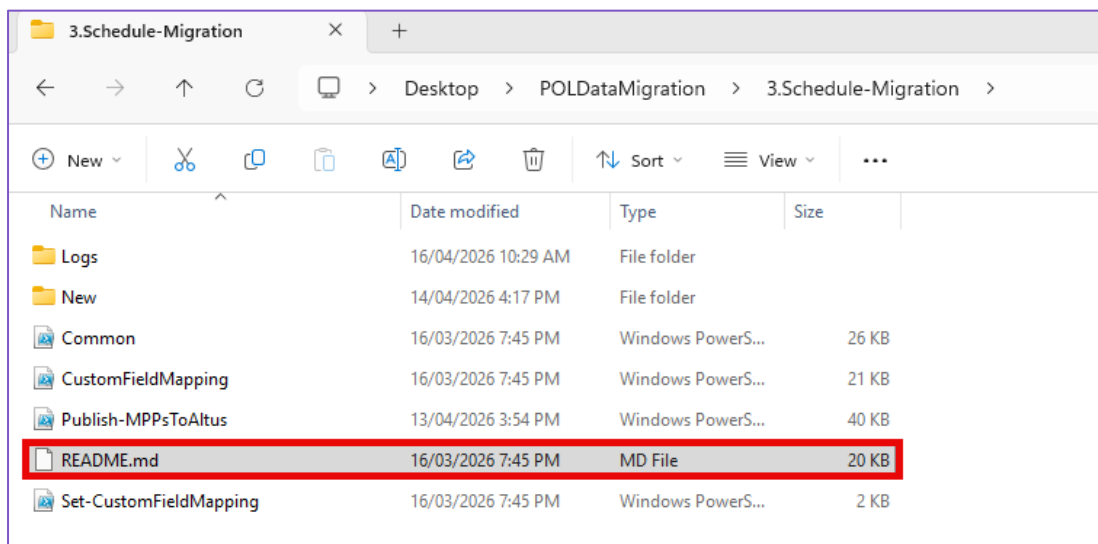
2.Project-and-Resource-Migr X +

Desktop > POLDataMigration > 2.Project-and-Resource-Migration >

Name	Date modified	Type	Size
AltusPOLMigration	1/04/2026 12:39 PM	Windows PowerS...	14 KB
CommonFunctions	16/03/2026 7:45 PM	Windows PowerS...	1 KB
Core	1/04/2026 12:39 PM	Windows PowerS...	8 KB
Defaults	13/04/2026 5:49 PM	Windows PowerS...	4 KB
Defaults_Original	16/03/2026 7:45 PM	Windows PowerS...	4 KB
ExtendedTableOperations	16/03/2026 7:45 PM	Windows PowerS...	3 KB
FinanceMigration	16/03/2026 7:45 PM	Windows PowerS...	12 KB
ImportProjects	2/04/2026 1:12 PM	Windows PowerS...	50 KB
ImportProjects	14/04/2026 4:55 PM	Compressed (zipp...	10 KB
ImportResources	13/04/2026 5:39 PM	Windows PowerS...	38 KB
ImportResources	14/04/2026 4:55 PM	Compressed (zipp...	7 KB
LoadLookupAndChoiceData	16/03/2026 7:45 PM	Windows PowerS...	15 KB
MetadataOperations	16/03/2026 7:45 PM	Windows PowerS...	69 KB
ProjectMappingHelpers	16/03/2026 7:45 PM	Windows PowerS...	25 KB
README.md	16/03/2026 7:45 PM	MD File	28 KB

3.Schedule-Migration

This readme provides additional information regarding the schedule migration process:



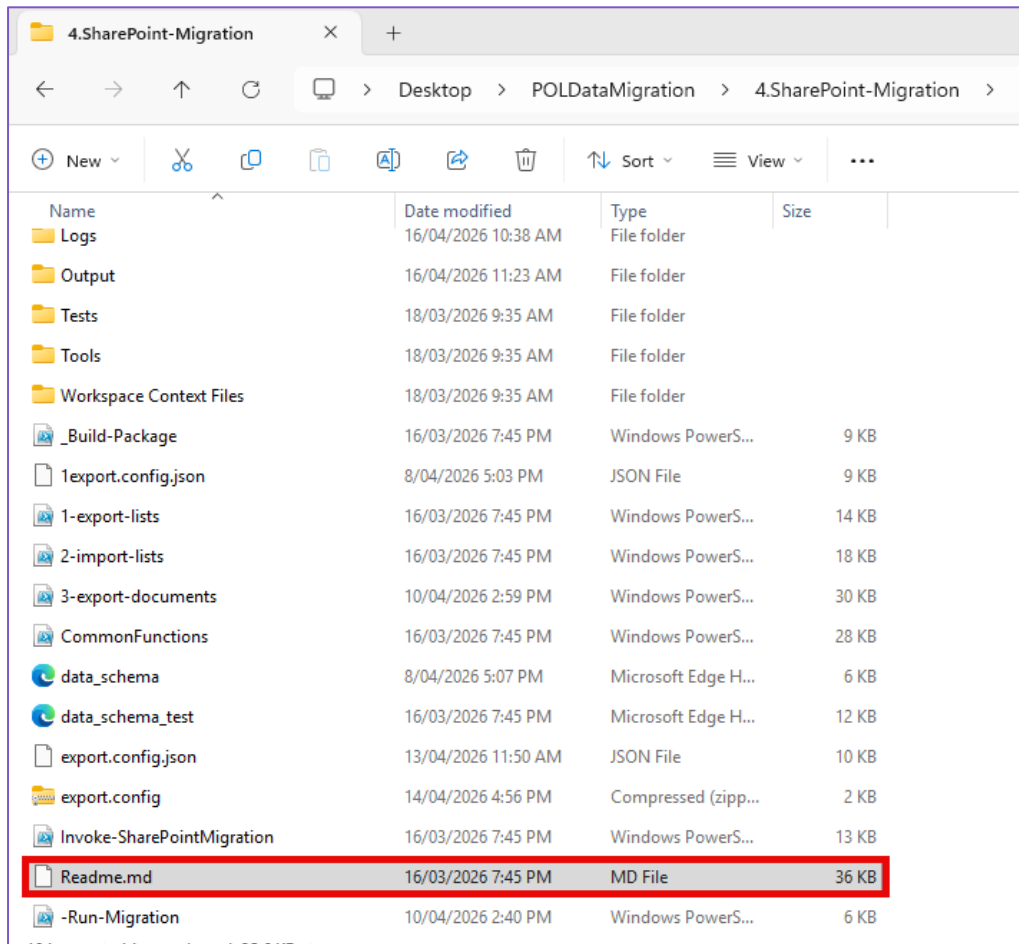
3.Schedule-Migration X +

Desktop > POLDataMigration > 3.Schedule-Migration >

Name	Date modified	Type	Size
Logs	16/04/2026 10:29 AM	File folder	
New	14/04/2026 4:17 PM	File folder	
Common	16/03/2026 7:45 PM	Windows PowerS...	26 KB
CustomFieldMapping	16/03/2026 7:45 PM	Windows PowerS...	21 KB
Publish-MPPsToAltus	13/04/2026 3:54 PM	Windows PowerS...	40 KB
README.md	16/03/2026 7:45 PM	MD File	20 KB
Set-CustomFieldMapping	16/03/2026 7:45 PM	Windows PowerS...	2 KB

4.SharePoint-Migration

This readme provides additional information regarding the SharePoint list and document migration process:



Example Migration scripts

In addition to this document, there are the following videos which provide details of a demo end to end data migration.

- Project Online Data Migration
- SharePoint Online List Data Migration
- SharePoint Online Document Migration

As part of this demo migration, the following scripts were updated. They have been zipped up as a reference point to the user when running their own data migrations.

Script	Updates made
 LookupTablesToImport.zip	Added the details of the lookup tables and choice sets to import to Altus.

 Defaults.zip	Updated to include the lookup tables and choice sets to load. These are aliases and referenced in the ImportResources.ps1 and ImportProjects.ps1 scripts.
 ImportResources.zip	Updated the script to map the resource custom fields between Project Online and Altus.
 ImportProjects.zip	Updated the script to map the project custom fields between Project Online and Altus.
 export.config.zip	Updated the JSON to map the SharePoint list fields between Project Online and Altus.
 -Run-Migration.zip	Updated to include the specific parameters required.

Upcoming Changes

Overview

In addition to the process articulated in the earlier parts of this document, we are working on the following enhancements which have a direction link to the migration of data from Project Online.

This includes the following:

Change	Status	Details
MS Project Cost API	Complete	This enables users to publish all of their costs to Altus for all resource types including Work, Material, Cost, and Fixed Costs.
Task Custom fields mapping script	Complete	This mitigates the need to migrate custom fields one by one.
Project Online export filters	Complete	Enables the user to filter the export script using the Project Online ODATA api.
Bulk migration task custom fields to Altus, bypassing Project Desktop	Complete	We have developed a process to enable custom fields to be migrated directly from Project Online to Altus.

Project Online export filters (Complete)

Enables the user to filter the export script using the Project Online ODATA api.

This has now been completed, details of which are listed below:

Operators

These can be used to determine how fields are filtered:

Operator	Syntax
Equals	eq
Not Equals	ne
Greater Than	gt
Greater Than or Equal To	ge
Less Than	lt
Less Than or Equal To	le

Examples

Filter Value	Syntax
Text	EnterpriseProjectTypeName eq 'Major Project'
Numbers	ProjectFixedCost gt 0
Guids	EnterpriseProjectTypeId eq guid'7473ef8c-9f25-e911-afb0-00155da06b17'
Booleans	ProjectEnterpriseFeatures eq true
Date	ProjectStartDate gt datetime'2024-09-01T00:00:00Z'

Task Custom fields mapping script (Complete)

The original task custom field migration script required users to individually map enterprise Project Online task custom fields to local MS Project fields.

This is a workable solution for customers with only a handful of task-level custom fields, but it not suitable for customers with dozens of task-level custom fields.

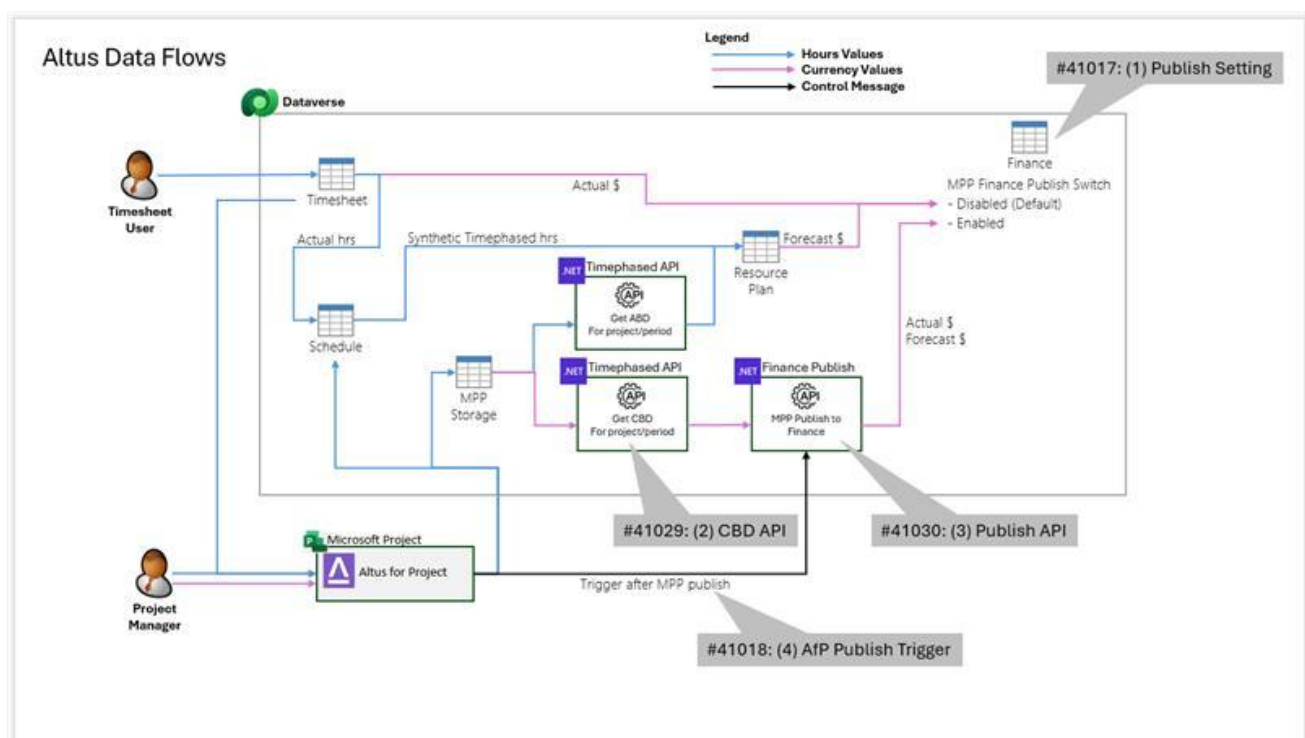
As such, we are working to modify the script so that it only requires a single script run to map and migrate all enterprise project online task custom fields to Altus.

The procedural instructions in [this section](#) of the document will be updated to reflect these changes once this has been deployed.

MS Project Cost API (Complete)

This enables users to publish all of their costs to Altus for all resource types including Work, Material, Cost, and Fixed Costs.

The diagram below provides further information about how this new API works.



The table below covers the key elements of this new API that we are currently building:

Element	Details
Work Resources	Work resource costs are determined using the Rate x Work. The costs will be written to the finance app for each work resource in the schedule.

Material Resources	Material resource costs are determined using the Units x Cost. The costs will be written to the finance app for each work resource in the schedule.
Cost Resources	Cost resource costs are determined using the Cost that has been associated with the task. The costs will be written to the finance app for each work resource in the schedule.
Fixed Costs	Fixed costs are specified using the fixed cost entry table. These will be picked up and written to the finance application for each task with fixed costs.
Granularity	The costs will be aggregated to a monthly level of granularity in the finance application.

Below is a mock-up of how the costs in the finance application are likely to appear once this change has been made.

Details Evaluation Financials Information Tracking Resources Strategy											
Form assist											
+ New ▾											
◀ Mar 2026 - Sep 2026 ▶											
Search											
Category Filter (0) Data ▾ View ▾ More ▾											
Actions	Name	Category		Overall Total	Today 24 Mar 2026 Mar 2026	Apr 2026	May 2026	Jun 2026	Jul 2026	Aug 2026	Sep 2026
...	Cost Penetration Testing	Other	Forecast	12,500				5,000	5,000	2,500	
...	Cost Server Purchase	Hardware	Forecast	2,500	2,500						
...	Material Storage Consumption (GB used)	Hardware	Forecast	1,000	500	500					
...	Material User Licenses	Software	Forecast	5,000	5,000						
...	Work Business Analyst	Labor	Forecast	28,500	5,000	6,000	5,500	4,500	4,500	3,000	
...	Work Project Management	Labor	Forecast	36,000	5,000	6,000	5,000	6,000	5,000	4,500	
...	Work Test Analyst	Labor	Forecast	13,500			4,000	3,500	3,000	3,000	
	Totals		Forecast	99,000	18,000	12,500	14,500	19,000	17,500	13,000	

* Currency displayed is Australian Dollar (\$)
Budget - Estimate at Completion (EAC) = Variance

Save Cancel

Bulk migration task custom fields to Altus, bypassing Project Desktop (Complete)

We have developed a process to enable custom fields to be migrated directly from Project Online to Altus. This enables the following issues to be remediated:

1. It enables all Project Online task-level custom fields to be migrated to Altus, without having to cater for the [hard limits imposed by Project Desktop](#).
2. It enables all text-based task lookup tables to be migrated to Altus along with the task values. This includes flat, hierarchical, and multi-select values. The process via MS Project breaks the lookup tables and converts these to single line of text fields.